SYLLABUS

B.Tech.

Electronics and Communication Engineering

IIITUGECE24



SCHOOL OF ELECTRONICS

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA

HIMACHAL PRADESH

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Institute Vision and Mission

Institute Vision:

To build a vibrant multicultural learning environment with value based academic principles and to achieve excellence in teaching and research and to contribute effectively and responsibly to the national and global community.

Institute Mission:

M1: To provide quality education to the students with practical orientation.

M2: To collaborate with industries and research institutions to solve socially relevant problems.

M3: To inspire students to become responsible citizens and competent professionals with ethical values.

School of Electronics: Vision and Mission

School Vision:

To impart up-to-date technical knowledge of electronics and communication engineering and equip the students to solve socially relevant global problems.

School Mission:

M1: To educate the students with theoretical and practical knowledge by imparting quality education.

M2: To increase research activities in emerging areas through continuous interaction with research organizations and industries.

M3: To encourage students to entrepreneurship, teamwork, and professional ethics in solving real world problems.

B Tech. in ECE

Program Educational Objectives (PEO):

PEO1: Technical Competence: To enable the graduates with the competence of technical skills, entrepreneurship abilities or research attitude for a successful career in electronics and communication engineering.

PEO2: Professionalism: To collaborate with industries and research institutions to solve socially relevant problems.

PEO3: Life-long Learning: To inspire students to become responsible citizens and competent professionals with ethical values.

PEO to Mission Mapping:

Mission / PEO	M1	M2	M3
PEO1	3	3	2
PEO2	2	2	3
PEO3	3	2	3

PEO	Mission	Mapping Level	Justification
PEO1	**M1**	3	Mapped strongly as the students will gain technical knowledge in the field of electronics and communication engineering that will enable them to have successful careers.
	M2	3	Mapped strongly as, with the interaction with the research organizations and industries, students will be exposed to state-of-art offerings in the field of ECE and will be motivated to pursue successful professional lives.
	M3	2	Mapped moderately as students will work on real-life problems as a part of their careers keeping in view the various ethical concerns.
PEO2	**M1**	2	Mapped moderately as the graduates will practice the various ethical and social responsibilities in their professional lives.
	M2	2	Mapped moderately as the graduates will practice the various ethical and social responsibilities in their professional lives by being exposed to the working of the organizations.
	M3	3	Mapped strongly as the graduates will be practicing the various ethical practices in their field of employment while solving various societal problems.
PEO3	**M1**	3	Mapped strongly as the graduates will be requiring state-of-art technical knowhow for lifelong learning.
	M2	2	Mapped moderately as graduates will be required to keep themselves updated with the latest technologies to be successful in their professional careers.
	M3	3	Mapped strongly as the graduates will be requiring strong technical and ethical know how to be successful entrepreneurs while solving socially relevant problems.

Program Specific Outcome (PSO):

PSO1: To solve problems and develop innovative solutions pertaining to next generation communication systems, artificial intelligence and high-speed networks.

PSO2: To identify processes and components for the design and development of eco-friendly and energy efficient embedded and VLSI systems.

PSO3: To apply the fundamentals of mathematics, basic sciences and electronics and communication engineering to develop ingenious solutions in multidisciplinary fields using the ethical practices.

PO/ PSO to PEO Mapping:

PO's		PEO1	PEO2	PEO3
PO1	Engineering Knowledge	3	1	3
PO2	Problem Analysis	3	1	3
PO3	Design/Development of solutions	3	2	3
PO4	Conduct investigations of complex problems	3	1	3
PO5	Modern tool usage	3	2	3
PO6	The engineer and society	1	3	1
PO7	Environment and sustainability	1	2	3
PO8	Ethics	2	3	3
PO9	Individual and Teamwork	3	3	2
PO10	Communication	2	3	3
PO11	Project management and finance	1	2	2
PO12	Lifelong Learning	3	2	3
PSO1	To solve problems and develop innovative solutions pertaining to next generation communication systems, artificial intelligence and high-speed networks.	3	2	3
PSO2	To identify processes and components for the design and development of eco-friendly and energy efficient embedded and VLSI systems.	2	3	2
PSO3	To apply the fundamentals of mathematics, basic sciences and electronics and communication engineering to develop ingenious solutions in multidisciplinary fields using the ethical practices.	1	2	3

Design of Curriculum

The B.Tech. Course Curriculum has been designed conforming to the recommendations of Senate and guidelines of AICTE/UGC, including NEP 2020.

CONFORMANCE TO NEP 2020

As per the approved ordinance of the institute.

Highlights of Curriculum

The students will study the curriculum with the following features:

- More Practical Oriented Teaching and Learning
- More hands-on Projects
- Multi-Disciplinary Electives/Projects
- Relative Grading
- Soft Skill Enhancement
- Overall Personality Development
- Employable Graduates for Industries
- Excellent Placements

Curriculum Components

S.No.	Category	Credits
1.	Major (Core)	75
2.	Minor (Stream)	14
3.	Multi-disciplinary	19
4.	Ability Enhancement	09
5.	Skill Enhancement	16
6.	Valued Added Courses	08
7.	Project	19
8.	Honours/Online Optional Courses	12
Total[#]		160

Total credits for degree with Honors is 160+12 = 172.

Coding Scheme

Example

E	C	M	C	1	0	1
1	2	3	4	5	6	7

The 7 letter code represents the following:	
Letter 1 & 2:	Academic Unit such as IC (Institute Core), CS (Computer Science), EC (Electronics Communication), DS (Data Science), CY (Cyber Security), IT (Information Technology), etc.
Letter 3 & 4:	Curriculum Component such as MC (Major Core), VA (Value Added), etc.
Letter 5:	Year (level) of the course such as 1,2,3... etc.
Letter 6 & 7:	Course identification number such as 01, 02, etc.

Academic Unit	
Name	**Code**
Computer Science and Engineering	CS
Computer Science and Engineering (Cyber Security)	CY
Computer Science and Engineering (Data Science)	DS
Electronics and Communication Engineering	EC
Information Technology	IT
Institute Core	IC
School of Computing	SC
School of Electronics	SE

Curriculum Component	
Name	**Code**
Major (Core)	MC
Minor (Stream)	MS
Multi-disciplinary	MD
Ability Enhancement	AE
Skill Enhancement	SE
Value Added Courses	VA
Internship	IN
Project	PR
Honours Online Courses	HC
Optional Online Courses	OC

B. Tech. (ECE)

Major Core (MC) Courses (75 credits)

S.No.	Course Code	Course Name	Credits
1.	ECMC101	Electronics Workshop	02
2.	CSMC101	Basics of Programming in C	04
3.	CSMC102	Computer Workshop	02
4.	CSMC103	Data Structures and Algorithms	04
5.	ECMC102	Basic Electrical and Electronics Engineering	04
6.	ECMC201	Electronic Devices and Circuits	04
7.	ECMC202	Digital Circuits and Systems	04
8.	ECMC203	Signals and Systems	04
9.	ECMC204	Electromagnetic Field Theory	04
10.	ECMC205	Network Analysis and Synthesis	04
11.	CSMC209	Python Programming	04
12.	ECMC206	Design Engineering	02
13.	ECMC207	Microwave Engineering	04
14.	ECMC208	Analog Communication	04
15.	ECMC209	Microprocessors and Microcontrollers	04
16.	ECMC210	Linear Integrated Circuits	04
17.	ECMC211	Control Systems	04
18.	ECMC212	Digital Signal Processing	04
19.	ECMC213	Product Development Lab	01
20.	ECMC301	Digital Communication	04
21.	ECMC302	VLSI Design	04
Total			75

Minor Stream (MS) / Open Elective# (14 credits)

S.No.	SoC		SoE		Credits
	Course Code	Course Name	Course Code	Course Name	
1.	SCMS301	Artificial Intelligence	SEMS301	VLSI Technology	03
2.	SCMS302	Soft Computing*	SEMS302	Speech Signal Processing*	04
3.	SCMS401	Deep Learning*	SEMS401	Introduction to IoT*	04
4.	SCMS402	Cyber Physical Systems	SEMS402	Biomedical Signal Processing	03
	SCMS403	Introduction to Federated Learning	SEMS403	Quantum Computing	
Total					14

*Includes Lab
#A faculty member may also offer a self-designed course as an open elective subject to the approval of the competent authority.

Multi-Disciplinary (MD) Courses (19 credits)

S.No.	Course Code	Course Name	Credits
1.	ICMD101	Introduction to Bioinformatics	03
2.	ICMD102	Engineering Chemistry	04
3.	ICMD103	Engineering Physics	04
4.	ICMD104	Engineering Mathematics-I	04
5.	ICMD105	Engineering Mathematics-II	04
Total			19

Ability Enhancement(AE) Courses (09 credits)

S.No.	Course Code	Course Name	Credits
1.	ICAE101	Communication Skills	04
2.	ICAE301	Professional Communication and Soft Skills	02
3.	ICAE302	Entrepreneurship and Start-ups	03
Total			09

Skill Enhancement (SE) / Program Elective (16 Credits)

S.No.	Course Code	Course Name	Credits
1.	ECSE301	Data Communication and Networks	04
	ECSE302	Embedded Systems	04
2.	ECSE303	Digital Design and Synthesis	04
	ECSE304	Fiber Optic Communication	04
3.	ECSE305	Artificial Neural Networks	04
	ECSE306	Digital Image Processing	04
4.	ECSE401	Wireless Communication	04
	ECSE402	Antenna and Wave Propagation	04
Total			16

Value Added (VA) Courses (8 credits)

S.No.	Course Code	Course Name	Credits
1.	ICVA101	Basic Environmental Science and Engineering	03
2.	ICVA102	Universal Human Values - I (Sustainability Development/Cultural Diversity)	01
3.	ICVA103	Universal Human Values - II (Personal Social Responsibility/Vedic Sciences/ Spiritual Development)	01
4.	ICVA301	Professional Ethics	03
5.	ICVA104	Yoga and Sports	00
6.	ICVA105	Extra-Curricular Activities	00
7.	ICVA201	Design Thinking	00
Total			08

Project (19 Credits)

S.No.	Course Code	Course Name	Credits
1.	ICPR301	Minor Project	04
1.	ICPR401	Major Project - I	05
2.	ICPR402	Major Project – II	10
Total			19

Honours/Optional Online Courses (HC/OC))

S.No.	Course Code	Course Name	Credits
1.	ICHC3XX	Honours On-line Courses (4 Courses)	12
2.	Optional Online Courses (4 Courses)	Internship	0-12*
***Based on number of courses and credits audited.**			

SEMESTER-WISE CURRICULUM

L-T-P-C Notation

L-T-P-C ⇒ Lecture-Tutorial-Practical-Credit.

Credit structure of each course is given in L-T-P-C form (e.g., 2–1–0–3). The numbers corresponding to L, T and P denote the contact hours per week for Lecture, Tutorial and Practical respectively, and that of C denotes the total number of credits for that course in a semester.

SEMESTER I

S.No.	Code	Course Name	L	T	P	C
1	CSMC101	Basics of Programming in C	3	0	2	4
2	ECMC101	Electronics Workshop	0	0	4	2
3	ICMD103	Engineering Physics	3	0	2	4
4	ICMD104	Engineering Mathematics-I	3	1	0	4
5	ICAE101	Communication Skills	3	0	2	4
6	ICVA101	Basic Environmental Science and Engineering	3	0	0	3
7	ICVA102	Universal Human Values - I	0	0	2	1
8	ICVA104	Yoga and Sports	0	0	4	0
Total			15	1	16	22

SEMESTER II

S.No.	Code	Course Name	L	T	P	C
1	CSMC102	Computer Workshop	0	0	4	2
2	CSMC103	Data Structures and Algorithms	3	0	2	4
3	ECMC102	Basic Electrical and Electronics Engineering	3	0	2	4
4	ICMD101	Introduction to Bioinformatics	3	0	0	3
5	ICMD102	Engineering Chemistry	3	0	2	4
6	ICMD105	Engineering Mathematics - II	3	1	0	4
7	ICVA103	Universal Human Values - II	0	0	2	1
8	ICVA105	Extra-Curricular Activity	0	0	4	0
Total			15	1	16	22

SEMESTER III

S.No.	Code	Course Name	L	T	P	C
1	CSMC209	Python Programming	3	0	2	4
2	ECMC201	Electronic Devices and Circuits	3	0	2	4
3	ECMC202	Digital Circuits and Systems	3	0	2	4
4	ECMC203	Signals and Systems	3	0	2	4
5	ECMC204	Electromagnetic Field Theory	3	1	0	4
6	ECMC205	Network Analysis and Synthesis	3	1	0	4
7	ECMC206	Design Engineering	0	0	4	2
Total			18	2	12	26

SEMESTER IV

S.No.	Code	Course Name	L	T	P	C
1	ECMC207	Microwave Engineering	3	0	2	4
2	ECMC208	Analog Communication	3	0	2	4
3	ECMC209	Microprocessors and Microcontrollers	3	0	2	4
4	ECMC210	Linear Integrated Circuits	3	0	2	4
5	ECMC211	Control Systems	3	1	0	4
6	ECMC212	Digital Signal Processing	3	1	0	4
7	ECMC213	Product Development Lab	0	0	2	1
Total			18	2	10	25

SEMESTER V

S.No.	Code	Course Name	L	T	P	C
1.	ECMC301	Digital Communication	3	0	2	4
2.	XXMS3XX	Open Elective – I	3	0	0	3
3.	ICAE301	Professional Communication and Soft Skills	0	0	4	2
4.	ECSE3XX	Program Elective-I	3	0	2	4
5.	ECSE3XX	Program Elective-II	3	0	2	4
7.	ICVA301	Professional Ethics	3	0	0	3
8.	ICHC301	Honours Online Course-I*	X	X	X	3
	ICOC301	Optional Online Course-I*	X	X	X	0-3
Total			15	0	10	20

***NPTEL/SWAYAM/MOOCs/etc.**

SEMESTER VI

S.No.	Code	Course Name	L	T	P	C
1.	ECMC302	VLSI Design	3	0	2	4
2.	XXMS3XX	Open Elective - II	3	0	2	4
3.	ICAE302	Entrepreneurship and Start-ups	3	0	0	3
4.	ECSE3XX	Program Elective-III	3	0	2	4
5.	ICPR301	Minor Project	0	0	8	4
6.	ICHC302	Honours Online Course-II*	X	X	X	3
	ICOC302	Optional Online Course-II*	X	X	X	0-3
Total			12	0	14	19

***NPTEL/SWAYAM/MOOCs/etc.**

SEMESTER VII

S.No.	Code	Course Name	L	T	P	C
1.	XXMS4XX	Open Elective - III	3	0	2	4
2.	XXMS4XX	Open Elective - IV	3	0	0	3
3.	ECSE4XX	Program Elective-IV	3	0	2	4
4.	ICPR401	Major Project-I	0	0	10	5
5.	ICHC401	Honours Online Course-III*	X	X	X	3
	ICOC401	Optional Online Course-III*	X	X	X	0-3
Total			9	0	14	16

***NPTEL/SWAYAM/MOOCs/etc.**

SEMESTER VIII

S.No.	Code	Course Name	L	T	P	C
1.	ICPR402	Major Project-II	0	0	20	10
2.	ICHC402	Honours Online Course-IV*	X	X	X	3
	ICOC402	Optional Online Course-IV*	X	X	X	0-3
Total			0	0	20	10

The students may be allowed to go for industry internship as per the institute rules.

***NPTEL/SWAYAM/MOOCs/etc.**

Summary									
Semester	I	II	III	IV	V	VI	VII	VIII	Total
Credit	22	22	26	25	20	19	16	10	160

SEMESTER I

Course Code	CSMC101
Course Title	Basics of Programming in C
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn algorithmic problem-solving techniques.
- To learn the fundamentals of C programming.
- To compose programs in C using conditions, iterations and decompose a problem into functions.
- To construct programs in C using array, functions and pointers.

COURSE CONTENT

Unit I: Introduction to Computers and Problem Solving Tools
Evaluation of Computers, History of Computers, Computer Organization, Characteristics, Hardware and Software, Modes of operation, Types of programming languages, developing a program. Algorithms, Characteristics, Flowcharts, Principles of Structured and unstructured programming, Functional Languages, Sequential, selective structures, Repetitive structures, Bounded, Unbounded and Infinite iterations – Examples for each.

Unit II: Introduction to C Programming
C character set, Identifiers and Keywords, Data types, Constants, Variables, Declarations, Expressions, Statements, Symbolic constants, Operators, Library functions, Data input and output: Single character input and output, entering input data, writing output data, gets and puts functions, Control and Conditional statements, Modular Programming.

Unit III: Functions in C Programming
Defining a function, accessing a function, Function prototypes, passing arguments to a function, passing arrays to a function, Functions and Procedures, Examples, Parameter passing methods. Arrays, defining an array, processing an array, Multidimensional arrays.

Unit IV: Pointers and Memory Management
Introduction, Passing pointers to a function, Recursion. String Handling, Introduction to Strings, Sample Program, Standard String Library Functions, Array of String.

Unit V: Structures, Unions and File Handling
Structures and Unions, Declaring and Instantiating Structures, Structures as Parameter and Pointer to Structures, Enumerated Data Types, Union, Bit Fields File Processing. Concept of Files, File Opening in Various Modes and Closing of a File, reading from a File, Writing onto a file.

LIST OF EXPERIMENTS

1. Implementation of basics of C programming.
2. Implementation of concepts of conditional statements in C programming.
3. Implementation of concepts of control statements in C programming.
4. Implementation of concepts of array and pointers in C programming.
5. Implementation of concepts of functions in C programming.
6. Implementation of concepts of strings handling functions in C programming.
7. Implementation of concepts of structures and union in C programming.
8. Implementation of concepts of file handling in C programming.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the computer fundamentals and design an algorithm, flowchart, and pseudo-code for a given problem	K2
CO2	Explain the basics and different constructs used in C programming	K2
CO3	Apply the concepts of control structures, arrays, and pointers to implement various algorithms and practice the skill of algorithmic thinking	K3
CO4	Apply the concepts of functions and string handling using C programming	K3
CO5	Apply the concepts of structures, unions, and file handling in C programming	K3
CO6	Solve Real-world problems by utilizing different concepts in C programming.	K5

TEXT BOOKS

1. Deital Paul and Deital Harvey, “C How to Program”, Prentice Hall London, Eighth Edition, 2015.
2. Kernighan Brian W. and Ritchie Dennis M., “The C Programming Language”, Prentice Hall, Second Edition, 2012.
3. “Programming with C”, by Byron S Gottfried, Schaum’s Outlines, Second Edition, Tata McGraw-Hill, 2016.

REFERENCE BOOKS

1. Dromey R.G., “How to solve it by Computer”, Pearson Education, Fourth Reprint, 2007.
2. Y. Kanetkar, “Let us C”, BPB Publication.
3. Hanly J.R. and Koffman E.B., “Problem Solving and Program Design in C”, Pearson Education, Sixth Edition, 2009.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	1	3	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	1	3	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	1	3	-	-	-	-	-	-	-	-	2	-	-
CO6	3	2	3	3	-	-	-	-	-	-	-	-	2	-	-
Score	18	12	6	15	-	-	-	-	-	-	-	-	10	-	-
COM	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-

Course Code	ECMC101
Course Title	Electronics Workshop
L T P C	0-0-4-2
Course Type	MC

COURSE OBJECTIVES

- To familiarize students with basic electronic components.
- To teach them to interface various sensors and actuators with Arduino.
- To implement a real life application using Arduino

LIST OF EXPERIMENTS

1. Introduction to Arduino board and various sensors and actuators.
2. Interfacing of LED with Arduino.
3. Interfacing of Buzzer with Arduino.
4. Interfacing of Ultra-sonic sensor with Arduino.
5. Interfacing of LCD with Arduino.
6. Interfacing of Seven Segment display with Arduino.
7. Interfacing of DC Stepper motor with Arduino.
8. Arduino based Mini Project.

Total Periods: 24

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Acquire knowledge about basic circuits and Arduino module.	K2
CO2	To interface various sensors with Arduino.	K3
CO3	To interface various actuators with Arduino.	K3
CO4	To interface both sensors and actuators with Arduino.	K5
CO5	Make a project targeting some real life application.	K6

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-	-	-	3	-	-	3	2	-	-
CO2	3	2	2	3	-	-	-	-	3	3	3	3	2	-	-
CO3	3	2	2	3	-	-	-	-	3	3	3	3	2	-	-
CO4	3	2	2	3	1	2	3	-	3	3	3	3	2	-	-
CO5	3	2	2	3	1	3	3	3	3	3	3	3	2	-	-
Score	15	10	10	15	2	5	6	3	15	12	12	15	10	-	-
COM	3	2	2	3	1	3	3	3	3	3	3	3	2	-	-

Course Code	ICMD103
Course Title	Engineering Physics
L T P C	3-0-2-4
Course Type	MD

COURSE OBJECTIVES

- To impart knowledge about the limitations of Newtonian Mechanics and alternate formalism of Lagrange and Hamilton and the concept of special relativity and its applications to physical sciences and engineering.
- To study the basic principles of quantum mechanics: Learn how to solve the Schrödinger's equation and its applications.
- To understand the basic framework of solid state physics.
- To apprise the students regarding the concepts of electromagnetic field theory and its applications in engineering.
- To recognize and classify the structures of optical fiber and types.

COURSE CONTENT

Unit I: Classical Mechanics

1. Review of Newtonian Mechanics in Rectilinear Coordinate System, Motion in Plane Polar Coordinates, Conservation Principles, Inertial and Non-inertial Frames, Rigid Body Dynamics. Introductory ideas about Lagrangian and Hamiltonian and their simple applications.
2. Special Theory of Relativity (STR)Michelson-Morley Experiment, Postulates of STR, Galilean Transformation, Lorentz Transformation, Simultaneity, Length Contraction, Time Dilation, Relativistic Addition of Velocities, Mass-Energy Equivalence, Energy-Momentum Relationships.

Unit II: Modern Physics

1. Basics of Quantum Physics: Origin of Quantum Theory, Planck's Quantum Theory, Black Body Radiation, Photoelectric Effect, Compton Effect, Wave-Particle Duality: De Broglie Wavelength, Group and Phase Velocity, Heisenberg's uncertainty Principle, Double Slit Experiment, Schrödinger Equation, Physical interpretation of Wave Function, Elementary Idea of Operators, Eigen-Value Problem, Solution of Schrödinger Equation for simple boundary value problems, Reflection and Transmission Coefficients, Tunneling, Particle in a three Dimensional Box, Degenerate States.

2. Quantum Statistics: Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Statistics. Density of States. Applications of B-E statistics: LASER (Spontaneous and Stimulated Emissions & Absorption, Einstein's Coefficients, Population Inversion, LASER Systems: Ruby LASER, He-Ne LASER, Semiconductor LASER-applications), Bose-Einstein Condensation. Applications of F-D Statistics: Free Electron Model of electrons in metals. Concept of Fermi Energy. Elementary Ideas of Band Theory of Solids, Quantum Computing.

Unit III: Physics of Materials

1. Structure of Materials: Space Lattice and Unit Cells, Crystal System, Symmetry Operation, Miller Indices, Packing Fractions, Structure Determination using X-ray Diffraction, Bragg's Law.

2. Semiconductor: Direct and indirect band gap semiconductors, Electron and Hole concentrations, Doping n-type, p-type, temperature variation of carrier concentration, Fermi level, Zener Diode, Tunnel Diodes, Photodiodes, Light Emitting Diodes. Hall Effect, Superconductors.

3. Magnetic and Dielectric Properties of Materials: Origin of Magnetism, Dia, Para, Ferro, Anti-Ferro and Ferrimagnetism, Soft and Hard Magnetic Materials, Dielectric Properties.

Unit IV: Electromagnetic Field Theory

1. Electrostatics: Coulomb's Law, Gauss's Law, proof of Gauss's Law, Electrostatic Field in Matter: Dielectric Polarization, polarizability and susceptibility, Types of Polarization, Claussius-Mosotti Equation.

2. Magetostatics: Lorentz Force, Equation of Continuity, Biot-Savart Law, Ampere's Law, Magnetostatic field in matter: Torques and forces on magnetic dipoles, Magnetization, Induction.

3. Electrodynamics: Maxwell's Equations, Propagation of EM Waves in Free Space.

Unit V: Fiber Optics Fermat's principle and Snell's law-optical fiber – principle and construction – acceptance cone - numerical aperture – V-Number - types of fibers, Fabrication: Double Crucible Technique-fiber optic communication principle – fiber optic sensors.

LIST OF EXPERIMENTS

1. To find the moment of inertia of a given flywheel.

2. To find the value of charge carrier concentration and Hall coefficient.

3. To determine the value of Planck's constant.

4. To find the value of wavelength of a given light source using Michelson Interferometer.

5. To find the value of wavelength of a given light source using Newton's rings.

6. To verify the Biot-Savart Law using the circular coil carrying current.

7. To find the resonance frequency in a series LCR circuit.

8. To find the energy gap of a material of p-n junction.
9. To verify the de-Broglie relation using electron diffraction.
10. To determine the plateau and optimal operating voltage of a Geiger-Müller
11. To determine the wavelength of a given LASER source using Diffraction Grating.
12. To study the interaction of high energy photons with matter (Compton Effect).
13. To study the B-H hysteresis curve.
14. To determine the value of e/m ratio.
15. To determine the angle, and refractive index of the prism using a spectrometer.

Total Periods: 42 + 24 = 66

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Identify the role of virtual work, Lagrange's and Hamilton's approach to the mechanics and develop skills to impart practical knowledge and apply the knowledge of Special Theory of Relativity.	K3
CO2	Describe and analyze the dynamics of systems that move under the influence of given potential, and to make use of the basics of Quantum Mechanics.	K4
CO3	Apply principles to determine crystal structure, thermal behavior of solids, dielectric, electric and magnetic behavior of solids, and develop skills to impart practical knowledge.	K3
CO4	Apply the concepts related to Faraday's law, induced emf and Maxwell's equations.	K3
CO5	Demonstrate optical fiber communication link, structure, propagation and transmission properties of an optical fiber.	K3

TEXT BOOKS

1. Kleppner, D., and Kolenkow, R. J. "An Introduction to Mechanics", Tata McGraw-Hill, New Delhi, 2000.
2. Griffiths, David J. "Introduction to Quantum Mechanics, 2nd edition", Pearson Education Ltd, 2014.
3. Kittel, Charles "Introduction to Solid State Physics, 8th edition", John Wiley & Sons, Inc, USA, 2005.

4. Griffiths, David J. " Introduction to Electrodynamics, 3rd edition", Prentice-Hall of India, 2005.

5. Kaye, Phillip, Laflamme, R. and Mosca M. "Introduction to Quantum Computing", Oxford University Press, New York, 2007.

6. Ghatak, Ajoy, and Thyagarajan, K., "Introduction to fiber optics", Cambridge University Press, 2000.

REFERENCE BOOKS

1. Goldstein, Herbert, Poole, Charles and Safko, John "Classical Mechanics, 2nd edition", Narosa, 1985.

2. Puri, R. K. and Babbar V. K. "Solid State Physics", S. Chand & Co. Pvt. Ltd, New Delhi, 2000.

3. Beiser, Arthur "Concepts of Modern Physics", Tata McGraw-Hill, New Delhi, 1995.

4. Resnick, R. "Introduction to Special Relativity" John Wiley, Singapore, 2000.

5. Avadhanulu, M. N. and Kashirsagar, P. G. "A Text Book of Engineering Physics", S. Chand & Co. Pvt. Ltd, New Delhi, 2008.

6. Ida, Nathan "Engineering Electromagnetics", Springer, 2005. Feynman, R. P., Leighton, R. B. and Sands, M."The Feynman Lectures on Physics, Vol. I" Narosa Publishing House, 1998.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	-	2	-	-	-	-	-	-	-	-	-	-	2
CO2	2	2	-	2	-	-	-	-	-	-	-	-	-	-	2
CO3	3	2	-	3	-	-	-	-	-	-	-	2	-	-	2
CO4	2	-	-	2	-	-	-	-	-	-	-	-	-	-	2
CO5	2	-	-	2	-	-	-	-	-	-	-	-	-	-	2
Score	11	4	-	11	-	-	-	-	-	-	-	2	-	-	10
COM	3	2	-	3	-	-	-	-	-	-	-	2	-	-	2

Course Code	ICMD104
Course Title	Engineering Mathematics-I
L T P C	3-1-0-4
Course Type	MD

COURSE OBJECTIVES

- To learn mathematical concepts and methods required for solving engineering problems.
- To acquire fundamental knowledge of linear algebra and calculus
- To understand fundamental concepts of mathematics such as convergence of series, continuity and differentiability of functions of two or more variables.
- To be equipped with knowledge of definite, indefinite integrals and multiple integrals and apply them in various engineering problems.

COURSE CONTENT

Unit I: Matrices
Matrices, Related matrices, Complex matrices, Rank of a matrix, Normal form of a matrix, Characteristic equation, Eigenvalues, Eigen vectors, Properties of Eigen values, Cayley- Hamilton theorem, Reduction to diagonal form, Quadratic form and their reduction to canonical form, Consistency of a linear system of equations, Rouche's theorem, Solution of linear system of equations (homogeneous and non-homogeneous), Gauss Elimination and Gauss Jordan methods.

Unit II: Infinite Series
Convergence and divergence of infinite series, Geometric series test, Positive term series, p-series test, Comparison test, D'A1embert's ratio test, Cauchy's root test (Radical test), Integral test, Raabe's test, Logarithmic test, Gauss's test, Alternating series and Leibnitz's rule, Power series, Taylor series, Radius and interval of convergence.

Unit III: Elementary calculus
Function of several variables, Limit, Continuity and Differentiability, Uniform continuity, Maxima and Minima, Mean value theorem, Partial Derivatives, Integration, Gauss's, Green's and Stokes' theorems.

Unit IV: Vector Spaces
Vector spaces, Sub Spaces, Linear Dependence and Independence of Vectors, Span, Bases and Dimensions, Direct Sum.

Unit V: Linear Transformations
Linear Transformations, Range Space and Rank, Null Space and Nullity, Matrix of Linear Transformations, Matrix Representation of a linear transformation, Change of bases.

Total Periods: 48

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate qualitative problems based on matrix analysis such as rank, Eigen values, and Eigen vectors etc.	K3
CO2	Test the convergence of the series by approximating complicated functions appearing in different engineering models.	K2
CO3	Simplify the problems on differentiation of functions of two variables and know about the maximization and minimization of these functions.	K4
CO4	Make use of the concepts of vector analysis such as linear independence and dependence of vectors etc.	K3
CO5	Interpret the use of linear transformation in real world problems.	K2

TEXT BOOKS

1. Jain R.K., lyengar S.R.K., “Advanced Engineering Mathematics”, Narosa Pub. House, Fifth Edition, 2016.
2. K. Hoffman and R. Kunze, “Linear Algebra ”, Prentice Hall, 2008.
3. G.Strang, “Linear Algebra and its Applications”, 4th Edition, Thomson, 2006.
4. Thomas, G. B., Finney, R. L., Weir, M. D., & Giordano, F. R. (2003). Thomas’ calculus. Reading: Addison-Wesley.

REFERENCE BOOKS

1. Wilfred Kaplan, “Advanced Calculus ”, Pearson, 2003.
2. Wylie, C.R. and Barrett, L.C., “Advanced Engineering Mathematics ”, 6th Edition, McGraw-Hill Inc.US, 1995.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
Score	15	10	-	-	-	-	-	-	-	-	-	-	10	-	-
COM	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-

Course Code	ICAE101
Course Title	Communication Skills
L T P C	3-0-2-4
Course Type	AE

COURSE OBJECTIVES

- To identify, rectify, and overcome mother tongue influence and sensitize usage of native English speech sounds, word accent, intonation and rhythm.
- To develop awareness about different forms of professional communication & social behaviour.
- To empower students with appropriate language usage for presentation delivery, interviews, group discussions and public speaking.

COURSE CONTENT

Unit I: The Process of Communication

1. Grammar Refresh: Synonyms and Antonyms, Homophones, Homonyms and Homographs, Tenses, Active Voice and Passive Voice, Idioms and Phrasal Verbs, Reported Speech.
2. Introduction to Communication, Communication Models, Noise/Barriers in Communication, Nonverbal Communication, Channels of communication, Technical Communication, Downward-Upward Communication, Internal-External Communication, Horizontal-Diagonal Communication, Written vs. oral Communication, Conversational problems of second language users, Difference between conversation and other speech events.
3. How to write Accurately, Briefly, clearly. Precis writing.
4. How to Read, Introduction to Comprehension Skills, Skills to improve Comprehension Skills.
5. Telephonic Communication, Templates for Telephonic Conversation, Do's and Don'ts of Telephonic Communication, Leaving a message.

Unit II: Job Applications and Interviews

1. Format of Resume and Cover Letter, how to make a great Resume, How to write a Covering Letter to Resume.
2. Preparing for an Interview, Self-Introduction in Interview, Select Questions and how to answer them, Mock Interview.
3. What is Group Discussion, how to ace you GD, Do's and Don'ts of GD, Mock GD.

Unit III: Managing Organizational Structure

1. Organizational Roles, Leadership and Management, Ad Hoc Committee, Roles and Responsibilities of Committee & its members.
2. Eustress & Distress, Regulating stress.
3. Simulated Conversation Template.
4. Drafting Formal/Corporate Emails.

Unit IV: Taking Notes and Preparing Minutes

1. Planning a Meeting, Roles of the members, Meeting Etiquettes, How to draft Notice of a Meeting, How to draft Agenda of a Meeting, How to draft Minutes of a Meeting.
2. Elements of Report Writing, Procedure & Guidelines, Types and Format.
3. Taking notes, Note-taking skill: essential components.

Unit V: Presentation Skills and Negotiation Skills

1. Parts of a Presentation Delivery, starting a Presentation Delivery, Introduction: Hooking the Audience, Body of a Presentation Delivery, structuring a Presentation Delivery, Conclusion of a Presentation, how to tackle Q & A form Audience, Podium Panic, Body Language, Do's and Don'ts of PD, Mock PD.
2. Types of Corporate Conversations, Negotiation, Mediation & Arbitration, Resolving arguments, Models of Negotiation Process, Types of Negotiation, Skills of a Negotiator, Steps of the Negotiation Process, Skills to improve Negotiation Process.

LIST OF EXPERIMENTS

1. Introduction to Phonetics, Phonetic alphabet.
2. Introduction to Speech Sounds: Vowels & Consonants, , Structure of Syllables & Stress
3. Extempore, Public Speaking.
4. Words and Phrasal Stress.
5. Mock Telephonic Conversation.
6. Resume and Presentation Skills.
7. Group Discussion.
8. Interview Skills.

Total Periods: 42 + 28 = 70

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Develop skills of comprehension, writing and speaking in professional English and learn strategies to enhance independent language learning.	K3
CO2	Use the appropriate template, language and body language for group discussions, interviews and public speaking.	K3
CO3	Explain and apply the nuances of English professional communication in an organisation.	K3
CO4	Plan and execute Meetings, and draft minutes, reports and relevant documents.	K3
CO5	Develop public speaking skills essential for presentation deliveries, negotiations and corporate communications.	K3

TEXT BOOKS

1. Rizvi, M. A., “Effective Technical Communication”, 2nd edition, McGraw Hill Education, 2017.
2. Mohan, K. and Banerji, M., “Developing Communication Skills”, 2nd edition, Laxmi Publications, 2009.

REFERENCE BOOKS

1. Bhatacharya, I., “An Approach to Communication Skills”, Dhanpat Rai & Co., 2007.
2. Evans, D., “Decision maker”, Cambridge University Press, 1997.
3. Thorpe, E., and Thorpe, S., “Objective English”, Pearson Education, New Delhi, 2007.
4. Fisher, D., “Communication in Organizations”, Jaico Publishing House, 2004.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	2	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-	3
Score	-	-	-	-	-	-	-	-	3	11	-	-	-	-	3
COM	-	-	-	-	-	-	-	-	3	3	-	-	2	-	3

Course Code	ICVA101
Course Title	Basic Environmental Science and Engineering
L T P C	3-0-0-3
Course Type	VA

COURSE OBJECTIVES

- Understand the environmental impact of various energy sources.
- Identify and analyze environmental problems and climate change.
- Impacts of environmental biodiversity in our daily life.
- Comprehend environmental impact assessment and sustainability drivers.
- Understand environmental laws and sustainable practices for environmental protection.

COURSE CONTENT

Unit I: Energy Sources and its Environmental Impact
Energy resources in India and their sustainability, basics of solar thermal energy, and wind turbines, different types of conventional power plants, energy demand scenario in India, introduction to energy sources such as fossil fuel, biomass, ocean, chemical and thermonuclear, advantages and disadvantages of conventional vs non-conventional power generation, environmental issues of various power plants, industrial and transport emissions- impacts.

Unit II: Environment, Pollution, and Climate Change
Introduction to the environment: physical environment, biotic environment, biotic and abiotic interactions, environmental pollution (water, air, soil, and noise): sources, effects, control, air quality standards, international standards for drinking water, greenhouse gases effect, acid rain, causes and effects of climate change, greenhouse gases and global warming, international agreements and policies.

Unit III: Impact of Organisms on the Environment
History (scientists and discoveries), classification and nomenclature of microorganisms, structural organization and multiplication of microbes, microscopic examination of microorganisms: light, fluorescent, dark field, phase contrast, and electron microscopy, stains and staining techniques, microbial nutrition and growth, control of microorganisms.

Unit IV: Environmental Impact Assessment and Ecosystem Sustainability
Introduction to Environmental Impact Assessment (EIA), Steps in EIA, Methods of Impact prediction and assessment, Case studies of EIA, Structure, and function of ecosystems, biodiversity: its conservation and importance, Definition and principles of Sustainable development, indicators of sustainability, the role of engineering in sustainable development, Sustainable technologies and practices.

Unit V: Environmental Laws, Policies, and Applications
Overview of major environmental laws and regulations, National and international environmental policies, Role of government and non-governmental organizations, Case studies of environmental policies, aerobic wastewater treatment, Anaerobic wastewater treatment, Bioremediation of contaminated land and water, Bio fertilizers, Bio pesticides, Biosensors.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Classify the environmental impact of various energy sources.	K2
CO2	Analyze different types of pollution and the guidelines for their control in the context of public health.	K3
CO3	Classify microbes according to energy source and carbon source.	K2
CO4	Understand the Environmental Impact Assessment, Monitoring, and Policy Frameworks.	K4
CO5	Apply scientific solutions to preserve water and land from contamination.	K3

TEXT BOOKS

1. Khan, B.H., "Non-Conventional Energy Resources," 3rd Edition, The McGraw Hill Education, 2017.
2. Rai, G. D., "Non-conventional Energy Sources," 6th Edition, Khanna Publishers, 2018.
3. Balasubramanian D., Bryce, C.F.A., Jayaraman K., Green J., and Dharmalingam K., Concepts in Biotechnology, 6th Edition. Hyderabad: Universities Press, 2005.
4. Pelczar M.J., Chan E.C.S., Krieg N. R., Microbiology, 6th Edition. McGraw Hill, India, 2018.
5. Thakur I.S., Environmental Biotechnology: Basic Concepts and Applications, 2nd Edition, I.K. International Publishing House Pvt. Ltd., 2019.
6. Glasson, J., and Therivel, R. "Introduction to environmental impact assessment." Routledge, 2013.
7. Bodansky, D., and van Asselt, H. "The art and craft of international environmental law." Oxford University Press, 2024.
8. Pozzo, B. Private Law and Environmental Sustainability. In Routledge Handbook of Private Law and Sustainability (pp. 3-27). Routledge, 2024.

REFERENCE BOOKS

1. Sargsyan G., Bhatia M., Banerjee S.G., Raghunathan K., and Soni R., Unleashing the Potential of Renewable Energy in India, World Bank Report, 2011.

2. Everett, G., Boyle, S., Peake, and Ramag J., Energy Systems and Sustainability. Power for a sustainable future. 2nd Edition, Oxford University Press, 2011.

3. Wang L.K., Tay J.H., Tay S.T.L., and Hung Y.T. Environmental Bioengineering, 1st Edition, Humana Press, 2010.

4. Evans G.G., and Furlong J., Environmental Biotechnology: Theory and Application, 2nd Edition, Wiley, 2018.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	-	-	-	-	2	-	-	-	-	-	-	2	-
CO2	3	2	2	1	1	-	2	-	-	-	-	2	-	2	2
CO3	-	2	-	-	-	2	2	3	-	-	-	-	-	-	2
CO4	-	-	2	2	1	-	2	-	-	-	-	-	-	-	2
CO5	3	-	2	2	1	2	2	3	-	-	-	2	-	-	2
Score	9	6	6	5	3	4	10	6	-	-	-	4	-	4	8
COM	3	2	2	2	1	2	2	3	-	-	-	2	-	2	2

Course Code	ICVA102
Course Title	Universal Human Values - I
L T P C	0-0-2-1
Course Type	VA

COURSE CONTENT

Module 1: Introduction to Value Education

- Understanding Value Education
- Self-exploration as the Process for Value Education
- Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment
- Right Understanding, Relationship and Physical Facility
- Happiness and Prosperity – Current Scenario
- Method to Fulfil the Basic Human Aspirations

Module 2: Harmony in the Human Being

- Understanding Human being as the Co-existence of the Self and the Body
- Distinguishing between the Needs of the Self and the Body
- The Body as an Instrument of the Self
- Understanding Harmony in the Self
- Harmony of the Self with the Body
- Programme to Ensure self-regulation and Health

Module 3: Harmony in the Family and Society

- Harmony in the Family – the Basic Unit of Human Interaction
- 'Trust' – the Foundational Value in Relationship
- 'Respect' – as the Right Evaluation
- Values in Human-to-Human Relationship

Total Periods: 32

Course Code	ICVA104
Course Title	Yoga and Sports
L T P C	0-0-4-0
Course Type	VA

COURSE OBJECTIVES

- Learning and practicing yoga bring the mind and body together and to lead a whole some, healthy and disease free life.
- To learn about the importance and basic techniques of fitness activities.
- To understand the discipline and coordination among the team.
- To know the basic strategies, rules and regulations of sports.
- To learn about the Self-confidence and self-worth.

COURSE CONTENT

Module 1: Yoga
Yoga: Yogic Posture (Tadasana, Vrksasana, Trikonasana, Virabhadrasana, Urdhva Mukha Svanasana, Savasana), Pranayama, Surya Namaskar, Medidation

Module 2: Physical Fitness
Physical Fitness: Gym and Athletics and Parade Marching Skills

Module 3: Indoor Games
Indoor Games: Any one game from Badminton, Table Tennis.

Module 4: Outdoor Games
Outdoor Games: Anyone game from Cricket, Volley Ball, Basket Ball, Kabaddi, Kho-Kho.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Students will exhibit a physically active lifestyle and will understand that physical activity provides opportunities for enjoyment, challenge and self-expression.	-
CO2	Students will achieve and maintain a health-enhancing level of physical fitness.	-
CO3	Improved understanding of discipline and coordination.	-
CO4	Improved knowledge of rules and strategies of particular games and sports.	-
CO5	Self-confidence and self-worth as they relate to physical education recreation programs.	-

SEMESTER II

Course Code	CSMC102
Course Title	Computer Workshop
Number of Credits (L-T-P-C)	0-0-4-2
Course Type	MC

COURSE OBJECTIVES

- To impart the knowledge of various hardware components of a computer system.
- To provide the skill of assembling the computer system.
- To impart knowledge about the troubleshooting and fault finding the computers and the peripherals.
- To impart the knowledge and usage of various tools such as Power Point, Word Excel, MS Outlook and Latex.
- To learn the basic commands in Linux operating system.
- To learn the basics of computer networks and different networking devices.

LIST OF EXPERIMENTS

1. Introduction to Von-Neumann Architecture, study of hardware components
2. To study and demonstrate the working of Chip Set, BIOS chip, Capacitors, Inductors, Resistors, Hub and Switch, Repeater and Bridges, Router and NIC.
3. To assemble a PC.
4. Dual Booting (Warm Booting and cold booting)
5. Installation process of Windows, Linux operating system(Ubuntu)
6. Study of Device Drivers and Installation process of Device Drivers
7. Hardware Troubleshooting (Demonstration): Students have to be given a PC which does not boot due to improper assembly or defective peripherals. Identifying problem and fixing it for getting to working condition.
8. Software Troubleshooting (Demonstration): Students have to be given a malfunctioning CPU due to system software problems
9. Introduction to MS Word, MS Excel, Power Point Presentation, MS Outlook, Latex.
10. Exposure to Basic commands and system administration in Linux.
11. Introduction to Cyber Security.

Total Periods: 48

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Describe various hardware components of a computer system.	K2
CO2	Identify existing configuration of the computer and peripherals and to troubleshoot common problems	K3
CO3	Make use of various Microsoft tools to solve the problems.	K3
CO4	Experiment with basic commands in Linux operating system.	K3
CO5	Explain the basics concepts related to computer networks and identify different networking devices.	K3

TEXT BOOKS

1. Mueller, Scott, *"Upgrading and Repairing PCs, 22nd Edition"*, QUE, Pearson Education, 2015.
2. Meyers, Mike, *"Introduction to PC Hardware and Troubleshooting"*, Tata McGraw Hill, New Delhi, 2003.
3. Zacker, Craig and Rourke, John, *"The complete reference: PC hardware"*, 1st Edition, Tata McGraw Hill, New Delhi, 2001.

REFERENCE BOOKS

1. Govindarajulu, B., *"IBM PC and Clones hardware troubleshooting and maintenance"*, 2nd Edition, Tata McGraw-Hill, New Delhi, 2008.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	3	3	-	-	-	-	-
CO3	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	3	3	-	-	-	-	-
Score	15	4	-	6	-	-	-	-	6	6	-	-	2	-	-
COM	3	2	-	3	-	-	-	-	3	3	-	-	2	-	-

Course Code	CSMC103
Course Title	Data Structures And Algorithms
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To impart the basic concepts of data structures and algorithms.
- To understand writing algorithms and step by step approach to solving problems with the help of fundamental data structures.
- To understand concepts about searching and sorting techniques.
- To be familiar with basic techniques of algorithm analysis.
- To learn and implement various data structures and algorithms.

COURSE CONTENT

Unit I: Complexity of algorithms and Algorithmic paradigms
Introduction to Algorithms, Introduction to Asymptotic notations and their significance, complexity analysis of algorithms, worst case and average case. Arrays and Recursion functions.

Unit II: Basic Data Structures
Stacks and Queues- Abstract data types, sequential and linked implementations, and applications of stacks and queues. Linked Lists- Abstract data type, sequential and linked representations, comparison of insertion, deletion, and search operations for sequential and linked lists, doubly linked lists, circular lists, skip lists, and applications of lists.

Unit III: Trees and Heaps
Trees- Abstract data type, sequential and linked implementations, tree traversal methods and algorithms, Binary trees. Search Trees- Binary search trees, search efficiency, insertion, and deletion operations, importance of balancing, AVL trees, searching, insertion, and deletions in AVL trees, 2-3 tree, B-tree, B+ Trees. Heaps- Heaps as priority queues, heap implementation, insertion and deletion operations, binary heaps.

Unit IV: Graphs
Definition, terminology, directed and undirected graphs, properties, implementation – adjacency matrix and adjacency linked representation, connectivity in graphs, graph traversal – breadth first and depth first, spanning trees, Graph algorithms: Depth First Search and Breadth First Search with applications, Minimum Spanning Trees and shortest paths.

Unit V: Basic algorithmic techniques
Sorting and Searching techniques with analysis, Greedy algorithms, divide and conquer for sorting problems, dynamic programming for longest common subsequence and Introduction to hashing.

Total Periods: 40

LIST OF EXPERIMENTS

1. Implementation and Operations on Arrays.
2. Implementation of ADT of Stacks.
3. Implementation of Applications of Stacks.
4. Implementation of ADT of Queues.
5. Implementation of Applications of Queues.
6. Implementation of ADT of Lists.
7. Implementation of Applications of Lists.
8. Implementation of Binary Trees.
9. Implementation of Search Trees.
10. Implementation of Heaps.
11. Implementation of Applications of Heaps.
12. Implementation of algorithms of graphs.
13. Implementation of sorting algorithms.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the fundamentals of data structures and algorithms.	K2
CO2	Build data structures for a given problem.	K3
CO3	Illustrate applications and use of tree data structures.	K3
CO4	Compare algorithms for graph data structures.	K4
CO5	Compare the basic algorithmic techniques and choose a suitable one for a given problem.	K5

TEXT BOOKS

1. Skiena Steven S., *"The Algorithm Design Manual"*, Springer, 2nd edition, 2008.
2. Cormen, T., Lieserson, C., Rivest, R., and Stein, C., *"Introductions to Algorithms"*, Prentice-Hall India, 3rd edition, 2009

REFERENCE BOOKS

1. Dasgupta, Sanjoy, Papadimitriou, Christos H. and Vazirani, Umesh V. *"Algorithms"*, Tata McGraw-Hill, 2008.

2. Kruse, Tondo and Leung, *"Data Structures and Program Design in C"*, 2nd edition, Prentice-Hall, 1997.

3. Lipschutz, Seymour, *"Data structures"*, McGraw Hill revised first edition, 2014.

4. Skiena Steven S., *"The Algorithm Design Manual"*, Springer, 2nd edition, 2008.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	2	2	-	3	-	-	-	-	-	3	-	-	3	-	3
CO2	-	-	-	-	3	-	-	-	-	-	-	-	3	-	3
CO3	2	2	-	3	1	-	-	-	-	-	-	-	-	-	-
CO4	2	2	-	3	3	-	-	-	3	-	-	-	-	-	-
CO5	2	2	-	-	1	-	-	-	-	-	-	-	-	-	3
Score	8	8	-	9	8	-	-	-	3	3	-	-	6	-	9
COM	2	2	-	3	2	-	-	-	3	3	-	-	3	-	3

Course Code	ECMC102
Course Title	Basic Electrical and Electronics Engineering
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn the problem-solving techniques in RLC circuits and power measurements.
- To learn the fundamentals of alternating current and direct current.
- To study the characteristics and use of PN junction diode and Zener diode.
- To study the various configurations of NPN and PNP transistors and their applications.
- To understand the JFET characteristics and its use as an amplifier.

COURSE CONTENT

Unit I: Electrical Circuits
Dependent and independent voltage sources and current sources, charging factor, power factor, Kirchoff's Laws: KVL and KCL, Nodal and Mesh analysis, delta to wye and wye to delta transformations, Linear and non-linear devices, RL, RC & RLC circuits, sinusoids, ac fundamentals, power and energy, self and mutual inductances, and energy in coupled circuit, Q factor and bandwidth.

Unit II: Network Theorems
Source transformation, Superposition Theorem, Thevenin's theorem, Norton's theorem, Millman's theorem, Reciprocity theorem and Maximum power transfer theorem Compensation theorem, Tellegen's theorem, and their applications.

Unit III: Semiconductors Diodes
Basics of semiconductors, intrinsic and extrinsic semiconductors, basic operation of a p-n junction diode, DC load line analysis, Rectifier circuits, filters (L, C, LC, π), Zener diode and its characteristics, Zener diode as a voltage regulator.

Unit IV: Bipolar Junction Transistor (BJTs)
Simplified structure, operation of n-p-n and p-n-p transistors, BJT as an amplifier and as a switch, Input and Output Characteristics of CE, CB, and CC configurations, DC load line analysis, selection of operating point (Q-point), gain (α, β, and γ), biasing techniques: fixed, emitter, voltage divider, collector feedback etc.

Unit V: Field Effect Transistor
Structure and basic operation of JFET, types (n-channel and p-channel), drain and transfer characteristics, structure and basic operation of MOSFET, types (depletion and enhancement), Drain and Transfer Characteristics, Comparison of BJT and FET.

Total Periods: 40

LIST OF EXPERIMENTS

1. To calibrate a given wattmeter by direct loading, verify ohm's law for BPLL element, calibrate a voltmeter & ammeter, calibrate a single-phase energy meter by direct loading.
2. To verify the Kirchoff's laws.
3. To verify network theorems, polarity test, voltage ratio test, open circuit test, short circuit test, load test on single phase transformer.
4. To study DSO, Function generator, Multimeter, and DC power supply.
5. To observe the V-I characteristics of PN Junction and Zener diode.
6. To study the half-wave and full-wave rectifier circuits without and with capacitor filter.
7. To observe the input and output characteristics of a transistor, DC biasing the transistor in common- emitter configuration and determine its operating point (i.e., various voltages and currents).
8. To draw the Transfer and Drain Characteristics of JFET.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Apply the various laws and methods to solve the electrical problems.	K2
CO2	Apply different network theorems to find the current and voltage in every branch of a given circuit.	K3
CO3	Explain various semiconductor diodes to develop an electronic circuitry.	K2
CO4	Develop different circuits using the BJT for various applications.	K3
CO5	Make use of different types of FETs for developing an amplifier.	K3

TEXT BOOKS

1. Bird, John. *"Electrical circuit theory and technology"*. 6th Edition, Routledge, 2017.
2. Charles, K.A. and Sadiku, N.O., *"Fundamental of Electric Circuits"*, 6th Edition Tata Mc-Graw Hill, 2018.
3. Hayt, W. H. and Kemmerly, J., *"Engineering Circuit Analysis"*, 8th Edition, McGraw Hill Education, 2013.
4. Boylestad, R. L. and Nashelsky, L., *"Electronic Devices and Circuits theory"*, 10th Edition, Pearson Education, 2013.

REFERENCE BOOKS

1. Sudhakar, A. and Palli, S. S., *"Circuits and Networks: Analysis and Synthesis"*, McGraw-Hill Education, 2017.

2. Sedra and Smith K. C., *"Microelectronics Circuits"*, 5th Edition, Oxford University, 2009.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-
CO3	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-
Score	15	10	8	15	-	-	-	-	-	-	-	-	10	-	-
COM	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-

Course Code	ICMD101
Course Title	Introduction to Bioinformatics
Number of Credits (L-T-P-C)	3-0-0-3
Course Type	MD

COURSE OBJECTIVES

- To learn the basics of molecular biotechnology for understanding the flow of information in the biological system.
- To get familiarized with the bioinformatics tools, biological databases, and resources.
- To know the applications of alignment algorithms for sequence studies.
- To comprehend the structure and evolutionary relatedness of bio-macromolecules using bioinformatics tools.
- To learn the bioinformatics applications in drug design and discovery.

COURSE CONTENT

Unit I: Basic concepts of Biotechnology
Basics of molecular biotechnology: Overview of cell structure, Biomolecules: structure and function, Molecular structure of genes and chromosomes, Nucleic acid: replication, transcription, splicing, genetic code, translation and their regulatory mechanisms, Non-coding and micro RNA, RNA interference, DNA damage and repair, Post-translation modification. Molecular and Analytical Tools: Gene Cloning and Vectors, PCR, DNA sequencing, Protein sequencing, Next-generation sequencing, Gene microarrays, Proteomics, Upstream and Downstream processing: Chromatography, Post-purification analysis methods.

Unit II: Biological Databases
Introduction to Bioinformatics and Computational Biology, Biological sequences, Classification of biological databases: Sequence databases, Structure databases, Genome specific databases, Special databases and applications: microarray, metabolic pathway, motif, and domain databases, Data file formats.

Unit III: Sequence Analysis
Sequence alignment: Homology vs Similarity, Similarity vs Identity, Types of sequence alignment: Pairwise and Multiple sequence alignment, Global alignment, Local alignment, Dotplot, alignment algorithms: Needleman-Wunsch and Smith-Waterman algorithm, Substitution matrices: PAM and BLOSUM, Multiple sequence alignment: Application of multiple alignments, Viewing and editing of MSA and Scoring function, Database similarity searching: Basic Local Alignment Search Tool (BLAST), FASTA, PHI BLAST, PSI BLAST.

Unit IV: Macromolecular structure and phylogenetic analyses
Gene prediction methods, Gene annotation: Structural and Functional gene annotation, Protein

structure visualization, Prediction of protein secondary structure, Tertiary structure prediction: Homology modeling, Threading, Ab-initio prediction, Validation of the predicted structure using Ramachandran plot, Steriochemical properties, Structure-structure alignment. Phylogenetic basics: Molecular clock theory, Ultrametric trees, Distance matrix methods: UPGMA, Neighbouring joining, Character based methods: Maximum Parsimony, Methods of evaluating phylogenetic methods: boot strapping, jackknifing.

Unit V: Bioinformatics approaches for drug discovery
Computer Aided Drug Design: Drug discovery process, Target identification and validation, lead optimization and validation, Computational ligand designing, Analog-Based drug design, Pharmacophores (3D database searching, conformation searches, deriving and using 3D Pharmacophore, Constrained systematic search, Genetic algorithm, Clique detection techniques, maximum likelihood method) and QSAR, Structure-based drug design: Docking, De Novo Drug Design (Fragment Placements, Connection Methods, Sequential Grow), Virtual screening.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate understanding of the biological processes and decipher them using advanced biotechnological tools.	K2
CO2	Use and describe bioinformatics data and information resources.	K3
CO3	Apply computational based solutions for biological perspectives.	K3
CO4	Understand the macromolecules structure prediction methods and analyses the evolutionary relationship among the organisms.	K3
CO5	Apply computational based solutions for drug discovery.	K3

TEXT BOOKS

1. Rastogi S.C., Rastogi P., Mendiratta N., *"Bioinformatics Methods and Applications"*, Prentice Hall India, Fifth edition (2022)

2. Arthur Lesk, *"Introduction to Bioinformatics"*, Oxford University Press, Fifth edition (2019).

3. Attwood TK, and Parry-Smith, *"Introduction to Bioinformatics"*, Pearson Education India, 1st Edition (2007).

REFERENCE BOOKS

1. William J. Thieman and Michael A. Palladino, *"Introduction to Biotechnology"*, Pearson New International Edition India, Fourth Edition (2021).

2. David W. Mount,*"Bioinformatics: Sequence and Genome Analysis"*, Second Edition, CSHL Press US (2004).

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	-	1	1	1	-	-	-	-	-	-	-	-	-	-
CO2	-	2	-	-	2	2	-	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	2	-	-	-	-	-	2	-	2
CO4	-	-	2	-	1	2	2	-	-	-	-	-	2	-	2
CO5	-	-	-	-	-	-	-	2	-	-	-	-	2	-	2
Score	6	4	5	1	4	4	4	2	-	-	-	-	6	-	8
COM	3	2	2	1	2	2	2	2	-	-	-	-	2	-	2

Course Code	ICMD102
Course Title	Engineering Chemistry
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MD

COURSE OBJECTIVES

- To learn about hard-soft water and solve problems based on hardness estimation.
- To comprehend the structure, properties, synthesis and applications of polymers.
- To investigate engineering materials such as fuels and lubricants.
- To acquire a thorough understanding of nanotechnology and practical applications in everyday contexts.
- To apply the electrochemical principles in batteries and semiconductor devices.
- To analyze the structure and properties of compounds using IR, UV, NMR, Thermal analysis and chromatography.
- To develop proficiency in conducting laboratory experiments, demonstrating safe and proper utilization of standard chemistry glassware and equipment.

COURSE CONTENT

Unit I: Water and its Treatment
Sources of water, Hard and Soft Water, Estimation of Hardness by EDTA Method, Softening of Water, Boiler Feed Water, Treatment Methods, Specifications for Drinking Water, BIS and WHO Standards, Desalination Processes- Reverse Osmosis and Electrodialysis.

Unit II: Polymer and Composites
Polymer: Introduction, Functionality and Classification of Polymers, Mechanism of Polymerization, Molecular Weight, Structure-Property Relationship, Moulding Techniques, Synthesis, Properties and Application of Commercially Important Polymers, Conducting Polymers. Composites: Introduction to Composites, Classification, Constituents, Advantages and applications of composites.

Unit III: Fuels and Lubricants
Fuels: Classification, Types of Coal, Determination of Calorific Value of Solid Fuels, Bomb Calorimeter, Theoretical Oxygen, Proximate and Ultimate Analysis of Coal, Manufacture of Metallurgical Coke, Flue Gas Analysis, Biofuels. Lubricants: Definition, Theories, Characteristics, Additives to Lubricants, Solid Lubricants.

Unit IV: Introduction to Nanotechnology
Nanotechnology, Concept of Nanochemistry, Synthesis, Characteristics and Applications of

Carbon Nanostructures- Graphene, Graphite, Fullerenes, Carbon Nanotubes, Nanowires, Nanorods, Porous Nanostructures, Drawbacks and futuristic aspects of Nanotechnology, Green Chemistry, Nanosensors.

Unit V: Basics of Electrochemistry and its Applications
Electrochemical Cells: Introduction, Electrolytic and galvanic cells, electrochemical series, Nernst equation, cell EMF. Batteries: Types, Working principle and uses. Corrosion: Types, Passivity, Polarization, Over-potential and its significance. Factors affecting corrosion. Protection from corrosion. Semiconductor Electrochemistry: Fuel Cells, Hydrogen Cell, Clean energy conversion and storage devices.

Unit VI: Characterization Techniques
Introduction to Spectroscopy, UV-Visible Spectroscopy: Principle, Instrumentation and Application; IR Spectroscopy: Principle and Applications; NMR: Principle, Instrumentation, Applications Of NMR; Thermal Method: Instrumentation, Fundamental Principles and Applications of TGA, DTA and DSC; Introduction to Chromatographic Techniques.

Total Periods: 42

LIST OF EXPERIMENTS

1. To determine the total hardness of the given hard water using EDTA titration method to estimate of total.
2. To determine the carbonate, non-carbonate and total hardness in the given water sample by EDTA method.
3. To determine the alkalinity of the given water sample by titrating it against 0.02 N H2SO4 solution.
4. To determine the strength of given solution of Mohr's salt.
5. To estimate amount of chlorine present in given sample of bleaching powder.
6. To determine the iron content in the given salt by using external indicator.
7. To estimate Dissolved oxygen (DO) level in the given water sample by Winkler's Method.
8. To determine the strength of an acid by pH – metric method.
9. To determine the strength of a mixture of acid solution (HCl and CH3COOH) by titrating against Sodium hydroxide solution conductometrically.
10. To identify the surface tension of an unknown liquid using Stalagmometer.
11. To identify the viscosity of an unknown liquid using Ostwald viscometer.
12. To separate the mixture of amino acids by thin layer chromatography.
13. To prepare urea formaldehyde resin.
14. To prepare polymer of Bakelite.

Total Periods: 42 + 28 = 70

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Distinguish between hard/soft water and solve the day-to-day problems associated with it.	K3
CO2	Explain the properties, structure, synthesis and applications of Polymers in engineering fields.	K2
CO3	Identify the different engineering materials and explain its usefulness in technological advancement.	K3
CO4	Assess the real-world applications of nanotechnology in revolutionizing our surroundings.	K3
CO5	Interpret the operation of electrochemical systems in batteries and semiconductor devices and identify solutions to prevent corrosion.	K4
CO6	Analyse the structures of known and unknown compounds using different characterization techniques.	K4
CO7	Apply the concepts of Engineering Chemistry to real-world situations.	K3

TEXT BOOKS

1. Vairam, S., *"Engineering Chemistry- A textbook of chemistry for engineers"*, Wiley India Pvt. Ltd., 2018.
2. Palanna, O. G., *"Engineering Chemistry"*, Tata McGraw-Hill Publishing Company Ltd., 2017.
3. Dara, S. S. and Umare, S. S., *"A Text book of Engineering Chemistry"*, S. Chand Publishing, 2011.

REFERENCE BOOKS

1. Poole, J.R, Charles, P., and Frank J. owens, *"Introduction to nanotechnology"*, John Wiley and Sons, 2009.
2. Pavia, D.L., Lampman, G.M., Kriz, G.S., *"Introduction to spectroscopy: a guide for students of organic chemistry"*, Philadelphia: W.B. Saunders Co., 1979.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	3	2	-	-	-	-	-	2	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	-	-	-	2	-	-	-	-	-	-	2	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	-	-	-	2	2	-	-	-	-	-	-	-	-
CO6	3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO7	3	2	-	3	-	-	-	-	-	-	-	-	-	-	-
Score	21	14	-	3	-	7	4	-	-	-	-	-	6	-	2
COM	3	2	-	3	-	3	2	-	-	-	-	-	2	-	2

Course Code	ICMD105
Course Title	Engineering Mathematics - II
L T P C	3-1-0-4
Course Type	MD

COURSE OBJECTIVES

- To understand probabilistic models that are employed in countless applications in all areas of science and engineering.
- To acquire knowledge of ordinary differential equations.
- To understand fundamental concepts of Laplace and Fourier transforms.
- To acquire knowledge of the basics of Fourier series and practical Harmonic analysis.

COURSE CONTENT

Unit I: Probability and Random Variable
Axioms of probability, Conditional probability, Total probability, Baye's theorem, Random variable, Probability mass function, probability density function, properties, Moments, Moment generating function and their properties, Joint distributions, Marginal and conditional distribution, Covariance.

Unit II: Standard Distributions
Binomial, Poisson, Geometric, Negative Binomial, Uniform, Exponential, Gamma, Weibull and Normal distributions and their properties, Central limit theorem.

Unit III: Ordinary Differential Equations
Brief review of first order ordinary differential equations, Exact equations, Equations reducible to exact equations, Equations of the first order and higher degree, Clairaut's equation, Applications of differential equations of first order (Orthogonal trajectories, Simultaneous linear first order equations), Linear differential equations with constant coefficients, Complimentary functions and particular integral, Method of variation of parameters, Equations reducible to linear equations with constant coefficients (Cauchy's and Legendre's linear equations), Applications of linear differential equations in engineering.

Unit IV: Laplace Transform
Laplace Transforms of standard functions and their properties, Inverse Laplace Transforms, General Properties of inverse Laplace transforms and Convolution Theorem, Laplace Transforms of periodic functions, Bessel functions, Error function, Dirac-delta Function, Heaviside's Unit Function, Applications to linear simultaneous differential equations.

Unit V: Fourier Series
Euler's formula, Conditions for a Fourier expansion, Functions having points of discontinuity, Change of interval, Odd and even periodic functions, Expansion of odd and even periodic functions, Half-range series, Typical wave-forms, Parseval's formula, Practical harmonic analysis.

Total Periods: 48

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the axiomatic formulation of modern Probability Theory and think of random variables as an intrinsic need for the analysis of random phenomena.	K2
CO2	Classify the function of random variables based on Discrete and Continuous Distributions, which can describe the real phenomenon.	K2
CO3	Analyze and solve system of differential equations which arise in real world problems.	K3
CO4	Apply Laplace transform technique to solve boundary value problems.	K3
CO5	Developing an understanding of Fourier analysis and compare them with harmonic analysis.	K3

TEXT BOOKS

1. Boyce W.E., Diprima R.C., "Elementary differential equations and boundary value problems", Wiley, Tenth Edition.
2. Veerarajan, T., "Probability, Statistics and Random Processes", 3rd edition, Tata McGraw Hill Education Pvt. Ltd., 2017.

REFERENCE BOOKS

1. Ross S., "A First Course in Probability", Pearson Education, Tenth Edition, 2015.
2. Kreyszig, E., "Advanced Engineering Mathematics", Wiley, Tenth Edition, 2015.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
Score	15	10	-	6	-	-	-	-	-	-	-	-	10	-	-
COM	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-

Course Code	ICVA103
Course Title	Universal Human Values - II
L T P C	0-0-2-1
Course Type	VA

COURSE CONTENT

Module I: Recap of UHVPE Part 1 and Harmony in the Society

1. Recap of Introduction to Value Education
2. Recap of Harmony in the Human Being
3. Recap of Harmony in the Family: Feeling of Trust
4. Recap of Harmony in the Family: Feeling of Respect and other Feelings
5. Understanding Harmony in the Society
6. Vision for the Universal Human Order

Module II: Harmony in the Nature (Existence)

1. Understanding Harmony in the Nature
2. Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature
3. Realizing Existence as Co-existence at All Levels
4. The Holistic Perception of Harmony in Existence

Module III: Implications of the Holistic Understanding – a Look at Professional Ethics

1. Basis for Universal Human Values
2. Definitiveness of (Ethical) Human Conduct
3. Professional Ethics in the light of Right Understanding
4. A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order
5. Holistic Technologies, Production Systems and Management Models-Typical Case Studies
6. Strategies for Transition towards Value-based Life and Profession

Total Periods: 32

Course Code	ICVA105
Course Title	Extra-Curricular Activities
L T P C	0-0-4-0
Course Type	VA

COURSE OBJECTIVES

- To actively the students in the institute extracurricular activities.
- To encourage students in participating the club activities.
- To motivate students in registering the student clubs/societies.
- To make the students understand the importance of service.
- To inculcate self-confidence and self-worth into students.

Modules

Module I:
The students need to participate in at least one major event conducted by the institute and provide the certificate as the proof.

Module II:
The student need to participate in at least one club activity organised by the societies.

Module III:
The student need to register in at least one society after joining the institute.

Module IV:
The student need to actively participate in the government initiatives organised by the institute.

Module V:
The student need to attend at least one self-growth sessions like yoga, mental counselling, etc. organised by the institute.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description
CO1	Foster a connection between extracurricular participation and academic pursuit.
CO2	Develop problem-solving and critical thinking skills through strategic planning and execution by participating in various events in clubs and societies.
CO3	Develop intrinsic motivation and a passion for lifelong learning and participation in various club events, govt. initiatives organized by the Institute.
CO4	Enhance the ability to work effectively within a team, understanding roles, responsibilities, and the importance of cooperation.
CO5	Maintain a healthy level of physical and mental fitness.

SEMESTER III

Course Code	CSMC209
Course Title	Python Programming
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn the basics of Python programming, including syntax, variables, data types, and control flow structures.
- To understand and apply modular programming using functions and modules for efficient code reuse.
- To learn the principles of object-oriented programming in Python, including classes, objects, inheritance, and polymorphism.
- To be able to work with Python's built-in data structures and perform file handling for data storage and manipulation.
- To explore basic data visualization techniques and libraries like NumPy and Matplotlib for analyzing and representing data.

COURSE CONTENT

UNIT I: Introduction to Python
Overview of Python Programming, Python Installation: Setting up Python, IDEs (Anaconda, Jupyter Notebook, PyCharm), Basic Syntax: Python keywords, variables, comments, and indentation, Data Types and Variables: Integers, floats, strings, type conversions, Operators: Arithmetic, comparison, logical, bitwise, and assignment operators, Input/Output Operations: input(), print(), formatting output

Unit II: Control Flow and Functions
Conditional Statements, Loops, Comprehensions, Functions: Defining functions, arguments (default, keyword, positional), return values, Lambda Functions: Anonymous functions, usage in functional programming, Recursion: Concept and examples, recursion vs iteration, Exception Handling: try, except, finally, raising exceptions.

Unit III: Data Structures Using Python
Lists: Creation, slicing, methods, list comprehensions, Tuples: Characteristics, tuple operations, immutability, Dictionaries: Key-value pairs, dictionary methods, nested dictionaries, Sets: Set operations, unique elements, set methods, Strings: String manipulation, methods, formatting, f-strings, File Handling: Opening, reading, writing, appending files, file modes, Modules and Packages: Importing modules, creating and using packages.

Unit IV: Object-Oriented Programming and Applied Computing
Introduction to OOP: Principles (Encapsulation, Inheritance, Polymorphism), Classes and Objects, Method, Constructors and Destructors, Inheritance, Polymorphism, Encapsulation, Abstract Classes and Interface, File I/O with CSV and JSON, Regular Expressions: Pattern matching, re-module, meta characters, search and match, Introduction to Libraries: NumPy, Pandas, Matplotlib, Web Scraping.

Unit V: Introductory Machine Learning
Introduction to Machine Learning, Types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning, Regression and Classification, Clustering and Dimensionality Reduction , Training, Testing, Overfitting, and Underfitting, Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Linear Regression, Logistic Regression, Decision Trees, k-Nearest Neighbors (k-NN), k-Means.

LIST OF EXPERIMENTS

1. Familiarize with Python programming, basic syntax, and working with Python in IDEs like Jupyter Notebooks.
2. Write Python scripts using loops, conditionals, and user-defined functions for modular programming.
3. Perform reading, writing, and manipulation of files (text and binary) using Python's built-in file handling techniques.
4. Explore and work with standard Python libraries like 'math', 'os', and 'datetime' for common programming tasks.
5. Implement error detection and handling in Python using 'try', 'except', and 'finally' blocks to make robust programs.
6. Implement sorting and searching algorithms using Python's list and dictionary data structures.
7. Develop programs using OOP principles like classes, inheritance, polymorphism, and encapsulation.
8. Create custom Python modules and use built-in modules for modular programming and code organization.
9. Perform array operations, matrix manipulations, and basic mathematical computations using the NumPy library.
10. Use Pandas for data manipulation, including data frames, filtering, grouping, and handling missing values.
11. Create basic plots, bar charts, histograms, and scatter plots to visualize data using Matplotlib.
12. Use Seaborn for advanced visualizations, including heatmaps, pair plots, and violin plots.

13. Implement pattern matching and text processing using Python's 're' module for regular expressions.
14. Learn how to interact with external APIs, fetch data, and process it using Python's 'requests' module.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the fundamental concepts of Python programming, including syntax, variables, and control structures.	K2
CO2	Apply modular programming techniques using functions and modules for code organization and reuse.	K3
CO3	Demonstrate object-oriented programming concepts like classes, inheritance, and polymorphism to solve problems.	K4
CO4	Utilize Python's data structures and perform file handling operations to manipulate and store data efficiently.	K3
CO5	Implement basic data analysis and visualization using Python libraries such as NumPy, Pandas, and Matplotlib.	K4

TEXT BOOKS

1. Python Crash Course (3rd Edition) by Eric Matthes, Publisher: No Starch Press, 2023.
2. Automate the Boring Stuff with Python (2nd Edition) by Al Sweigart, Publisher: No Starch Press, 2021.

REFERENCE BOOKS

1. Fluent Python: Clear, Concise, and Effective Programming (2nd Edition) by Luciano Ramalho, Publisher: O'Reilly Media, 2022.
2. Python Programming: An Introduction to Computer Science (3rd Edition) by John Zelle, Publisher: Franklin, Beedle & Associates Inc., 2023.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	-	3	1	1	-	1	1	0	0	0
CO2	3	3	2	-	-	2	-	1	2	1	-	2	0	0	0
CO3	3	2	2	1	1	1	1	1	2	2	-	1	0	0	0
CO4	2	1	2	2	3	1	2	1	-	3	-	1	0	0	0
CO5	2	1	2	1	2	1	1	2	-	1	-	1	0	0	0
Score	13	09	10	05	07	05	07	06	05	07	01	06	0	0	0
COM	3	3	2	2	3	2	3	2	2	3	1	2	0	0	0

Course Code	ECMC201
Course Title	Electronic Devices and Circuits
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To familiarize with the semiconductor properties and formation of PN Junction diode
- To study the operation and applications of BJT and FET
- To describe the functionality of power semiconductor devices
- To know the switching characteristics of components
- To describe performance analysis of electronic circuits

COURSE CONTENT

Unit I: Introduction to Semiconductors Devices
Classification of semiconductors, Conductivity of semiconductors, Energy distribution of Electrons, Fermi Level, Carrier concentration in an Intrinsic semiconductor, Drift and Diffusion currents, Poisson and Continuity Equations, Theory of PN junction diode, VI characteristics of a PN Junction Diode, Temperature dependence of VI characteristics of diodes, Breakdown in PN junction diodes.

Unit II: Rectifiers and Special Purpose Diodes
Rectifiers, Half Wave and Full Wave Rectifiers, Clippers (Single and Two Level), Clampers, Analysis with Ideal and Practical Diodes, Special Purpose Diodes: Zener Diode, Tunnel Diode, Varactor Diode, LED, Photo Diode, Schottky Diode.

Unit III: Bipolar Junction Transistor
Transistors: Construction, Operation, Characteristics, Parameters, Transistor as an Amplifier at Low Frequency, Hybrid Model and re-Model of BJT, Analysis of Amplifier using Hybrid Model and re Model, Amplifier Types: CE, CB, CC, DC Operating Point, Biasing Circuits - Fixed Bias, Emitter Bias, Voltage Divider Bias, Bias Stabilization.

Unit IV: Field-Effect Transistors
Junction FET: Construction, Operation, Characteristics, Parameters, Biasing of JFET, JFET as an Amplifier (Common Source and Common Drain). Construction and Operation of N-Channel and P-Channel MOSFET, Enhancement and Depletion Type MOSFET, Characteristics, Threshold Voltage, Channel Length Modulation, Applications of MOSFETs in CMOS Circuits.

Unit V: Power Semiconductor Devices
Power Semiconductor Devices: Construction, Principle of Operation, Characteristics and Applications of UJT, PNPN Diode, SCR, LASCR, DIAC, TRIAC, GTO Thyristors, Power BJT, Power MOSFET, DMOS, VMOS.

LIST OF EXPERIMENTS

1. Study on Oscilloscope, signal generator to view waveforms and measure amplitude and frequency.
2. Study of V-I characteristics of PN Junction and Zener diodes, and determine its DC and AC resistance.
3. Study of half-wave and full-wave rectifier circuits without and with capacitor filter.
4. Study of various clipper and clamper circuits.
5. Study of V-I characteristics in common-base transistor configuration.
6. Study of V-I characteristics, DC biasing, and operating point in common-emitter transistor configuration.
7. Study of BJT fixed bias, self-bias, and voltage divider bias configurations.
8. Study of Transfer and Drain Characteristics of JFET.
9. Study of FET fixed bias and self-bias configurations.
10. Study of Differential Amplifier using BJT.
11. Study of MOSFET in common source configuration

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Describe the basics of semiconductor physics and working principle of PN Junction diode.	K2
CO2	Apply the working principle of special semiconductor devices.	K3
CO3	Analyze the Bipolar Junction Transistor characteristics and the biasing techniques.	K4
CO4	Analyze the Field Effect Transistor characteristics and its applications.	K4
CO5	Explain the construction and working principle of power semiconductor devices.	K2

TEXT BOOKS

1. Milliman, J., Halkias, C., and Jit, S., “Electronics Devices and Circuits”, 4thedition, McGraw Hill Education, 2015.
2. Salivahanan, S., and Suresh, N. K., “Electronic Devices and Circuits”, 4th edition, McGraw Hill Education, 2017.

REFERENCE BOOKS

1. Neamen, D. A., and Biswas, D., “Semiconductor physics and devices: basic principles”, 4th edition, McGraw Hill Education, 2017.

2. Sedra, A. S., Smith, K. C., Carusone, T. C., and Gaudet, V. Microelectronic Circuits. Vol. 4. New York: Oxford University Press, 2017.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	3	-	-	-	-	-	-	-	-	-	-	-	2
CO2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	2
CO3	3	3	-	3	-	-	-	-	-	-	-	-	-	-	2
CO4	3	3	3	3	1	-	-	-	-	-	-	-	-	-	2
CO5	3	3	3	-	1	-	-	-	-	-	-	-	-	-	2
Score	15	14	9	6	2	-	-	-	-	-	-	-	-	-	10
COM	3	3	3	3	1	-	-	-	-	-	-	-	-	-	2

Course Code	ECMC202
Course Title	Digital circuits and Systems
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To describe the basic principles of digital circuits and solve logic expressions and circuits using Boolean laws and K-map.
- To impart knowledge for the design and implementation of combinational logic circuits.
- To impart knowledge for the design and implementation of sequential logic circuits.
- Classify and comprehend the working principle of logic family and data converters.
- To learn the basics of Verilog HDL for system modeling.

COURSE CONTENT

Unit I: Number System And Logic Gates
Introduction to various number systems and their Conversion. Binary Arithmetic (Addition, Subtraction, Multiplication and Division), BCD codes, Excess-3 code, Gray code, Hamming code, Error Detection and Correction. Basic Logic Operations, Basic Identities, Algebraic Laws, Useful Boolean Identities, Canonical Logic Forms, Algebraic Reductions, Complete Logic Sets, Karnaugh Maps.

Unit II: Combinational Logic Networks
Concept of a Digital Component, BCD Validity Detector, Binary Adders, Subtractors, Multiplexers, Demultiplexers, Line Decoders and Encoders, Binary Multiplication, Binary Comparators.

Unit III: Sequential Logic Circuits
Concept of a Sequential Network, Latches, Clock and Synchronization, Flip-Flops, Design of Synchronous & Asynchronous Sequential circuits. Shift registers: Principle of 4-bit shift registers. Shifting principle, Timing Diagram, SISO, SIPO, PISO and PIPO registers.

Unit IV: Introduction to State Machines and PLDs
Algorithmic State Machine, Finite State Machine: Mealy and Moore Machine, Sequence Detector, Vending Machine; Introduction to PLDs, PROM and its types, PAL, and PLA.

Unit V: First Concepts In Verilog HDL
Defining Modules in Verilog HDL; Gate, Data Flow and (Behavioral) Structural Modelling, Learning Verilog HDL. CMOS Logic Circuits: NOT Function in CMOS, Complex Logic Gates in CMOS.

LIST OF EXPERIMENTS

1. Introduction of Digital Logic Gates: Investigate logic behavior of NOT, AND, OR, NAND, NOR, EX-OR, EX-NOR gates.
2. Gate-level minimization: Two-level and multi-level implementation of Boolean functions.
3. Combinational Circuits design assemble and test: adders and subtractors.
4. Code Converter: BCD to Excess-3 code converter, gray code to binary converter, binary to gray code converter.
5. Design, implement and test a given design example with (i) NAND Gates only (ii) NOR Gates only and (iii) using minimum number of Gates.
6. Design of multiplexers and de-multiplexers.
7. Design of encoders and decoders.
8. Binary Multiplier: design and implement a circuit that multiplies 4-bit unsigned numbers to produce an 8-bit product.
9. Parallel adder and accumulator: design, implement and test.
10. Flip-Flop: assemble, test and investigate operation of S-R, D & J-K flip-flops.
11. Counters: Design, assemble and test various Asynchronous and Synchronous binary counter with parallel load.
12. Shift Registers: Design and investigate the operation of all types of shift registers with parallel load.
13. Design of Combinational circuits using the Verilog HDL module.
14. Design of Sequential circuits using the Verilog HDL module.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Describe the different number systems, solve logic expressions and circuits using Boolean laws and K-map.	K2
CO2	Develop competence in combinational Logic problem formulation and logic optimization.	K3
CO3	Develop competence in analysis of synchronous and asynchronous sequential circuits.	K3
CO4	Compare different types of logic families and data converters	K4

CO5	Demonstrate knowledge of Verilog HDL for system Modeling.	K2

TEXT BOOKS

1. Thomas L. Floyd. *"Digital Fundamentals"*, 11th Edition, Pearson International Education, 2017.

2. Mano, M. M. and Ciletti, M. D. *"Digital Design: With an Introduction to the Verilog HDL"*, 5th Edition, Pearson Education, 2013.

REFERENCE BOOKS

1. Palnitkar, S. "Verilog HDL: A Guide to Digital Design and Synthesis," Prentice Hall, 1996.

2. Perry, D. L. & quot; *"VHDL: Programming by Example"*, 4th Edition ", McGraw-Hill, 2002.

3. Roth, C. H. and John, L. K.,Digital System Design Using VHDL, 2nd Edition", Cengage Learning, 2008.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	2	-	-	-	-	3	-	3	-	-	2
CO2	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
CO3	3	3	-	3	2	-	-	-	-	3	-	3	3	-	2
CO4	3	2	2	3	2	2	3	3	2	3	-	3	-	-	2
CO5	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
Score	15	11	2	15	10	2	3	3	2	15	-	15	9	-	10
COM	3	3	2	3	2	2	3	3	2	3	0	3	3	0	2

Course Code	ECMC203
Course Title	Signals and Systems
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn various types of signals and systems with their properties.
- To compose periodic signals in different forms of Fourier series.
- To analyse the signals using Fourier Transform.
- To get familiar with the concepts of Laplace Transform.
- To analyse the signals using Z-Transform.

COURSE CONTENTS

Unit I: Classification of Signals and Systems
Signal Classification (Analog-Digital, Energy-Power, Even-Odd, Periodic-Aperiodic, Deterministic-Random, etc.), Standard Signals (Unit Step, Unit Impulse, Ramp, Exponential, Sinusoids, etc.), Transformations of the Independent Variable (Shifting, Scaling, Reversal), Discrete Unit Step and Impulse Signals and Their Properties. System Classifications (Linearity, Time Invariance, Memory, Causality etc.), Time-Convolution, System Properties via the Impulse Response (Causality, Memory, Stability, Invertibility, Unit Step Response).

Unit II: Fourier Series of Periodic Functions
History of Fourier Series, Euler to Fourier, Response of LTI Systems to Complex Exponentials and Fourier Series Representation of Continuous Time Periodic Signals, Fourier Series Representation of sawtooth, square and triangular Waves, Gibbs Phenomena, Properties of Fourier Series and Examples, Convergence of Fourier Series, Fourier Series and LTI Systems, Filtering.

Unit III: Fourier Transform
Introduction of the Continuous Time Fourier Transform by Taking the Limit of a Periodic Signal (i.e., Making it Aperiodic), Fourier Transforms of Periodic Signals, Properties of Continuous Time Fourier Transforms, Convolution Property of Fourier Transforms, Discrete Time Fourier Transform.

Unit IV: Laplace Transform
Introduction to Laplace Transform; Region of Convergence, Inverse Laplace Transform, Properties of Laplace transforms, initial/final Value Theorems, Laplace Transforms and LTI systems, causality/stability, Laplace Transforms and Block System Diagrams, Unilateral Laplace Transform and Initial Value Problems.

Unit V: Z-Transform
The z-Transform, Properties of z-Transforms, Inversion of z-Transform, One-sided z-Transform, Analysis of Linear Time Invariant Systems in the z-Domain, Convolution of two sequences. Solutions of Difference Equations with Constant Coefficients by Z- transform.

LIST OF EXPERIMENTS

1. To Write a program to generate the discrete sequences (i) unit step (ii) unit impulse (iii) ramp (iv)periodic sinusoidal sequences. Plot all the sequences.
2. Write a program to convolve two discrete time sequences. Plot all the sequences. Verify the result by analytical calculation.
3. Write a program to classify the systems.
4. Write a program to find the trigonometric Fourier series coefficients of a rectangular periodic signal. Reconstruct the signal by combining the Fourier series coefficients with appropriate weightings.
5. Write a program to find the polar and exponential Fourier series coefficients of a periodic rectangular signal. Plot the discrete spectrum of the signal.
6. Find the Fourier transform of a square pulse. Plot its amplitude and phase spectrum.
7. Write a program to find the Laplace transform of the signals and calculation of ROC.
8. Write a program to find the Z-transform and inverse Z-transform of some signals.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain different types of signals and their properties. Apply different conditions such as Causality, memory requirement, stability, and invertibility for checking the type of the system.	K2
CO2	Estimate the periodic signals in different forms of Fourier series.	K2
CO3	Determine the Fourier Transform of signals for finding their response at different frequencies.	K3
CO4	Analyse the unstable systems using Laplace transform.	K3
CO5	Apply concepts of z-transform for analysing the signal properties.	K4

TEXT BOOKS

1. Oppenheim, A. V., Willsky, A. S., and Nawab, S. H., “Signals and Systems,” 2nd Edition, Pearson Education, 2015.

2. Lathi, B. P., “Principles of Linear Systems and Signals,” 2nd Edition, Oxford, 2009.

REFERENCE BOOKS

1. Haykin, S., “Signals and Systems,” 2nd Edition, Wiley, 2008.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	2	-	-	-	-	3	-	3	-	-	2
CO2	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
CO3	3	3	-	3	2	-	-	-	-	3	-	3	3	-	2
CO4	3	2	2	3	2	2	3	3	2	3	-	3	-	-	2
CO5	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
Score	15	11	2	15	10	2	3	3	2	15	-	15	9	-	10
COM	3	3	2	3	2	2	3	3	2	3	0	3	3	0	2

Course Code	ECMC204
Course Title	Electromagnetic Field Theory
L T P C	3-1-0-4
Course Type	MC

COURSE OBJECTIVES

- To learn fundamentals of vector algebra.
- To study concepts of Electrostatics and Magnetostatics.
- To impart knowledge on Maxwell equations and their applications.
- To study how electromagnetic waves propagate in a medium.
- To study fundamentals of transmission lines and its various characteristics.

COURSE CONTENTS

Unit I: Vector Algebra
Introduction, Scalar fields, Vector fields; Cartesian coordinate system, cylindrical coordinate system, Spherical coordinate system, Transformation from Cylindrical to Spherical Coordinates, Gradient of a Scalar function, Divergence of a vector field, Circulation of a vector and Curl.

Unit II: Electrostatics and Magnetostatics
Electrostatics, Coulomb's Law, Electric fields of point charges, Gauss's Law and applications, Electric potential, Poisson's and Laplace equations, Method of Images, Multipole Expansion, Electrostatic fields in matter, Dielectrics and electric polarization, Capacitors with dielectric substrates, Magnetostatics: Magnetic fields of steady currents, Biot-Savart's and Ampere's Laws, Magnetic vector potential; Magnetic properties of matter, Faraday's Law.

Unit III: Maxwell's Equations
Equation of continuity for time varying fields. Inconsistency of Amperes law, Maxwell's equations and their physical interpretation, Maxwell's Equations in Differential form and Integral form, Interface Conditions for Electromagnetic field, Time-harmonic representation of Maxwell's equations, Source-free equations.

Unit IV: Electromagnetic Waves and Propagation
The Electromagnetic wave equation and its solution, Time-dependent wave equation, Time-Harmonic wave equation, One-Dimensional Wave equation in Free-space and perfect dielectrics; Poynting theorem and electromagnetic power, Complex Poynting vector, Propagation of Plane waves in different materials, Polarization of Plane waves, Normal Incidence on a general dielectric interface, Oblique incidence on dielectric interface for perpendicular and parallel polarization, Brewster's Angle.

Unit V: Transmission Lines
Distributed parameters, Time-domain transmission Line Equations, Types of transmission lines, Finite Transmission lines, Load reflection coefficient, Line impedance and generalized reflection coefficient, Standing wave ratio; Power relations on a generalized transmission line, The Smith chart, Impedance Matching, Stub Matching, Quarter wave transformer matching.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the fundamentals of vector algebra and apply those in the study of electromagnetic field theory.	K2
CO2	Analyze the principles of electrostatics and magnetostatics with the help of different laws.	K4
CO3	Analyze the inconsistency in Ampere's law and formulate Maxwell's equations in differential and integral forms.	K4
CO4	Identify how electromagnetic waves propagate in any material medium.	K3
CO5	Analyze the principle of flow of V/I waves in a transmission line and study concept of impedance matching.	K4

TEXT BOOKS

1. Griffiths, D. J., "Introduction to Electrodynamics", 4th Edition, Cambridge University Press, 2020.
2. Nathan, I., "Engineering Electromagnetics", 2nd Edition, Springer India, 2005.

REFERENCE BOOKS

1. Verma, H.C., "Classical Electromagnetism", 1st Edition, Bharati Bhawan Publishers, 2022.
2. Guru, B. and Hiziroglu, H., "Electromagnetic field theory Fundamentals", Cambridge University Press, 2005.
3. Jordon, E. C. and Balmain, K. G., "Electromagnetic waves and Radiating System", 2nd Edition, Prentice Hall New Delhi, 2007.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	2	-	-	-	-	3	-	3	-	-	2
CO2	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
CO3	3	3	-	3	2	-	-	-	-	3	-	3	3	-	2
CO4	3	2	2	3	2	2	3	3	2	3	-	3	-	-	2
CO5	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
Score	15	11	2	15	10	2	3	3	2	15	-	15	9	-	10
COM	3	3	2	3	2	2	3	3	2	3	0	3	3	0	2

Course Code	ECMC205
Course Title	Network Analysis and Synthesis
L T P C	3-1-0-4
Course Type	MC

COURSE OBJECTIVES

- To under the basic concepts of AC and DC analysis.
- To simplify any network using network theorems.
- To learn the basics of Laplace transform.
- To understand the transient response.
- To analyse any network any two-port network.
- To synthesize any network.
- To analyse a network using graph theory.
- To study the basic filter circuits.

COURSE CONTENTS

Unit I: Sinusoidal Steady State Analysis
Sinusoids, Phasors, Impedance and admittance, kirchhoff's law in frequency domain, impedance combinations, steady state analysis: nodal and mesh analysis, dependent, independent voltage and current sources, source transformation, Thevenin and Norton equivalent. AC power analysis: instantaneous and average power, max average power transfer, RMS value, apparent power and power factor, complex power, conservation of AC power. Three phase circuits: types of load and source connections, power in balanced three phase circuits, star delta transformations. Network theorems: compensation, superposition, reciprocity, millman's and tellegen's theorem.

Unit II: Transient Network Analysis
Complex frequency and Laplace transforms, circuits analysis in S domain, poles, zeros, transfer Functions and driving point impedances and convolution. Step and impulse response of RL, RC, LC, RLC circuits, initial and final conditions.

Unit III: Two Port Networks
Short circuit admittance parameter, open circuit impedance parameters, hybrid and transmission parameters, series parallel and tandem connection of two port networks, multi-port networks, multi terminal networks, indefinite admittance matrix and its properties, relationships among different network parameters.

Unit IV: Network Synthesis
Elements of realizability theory: causality and stability, Hurwitz polynomials, positive real

functions, elementary synthesis procedure, synthesis of one port network with two kind of element: L-C driving point immittances, synthesis of R-L, L-C functions.

Unit V: Graph Theory and Filters
Basic definitions of graph, tree, link, branches, node, topological equations for loop current and node voltages, duality, series and parallel resonance, single and double tuned circuits, passive filters: lowpass, highpass, bandpass and bandstop filters, frequency response.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Analyze AC and DC circuits (single and three phase) making use of various circuit techniques.	K2
CO2	Learn the frequency domain analysis.	K3
CO3	Acquire knowledge of the fundamentals of network analysis using matrices, two-port and multi-port networks, network synthesis and filter circuits.	K2
CO4	Synthesize a network using various techniques.	K3
CO5	Analyse a network using graph theory.	K3
CO6	Design and observe the characteristics of filter circuits.	K3

TEXT BOOKS

1. Alexander, Charles K., Matthew NO Sadiku, and Matthew Sadiku. Fundamentals of electric circuits. Boston: McGraw-Hill Higher Education, 2007.
2. Van Valkenburg, Mac E. Network analysis. 1974.
3. Circuit Theory (Analysis and Synthesis), authored by A. Chakrabarti
4. Engineering Circuit Analysis by William H. Hayt, Jr., Jack E. Kemmerly, and Steven M. Durbin

REFERENCE BOOKS

1. Van Valkenburg, Mac E. Network analysis. 1974.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	3	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-
CO3	3	2	3	3	-	-	-	-	-	-	-	-	2	-	-
CO4	2	3	2	2	-	-	-	-	-	-	-	-	3	-	-
CO5	3	2	-	3	-	-	-	-	-	-	-	-	2	-	-
Score	13	12	9	14	-	-	-	-	-	-	-	-	11	-	-
COM	3	2	2	3	-	-	-	-	-	-	-	-	2	-	-

Course Code	ECMC206
Course Title	Design Engineering
L T P C	0-0-4-2
Course Type	MC

COURSE OBJECTIVES

- Learn to use circuit design and simulation tools such as OrCAD, PSpice, and other relevant software.
- Gain hands-on experience in digital and analog circuit simulation, PCB design, and schematic capture.
- Develop engineering thinking for problem solving, design validation, and prototype simulation.
- Bridge the gap between theory and real-world system design using software-based experimentation.
- Familiarize students with key tools used in AI/ML, IoT, and Robotics design and simulation.
- Enable end-to-end prototyping with sensors, embedded systems, and ML models.
- Strengthen design thinking and interdisciplinary skills through hands-on mini projects.

LIST OF EXPERIMENTS

1. Introduction to Circuit Design Tools – Overview of OrCAD & PSpice:
   Students will be introduced to Electronic Design Automation (EDA) tools—OrCAD and PSpice—for designing, simulating, and analyzing electronic circuits. The session covers workspace setup, component libraries, and simulation flow.
2. DC and AC Circuit Simulation:
   Students will simulate basic DC and AC circuits using PSpice, observe voltage-current relationships, and verify Ohm's and Kirchhoff's laws. Frequency response and phase analysis of RLC circuits are also included.
3. Analog Components Simulation:
   Simulation of analog components like diodes, BJTs, op-amps, and MOSFETs. Emphasis is placed on understanding I-V characteristics, amplifier configurations, and nonlinear behavior.
4. Digital Logic Design:
   Using digital libraries in simulation tools, students design and test logic gates, combinational circuits (e.g., adders, multiplexers), and validate truth tables and Boolean equations.

5. Timing Analysis& Sequential Circuits:
   Focus on timing diagrams and sequential logic circuits such as flip-flops, counters, and registers. Timing simulation and clock pulse behavior are studied using PSpice.

6. Microcontroller Simulation Basics:
   Introduction to simulating microcontrollers in software (e.g., Proteus). Basics of memory, I/O control, and simple embedded routines using LEDs and switches are practiced.

7. PCB Design Principles:
   Students learn about schematic capture, PCB layout, netlist generation, routing, and design rule checks. The experiment emphasizes the complete process of translating circuit diagrams to physical PCBs.

8. Project Ideation & Block Diagram Design :
   Students brainstorm project ideas in groups and prepare functional block diagrams. Concepts of modularity, functional division, and signal flow are emphasized.

9. Design Project Implementation – Part 1:
   Initial phase of hardware implementation: component selection, microcontroller programming, breadboard testing, and partial circuit realization.

10. Design Project Implementation – Part 2:
    Final project integration: assembling the entire circuit/system, debugging, testing outputs, and validating functionality as per block diagrams.

11. Project Demo & Viva:
    Groups present and demonstrate their projects, followed by individual viva sessions. Evaluation includes innovation, working model, documentation, and presentation.

12. Introduction to AI/ML, IoT & Robotics – Concept Mapping, Group Ideation:
    Overview session on the convergence of AI, ML, IoT, and robotics. Concept mapping and ideation exercises are conducted to explore interdisciplinary project possibilities.

13. Python Basics for ML + Intro to Jupyter – Jupyter Notebook, scikit-learn:
    Hands-on with Python syntax, data types, loops, and functions. Introduction to Jupyter Notebook for coding and data visualization, and use of scikit-learn for basic ML tasks.

14. ML for Engineers – Classification Models – Teachable Machine or scikit-learn:
    Students train classification models using either Teachable Machine or scikit-learn. Topics include data collection, training, testing, and performance metrics like accuracy and confusion matrix.

15. IoT Introduction + Arduino Sensor Interfacing – Arduino IDE, DHT11/IR Sensors:
    Introduction to IoT architecture and basic interfacing of sensors (e.g., DHT11 for temperature/humidity, IR for object detection) using Arduino. Coding via Arduino IDE.

16. NodeMCU/ESP32 Programming + Cloud Integration – ThingSpeak, Blynk, Firebase:
    IoT data transmission using NodeMCU/ESP32 with real-time cloud platforms. Students learn to send data to ThingSpeak, visualize on Blynk dashboard, and log to Firebase.

17. Robotics: Actuators & Sensor Integration – Motor Drivers, Ultrasonic Sensors:
    Basics of robotics including actuator (DC/servo motor) control and sensor (ultrasonic) feedback. Students build simple obstacle detection setups using L298N driver modules.

18. Line Follower or Obstacle Avoider Bot – Arduino + IR Sensors:
Students design and implement a basic autonomous robot. Line following logic or obstacle avoidance using IR sensors and Arduino is developed and tested.

19. IoT Dashboard & Data Visualization:
Students create custom IoT dashboards using tools like Blynk, ThingSpeak, or Google Sheets. Emphasis is on real-time plotting, alerts, and controlling devices remotely.

20. ML + IoT: Smart Prediction System – ESP32 + onboard ML using TensorFlow Lite:
A complete project integrating machine learning models into IoT devices. ESP32 is used to run ML models (e.g., motion detection, air quality prediction) using TensorFlow Lite.

21. Introduction to AI/ML, IoT & Robotics – Concept Mapping, Group Ideation:
Final ideation session to consolidate all concepts learned throughout the course. Students brainstorm future project ideas that combine AI, ML, IoT, and robotics in practical use-cases.

Total Periods: 48

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	To introduce students to electronic circuit design and simulation tools such as OrCAD and PSpice	K1
CO2	To develop skills in microcontroller programming and sensor interfacing using platforms like Arduino, ESP32, and NodeMCU for embedded system applications.	K2
CO3	To impart knowledge of PCB design principles and guide students through the complete design cycle from ideation to hardware implementation.	K3
CO4	To introduce the fundamentals of Artificial Intelligence and Machine Learning using Python and scikit-learn	K3
CO5	To enable students to integrate IoT with AI/ML and robotics by building smart systems using cloud platforms	K4

TEXT BOOKS

1. S. W. Smith, The Scientist and Engineer's Guide to Digital Signal Processing, 1st ed. San Diego, CA, USA: California Technical Publishing, 1997.

2. A. Bahga and V. Madisetti, Internet of Things: A Hands-On Approach, 1st ed. Atlanta, GA, USA: VPT, 2014.

REFERENCE BOOKS

1. C. Platt, Make: Electronics: Learning Through Discovery, 2nd ed. San Francisco, CA, USA: Maker Media, 2015.

2. S. Raschka and V. Mirjalili, Python Machine Learning, 3rd ed. Birmingham, U.K.: Packt Publishing, 2019.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	0	0	0	0	3	0	0	3	2	0	0
CO2	3	2	2	2	0	0	0	0	3	0	0	3	2	0	0
CO3	3	2	2	2	0	0	0	0	3	0	0	3	2	0	0
CO4	3	2	2	2	1	2	0	0	3	0	0	3	2	0	0
CO5	3	2	2	2	1	3	0	0	3	0	0	3	2	0	0
Score	15	10	10	10	2	5	0	0	15	0	0	15	10	0	0
COM	3	2	2	2	1	3	0	0	3	0	0	3	2	0	0

SEMESTER IV

Course Code	ECMC207
Course Title	Microwave Engineering
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn fundamentals of microwaves.
- To represent microwave network using scattering matrix.
- To impart knowledge on microwave waveguides and components.
- To study design consideration and principle of operation of microwave tubes.
- To study various microwave semiconductor devices and their applications.

COURSE CONTENTS

Unit I: Introduction to Microwaves
History of Microwaves, Microwave Frequency bands, General Applications of Microwaves, Advantages of Microwaves, Mathematical model of Microwave Transmission, Microwave Propagation (Friis Free space propagation model), Losses associated with microwave transmission, Concept of Impedance in Microwave transmission.

Unit II: Microwave Waveguides and Components
Analysis of Parallel plane waveguides, Rectangular waveguides and Circular waveguides, Rectangular cavity resonator, Microwave Hybrid circuits, Waveguide Tees, Magic tees, Rat-race circuits, Directional coupler, Circulators and Isolators, Wilkinson power divider.

Unit III: Microwave Network Analysis
Equivalent voltages and currents, concept of impedance, impedance and admittance matrices of microwave junctions, scattering matrix representation of microwave networks, ABCD parameters, excitation techniques for waveguides.

Unit IV: Microwave Tubes & Semiconductor Devices
Design considerations and current status of microwave tubes, principle of operation of multicavity and reflex klystron, magnetron and traveling wave tube, Applications of Microwave tubes, Operation and circuit applications of Gunn diode, IMPATT diode, PIN Diode, and Schottky barrier diode, HEMT and their applications.

Unit V: Basics of Antennas
Introduction to antennas, Radiation mechanism and Current distribution of single wire, two wire, and dipole, Types of antennas and their applications, Isotropic radiators, Fundamental antenna parameters.

LIST OF EXPERIMENTS

1. To study I-V characteristics of Gunn diode.
2. To study the output power and frequency as a function of voltage characteristic of Gunn diode.
3. To study square wave modulation through Pin diode.
4. To study the characteristics of Reflex Klystron Tube.
5. Measurement of Directional Coupler parameters.
6. To study characteristics of Isolator and Circulator.
7. To study characteristics of Waveguide Tees.
8. Measurement of Frequency and Wavelength.
9. Measurement of Impedance.
10. Antenna Measurements.
11. Low, medium and high VSWR measurements.
12. To study the variable attenuator.
13. To study voice communication using Microwave test bench.
14. To study the square law behavior of microwave crystal detector.
15. To design Microstrip patch antenna using CST microwave studio.
16. To design Vivaldi antenna using CST microwave studio.
17. To design Anti-Podal Vivaldi antenna using CST microwave studio.
18. To design a Frequency selective surface for X-band using CST microwave studio.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the fundamentals of microwave engineering and outline the losses associated with microwave transmission.	K2
CO2	Analyze the operating principles of various microwave devices and recognize their usability in microwave network analysis.	K4
CO3	Analyze the basic mechanism behind the wave propagation in waveguides.	K4
CO4	Identify the effect of microwaves on different parts of human body and study operating principles of microwave tubes.	K3
CO5	Assess the methods used for generation and amplification of microwave power.	K5

TEXT BOOKS

1. Liao. S. Y., "Microwave Device and Circuits", 3rd Edition, Pearson Education, 2000.
2. Pozar, D. M., "Microwave Engineering", 4th Edition, Wiley, 2013.
3. Balanis, C. A., "Antenna Theory and Applications", 4th Edition, Wiley, 2021.

REFERENCE BOOKS

1. Collin, R. E., "Foundations for Microwave Engineering", 2nd Edition, Wiley, 2007.
2. Das, A. and Das, S. K., "Microwave Engineering", 3rd Edition, McGraw Hill Education, 2017.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	2	-	-	-	-	3	-	3	-	-	2
CO2	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
CO3	3	3	-	3	2	-	-	-	-	3	-	3	3	-	2
CO4	3	2	2	3	2	2	3	3	2	3	-	3	-	-	2
CO5	3	2	2	3	2	-	-	-	3	3	-	3	-	-	-
Score	15	11	4	15	10	2	3	3	5	15	-	15	6	-	8
COM	3	3	2	3	2	2	3	3	3	3	0	3	3	0	2

Course Code	ECMC208
Course Title	Analog Communication
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To understand the concepts of analog modulation and demodulation techniques with the performance of AM, FM and PM schemes.
- To know the design characteristics of Amplitude modulation transmitter and receiver systems.
- To understand the Frequency modulation transmitter and receiver systems.
- To introduce the concepts of various pulse modulation techniques and their demodulation.
- To study the different types of noises in communication systems.

COURSE CONTENTS

Unit I: Modulation Techniques
Various frequency bands used for communication, Types of Communication and need of modulation. Introduction to AM, FM, PM, frequency spectrum of AM Waves, Representation of AM, Power relation in AM waves, Need and description of SSB, Suppression of carrier, Suppression of unwanted side-bands, Independent sideband system, Vestigial sideband system, Mathematical representation of FM, Frequency spectrum of AM waves, wideband and narrow band FM, Phase Modulation.

Unit II: AM Transmitters and Receivers
AM Transmitters: Generation of AM, low level and high level modulation, AM transmitter block diagram, collector class C modulator, Base Modulator, DSB-SC modulator. Generator of SSB, balanced modulator circuit, filter method, phase shift method, third method, Phase cancellation method. AM Receivers: Tuned radio frequency (TRF) receiver. Super heterodyne receiver, RF section and characteristics, mixers, frequency changing and tracking, IF rejection and IF amplifiers. Detection and automatic gain control (AGC), AM receiver characteristics, Demodulation of SSB, product demodulator, Diode detection technique of SSB.

Unit III: FM Transmitters and Receivers
FM Transmitters: Generation of FM, FM Modulation methods: Direct methods (Variable capacitor Modulator, Varactor Diode Modulator, FET Reactance Modulator, Transistor Reactance Modulator, AFC in reactance modulator), Disadvantages of direct method, Indirect modulators (RC-phase shift modulators, Armstrong FM systems). FM Receivers: Limiters, single and double-tuned demodulators (balanced slope detector, Foster-Seeley or Phase Discriminator, ratio Detector, FM Detector using PLL), Pre-emphasis and De-emphasis, Block diagram of FM

Receivers, RF Amplifiers, FM Receiver characteristics, Phase modulator and demodulator circuits.

Unit IV: Pulse Modulation Techniques
Pulse amplitude modulation and demodulation, Pulse width modulation and demodulation, Pulse position modulation and demodulation, Sampling theorem, Time Division Multiplexing, Frequency Division Multiplexing.

Unit V: Noise
Thermal Noise, Shot noise, Partition noise, Flicker noise, Gaussian Noise, Noise in Bipolar Junction Transistors (BJTs), FET noise. Equivalent input noise, Signal to Noise Ratio (SNR), Noise Temperature, Noise equivalent Bandwidth, Noise Figure. Experimental determination of Noise Figure, Pulse Response and its elimination.

LIST OF EXPERIMENTS

1. To study Amplitude Modulation/demodulation using a transistor, depth of modulation, and observe the diagonal peak clipping effect of AM.
2. Frequency Modulation using Voltage Controlled Oscillator.
3. Generation of DSB-SC, SSB, and VSB.
4. Study of Phase Lock Loop (PLL) and detection of FM Signal using PLL.
5. Study functioning of Super-heterodyne AM Receiver, and study the limiter circuit and IF filter frequency response, study of the effect of image frequency.
6. Study functioning of Super-heterodyne FM Receiver
7. Measurement of Sensitivity, Selectivity, and Fidelity of radio receivers.
8. Study of PAM, PPM, and PPM.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Analyze and compare different types of modulation and demodulation schemes for their efficiency and bandwidth.	K2
CO2	Attain in-depth knowledge of different Transmitters and Receivers design of amplitude modulation	K3
CO3	Attain in-depth knowledge of different Transmitters and Receivers design of frequency modulation.	K3
CO4	Compare different pulse modulation and multiplexing techniques	K3
CO5	Apply concepts of noises in designing of communication systems.	K2

TEXT BOOKS

1. Haykin S., “Communications Systems”, John Wiley and Sons, 2001.
2. B. P. Lathi, “Modern Digital and Analog Communication Systems,” 4th Edition, Oxford, 2011.

REFERENCE BOOKS

1. Tomasi, W., “Advanced Electronic Communications systems” 6th Edition, Pearson Publishers, 2015.
2. Taub H. and Schilling D.L., ”Prnciples of Communication Systems”, Tata McGraw Hill, 2001.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	2	-	-	-	-	3	-	3	-	-	2
CO2	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
CO3	3	3	-	3	2	-	-	-	-	3	-	3	3	-	2
CO4	3	2	2	3	2	2	3	3	2	3	-	3	-	-	2
CO5	3	2	-	3	2	-	-	-	-	3	-	3	3	-	2
Score	15	11	2	15	10	2	3	3	2	15	-	15	9	-	10
COM	3	3	2	3	2	2	3	3	2	3	0	3	3	0	2

Course Code	ECMC209
Course Title	Microprocessor and Microcontroller
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To understand microprocessor evolution, architecture, and applications, focusing on Intel 8085.
- To learn the development of assembly language programming of 8085.
- To learn essential data transfer methods for microprocessor-based systems.
- To learn the Interfacing of 8085 microprocessor with various peripherals.
- To understand the fundamentals of 16-bit microprocessor architecture.

COURSE CONTENT

Unit I: Introduction to Microprocessors and MicroComputers
History and Evolution, Types of microprocessors, Microcomputer programming languages, Microcomputer architecture, Intel 8085 Microprocessor, Pin Diagram, Architecture Diagram, Addressing Modes, Instruction Format, Types of Instructions, Instruction Set of 8085.

Unit II: Assembly Language Programming, Timing Diagram
Assembly Language Programming in 8085, Macros, Labels and Directives, Microprocessor timings, Micro Instructions, Instruction Cycle, Machine Cycle, T-states, state transition Diagrams, Timing diagrams for different machine cycles. INTERRUPTS: Interrupts in 8085, Issues in implementing interrupts, Multiple Interrupts and priorities, Interrupt handling in 8085, enabling, disabling, and masking of interrupts.

Unit III: Data Transfer Techniques
Data transfer techniques, Introduction to PPI 8255, Programmable Interval Timer/Counter 8253, Programmable Interrupt Controller (8259), DMA Controller (8257), USART (8251).

Unit IV: Microprocessor Interfacing Techniques
Interfacing Traffic Light Interface, Stepper Motor, 4 Digit 7 Segment LED, Elevator, Musical Tone Generator and 8 Channel 12Bit ADC with Multiplexor and A/D converters, D/A converters.

Unit V: Introduction to 8051 Microcontroller
Architecture of 8051, Special Function Registers (SFRs), Pin Diagram, Addressing modes, Instruction set, Assembly language programming, Programming 8051 timers, Serial Port Programming.

LIST OF EXPERIMENTS

1. WAP using 8085 Microprocessor for Decimal, Hexadecimal addition and subtraction of two Numbers.
2. WAP using 8085 Microprocessor for addition and subtraction of two BCD number.
3. WAP to perform multiplication and division of two 8 bit numbers using 8085.
4. WAP to find the largest and smallest number in an array of data using 8085 instruction set.
5. WAP to write a program to arrange an array of data in ascending and descending order.
6. WAP to initiate 8251 and to check the transmission and reception of character.
7. WAP to interface 8253 programmable interval timer to 8085 and verify the operation of 8253 in six different modes.
8. Parallel communication between two microprocessors kits using 8255.
9. Serial communications between two microprocessors kits using 8251.
10. To interface DAC with 8085 to demonstrate the generation of square, saw tooth and triangular Wave.
11. Program for sorting an array for 8086.
12. Program to find the factorial of a given number.
13. Programs for searching a number or character in a string for 8086.
14. Programs for string manipulation for 8086.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	To familiarise with the fundamental concepts of microprocessors and its applications	K2
CO2	Apply programming techniques in developing the assembly language program for microprocessor applications.	K2
CO3	Demonstrate an understanding of various data transfer techniques in microprocessor systems.	K4
CO4	Develop practical skills for interfacing microprocessors with various real-world devices.	K4
CO5	Demonstrate a foundational understanding of 16-bit microprocessor architecture	K2

TEXT BOOKS

1. TRamesh S Gaonkar"Microprocessor Architecture- Programming and Applications with 8085/8080A", Penram International Publishing (India) Pvt. Ltd, 5th Edition.
2. B. Ram "Introduction of Microprocessors and Microcomputers", Dhanpat Rai Publisher (P) Ltd, 4th Edition.
3. K Bhurchandi, A. K. Ray, "Advanced Microprocessor and Peripherals", Tata McGraw-Hill Publishing Company, 3rd Edition (2012).

REFERENCE BOOKS

1. RodnayZaks and Austin Lesea "Microprocessor Interfacing Technique", BPB Publication, 1st Indian Edition (1988).
2. PJames L Antonakes"An introduction to Intel family of Microprocessors", Pearson Education, 3rd Edition.
3. Charles M Gilmore, "Microprocessor; Principles and Applications", McGraw Hill, 2nd Edition.
4. Brey, B. B., "Intel Microprocessors", 8th Edition, Pearson Education, 2013.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	3	-	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	1	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	3	3	3	-	-	-	-	-	-		-	2	-
CO5	3	3	3	3	3	3	-	-	-	-	-	-	-	2	-
Score	15	13	15	10	09	03	-	-	-	-	-	-	-	4	-
COM	3	3	3	2	-	2	3	3	-	-	-	-	-	2	-

Course Code	ECMC210
Course Title	Linear Integrated Circuits
L T P C	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To learn the fundamentals of BJT Amplifiers.
- To learn the fundamentals of multistage and differential amplifiers.
- To learn general operational amplifiers and linear applications of Op-Amp.
- To impart knowledge on active filters and oscillators.
- To study special IC applications using Op-Amp.

COURSE CONTENTS

Unit I: BJT Amplifiers and Frequency Response of Amplifiers
Introduction of Biasing of BJT, BJT Amplifier using re, π and Hybrid-π Model; Common Emitter, Common Collector, and Common base, Requirement of Frequency Response, Low and High Frequency Response of BJT Amplifiers.

Unit II: Multistage and Differential Amplifiers
Multistage Amplifiers, Types of multistage couplings. Feedback Amplifier; Concept of feedback, Analysis of various configurations of feedback in amplifiers; Current mirror and current sources, Current sources as active loads, BJT Differential amplifier with active loads.

Unit III: General Linear Applications of OP-AMP
Introduction to operational amplifiers, the ideal Op-Amp, Ideal Voltage Transfer curve, Open-loop Op-Amp configurations, Characteristics of practical Op-Amp; The peaking amplifier, Summing, Scaling and Averaging amplifiers, Voltage Follower, Differential input and Differential output amplifier, V-to-I with Floating load and Grounded load, I-to-V converters, Instrumentation amplifier, Integrator, Differentiator, Logarithmic amplifier, Antilogarithmic amplifier.

Unit IV: Active Filters and Oscillators Using OP-AMP
Active filters, First-order and second order Butterworth filters; Low pass and High pass, Band pass filters, Band-reject filters, All-pass filter, Oscillators; Principles, Types, Phase shift oscillator, Wein Bridge oscillator, Basic Comparator, Schmitt Trigger, Zero-crossing detector, Astable Multivibrator and Monostable Multivibrator (using op-amp).

Unit V: Special Function IC Applications
The 555 Timer; as a monostable multivibrator, as a astable multivibrator, IC Voltage regulators – Three terminal fixed and adjustable voltage regulators, IC 723 general purpose regulator,

Monolithic switching regulator, Frequency to Voltage and Voltage to Frequency converters, Phase-Locked loops.

LIST OF EXPERIMENTS

1. Design and testing of inverting and non-inverting operational amplifiers.
2. To verify the function of an operational amplifier as an adder and subtractor.
3. To demonstrate the integrator and differentiator circuits using Op-amp.
4. Design and testing Instrumentation amplifier.
5. Design and testing active low-pass, high-pass, band-pass and band-stop filters.
6. Design and testing of astable and mono-stable multi-vibrators using Op-amp.
7. Design and testing Schmitt Trigger and Comparator using Op-amp.
8. Design, Build and Test a Square wave and triangular wave generators using op-amp.
9. To study the operation of IC 565 as PLL.
10. Study of Multiplier IC.
11. To study the output characteristics of half-wave and full-wave rectifier using op-amp.
12. Above experiments are to be Simulated using SPICE.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the working of BJT Amplifiers and study its frequency response.	K2
CO2	Infer the DC and AC characteristics of multistage amplifiers and explain their effect on the output.	K2
CO3	Explain the working of general-purpose OP-AMP and design the general linear applications of an OP-AMP.	K6
CO4	Examine the functionality of various active filters and oscillators.	K4
CO5	Design application specific ICs such as Voltage regulators, PLL and determine their applications in modern day communication systems.	K6

TEXT BOOKS

1. Gayakwad, R. A., “OP-AMP and Linear Integrated Circuits”, 4th Edition, Pearson Education, 2015.

2. Franco, S., “Design with Operational Amplifiers and Analog Integrated Circuits”, 4th Edition, McGraw-Hill education, 2016.

REFERENCE BOOKS

1. Choudhary, D. R. and Jain, S. B., “Linear Integrated Circuits”, 5th Edition, New Age International, 2018.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	-	-	-	-	-	-	-	3	-	3	-	-	-
CO2	3	3	-	-	2	-	-	-	-	3	-	3	-	-	2
CO3	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3
CO4	3	2	2	3	2	-	-	-	-	3	-	3	-	-	2
CO5	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3
Score	15	13	8	9	10	-	-	-	6	15	-	15	6	4	10
COM	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3

Course Code	ECMC211
Course Title	Control Systems
L T P C	3-1-0-4
Course Type	MC

COURSE OBJECTIVES

- To acquire the knowledge of the dynamic modelling of the systems and their representations.
- To analyse the time domain behaviour of the control systems.
- To analyse the open and closed loop stability.
- To analyse the frequency domain behaviours and design control systems using different classical control theory.
- To understand the concept of state variable analysis.

COURSE CONTENTS

Unit I: Basic Concepts
Historical review, Definitions, Classification, Relative merits and demerits of open and closed loop systems, Linear and non-linear systems, Transfer function, Mathematical modelling of electrical, Mechanical and thermal systems, Analogies, Block diagrams and signal flow graphs, Control System Components.

Unit II: Time Domain Analysis
Importance of time response in transient and steady state analysis, typical test input signals, transient response of the first order and second order system, time response specifications, dominant closed loop poles of higher order systems, steady state error and error coefficients, PID Control-Analytical design for P, PI, PID control systems.

Unit III: Stability
Concepts of absolute and relative stability, pole zero location, Routh Hurwitz criteria. Relative Stability-Root Locus Concept-Guidelines for sketching root locus- Nyquist stability criterion.

Unit IV: Frequency Domain Analysis and Compensator Design
Closed loop frequency Response-Performance specification in frequency domain- Frequency response of standard second order system- Bode Plot — Polar Plot- Nyquist Plots. Design of compensators using Bode plots, lead compensation, lag compensation, lag-lead compensation.

Unit V: State Variable Analysis
Concept of State, State variables & state models, State space representation of linear continuous time systems, State models for linear continuous time systems, State variables and linear discrete time systems, Solution of state equations, Concept of controllability & observability

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain and distinguish between open and closed control systems.	K2
CO2	Compute the transfer function of the mathematical model of a given physical system.	K3
CO3	Distinguish the various type of control system hardware and their functionality.	K2
CO4	Analyze and measure the response and stability of the closed and open loop systems.	K5
CO5	Design various kinds of controllers and compensators and then compare their performance.	K6

TEXT BOOKS

1. Nagrath, I. J. and Gopal, M., "Control System Engineering," 7th Edition, New Age International, 2021.
2. Nise, Norman S., "Control Systems Engineering" 8th Edition, John Wiley& Sons, 2019.

REFERENCE BOOKS

1. Warwick, K., "An Introduction to Control Systems", 2nd Edition, World Scientific Publishing, 1996.
2. Åström, K. J. and Murray, R.M. "Feedback systems. An Introduction for Scientists and Engineers". 2008.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	–	–	–	–	–	–	–	–	3	–	–
CO2	2	2	2	3	2	–	–	–	–	–	–	–	3	–	3
CO3	2	2	2	3	–	–	–	–	–	–	–	–	3	–	–
CO4	2	3	3	3	2	–	–	–	3	3	–	2	3	–	3
CO5	2	3	3	3	2	–	–	–	3	3	–	3	3	–	3
Score	10	12	12	13	6	–	–	–	6	6	–	5	15	–	9
COM	2	3	3	3	2	–	–	–	3	3	–	3	3	–	3

Course Code	ECMC212
Course Title	Digital Signal Processing
L T P C	3-1-0-4
Course Type	MC

COURSE OBJECTIVES

- To study the basics of Signal representation and discrete systems.
- To understand the analytical tools such as Fourier transforms, Discrete Fourier transforms and Fast Fourier Transforms required for digital signal processing.
- To understand the digital filters for digital signal processing.
- To design and realize various IIR and FIR.

COURSE CONTENTS

Unit I: Introduction to Signals and Signal Processing
Signal and its classification, basic elements of digital signal processing, advantages of digital signal processing over analog processing, correlation of signals.

Unit II: Frequency Analysis of Discrete Time Signals
Discrete Fourier Transform: frequency domain sampling of DTFT, DFT as linear transformation, properties of DFT, circular convolution, discrete time signal analysis using DFT, introduction to FFT: minimum complexity space for DFT, decimation in time and decimation in frequency methods to compute FFT algorithm, Goertzel algorithm.

Unit III: Realization of LTI Discrete Time Systems
Introduction to recursive and non-recursive systems, structural look of LTI systems, recursive and non-recursive realization of FIR systems, basic FIR digital filter structures: direct and cascade form structures, lattice structures, linear phase FIR systems, IIR digital filter structures: direct and cascade form, parallel and lattice structures.

Unit IV: FIR Digital Filter Design
Introduction, General considerations: causality and Paley-Wiener theorem, characteristics of frequency selective filters, design of linear phase FIR filters using windowed Fourier series and frequency sampling method.

Unit V: IIR Digital Filter Design
Introduction, design of IIR filters from analog filters, IIR filter design via approximation of derivatives, impulse invariance, bilinear transformation and matched z-transformation.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Summarize the fundamentals of different signals and basic operations of Signal processing.	K2
CO2	Understand the handling of discrete/digital signals using MATLAB.	K2
CO3	Analyse the spectral components of discrete signals.	K4
CO4	Implement FIR/IIR systems using efficient structures.	K3
CO5	Design FIR/IIR systems for given specifications.	K3

TEXT BOOKS

1. Proakis, J. G. and Manolakis, D. G., "Digital Signal Processing: Principles, Algorithms and Applications", 4th Edition, Pearson education, 2014.

2. Mitra, S. K., "Digital Signal Processing: A computer-based approach", 4th Edition, Mc-Graw Hill Education, 2013.

REFERENCE BOOKS

1. Tan, L. and Jiang, J., "Digital Signal Processing: Fundamentals and Applications", 2nd edition, Academic Press, 2013.

2. Oppenheim, A. V. and Schafer, R. W., "Discrete-Time Signal Processing", 3rd edition, Pearson Education, 2014.

3. Andreas Antoniou (2006), "Digital Signal Processing", Tata McGraw Hill, New Delhi.

4. Ifeachor and Jervis, "Digital Signal Processing", Pearson Education India.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	-	-	-	-	-	-	-	-	3	-	-
CO2	2	2	2	3	2	-	-	-	-	-	-	-	3	-	3
CO3	2	2	2	3	-	-	-	-	-	-	-	-	3	-	-
CO4	2	3	3	3	2	-	-	-	3	3	-	2	3	-	3
CO5	2	3	3	3	2	-	-	-	3	3	-	3	3	-	3
Score	10	12	12	13	6	-	-	-	6	6	-	5	15	-	9
COM	2	3	3	3	2	-	-	-	3	3	-	3	3	-	3

Course Code	ECMC213
Course Title	Product Development Lab
Number of Credits (L-T-P-C)	0-0-2-1
Course Type	MC

COURSE OBJECTIVES

- To introduce students to PCB design tools and techniques.
- To familiarize students with the process of converting a schematic design into a physical prototype.
- To enhance students' understanding of CAD tools and their role in prototyping.
- To enable students to integrate and test basic electronic components using prototyping techniques.
- To develop the practical skills on designing the real-world circuits.

COURSE CONTENT

Module I: Introduction to PCB Design
Overview of PCB design, types of PCBs, applications, materials used, and design considerations. Introduction to PCB design software.

Module 2: Schematic Design
Exploration of the PCB design software: Symbols and footprints, component library management. Design and simulation of basic circuits.

Module 3: Layout Design
PCB stack-up, routing strategies, grounding techniques, component placement, and design rules. Designing single-layer and double-layer PCBs, performing ERC (Electrical Rule Check) and DRC (Design Rule Check).

Module 4: PCB Fabrication and Assembly
PCB manufacturing process, CAM files (Gerber files), via types, solder mask, silkscreen, soldering techniques, and safety. Generating Gerber files, preparing PCB for fabrication, printing and etching PCB prototypes, Hand soldering components, placing and soldering surface mount components, and assembling the PCB

Module 5: Testing and Troubleshooting
Testing procedures for continuity, voltage, and functionality; debugging and troubleshooting techniques. Functional testing of the PCB; identifying and correcting issues

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand and utilize PCB design software to create schematics and PCB layouts	K1
CO2	Design a PCB using CAD tools and prepare necessary documentation.	K4
CO3	Fabricate a PCB prototype by implementing design-for-manufacturing principles.	K5
CO4	Assemble and solder components on a PCB and perform functional testing.	K3
CO5	Identify and troubleshoot design and assembly errors effectively	K2

TEXT BOOKS

1. Khandpur, Raghbir Singh. ”Printed Circuit Boards Design, Fabrication, and Assembly.” (2006).
2. Mitzner, Kraig. Complete PCB design using OrCAD Capture and PCB editor. Newnes, 2009.

REFERENCE BOOKS

1. Wilson, Peter. The circuit designer’s companion. Newnes, 2017.
2. Eppinger, Steven D., and Karl Ulrich. ”Product design and development.” (1995).

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	-	-	1	-	-	-	-	-	-	-	-	-	-
CO2	2	3	1	-	2	-	-	-	-	-	-	-	-	-	2
CO3	-	2	2	-	-	-	-	-	-	-	-	1	-	-	-
CO4	-	1	3	2	-	-	-	2	-	-	-	-	-	-	-
CO5	1	2	2	1	-	-	-	-	1	-	-	1	-	-	-
score	5	11	8	3	3	-	-	-	3	-	-	2	-	-	2
COM	2	3	2	2	2	-	-	-	1	-	-	1	-	-	2

SEMESTER V

Course Code	ECMC301
Course Title	Digital Communication
Number of Credits(L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- Familiarize students with fundamental concepts in digital communications, including signal processing and basics in random process
- Introduce students to various digital transmission systems, such as baseband transmission and passband transmission.
- Discuss the effects of noise on digital communication systems and various channel characteristics.
- Describe the principles information theory and various coding techniques
- Provide practical, hands-on experience with simulation tools, software-defined radio platforms to enhance students' understanding of digital communication principles.

COURSE CONTENT

Unit I: Random Signal Theory
Geometric representation of signals, Gram-Schmidt orthogonalization, Introduction to random variables, joint probability density function, characteristic functions, derived distributions, Introduction to random processes, Gaussian process, Linear functional of random process, Stationary and wide sense stationary random process, Power spectral density, White Gaussian noise.

Unit II: Base Band Transmission
Pulse code modulation (PCM), PCM Generation and Reconstruction, Quantization Noise, Non-Uniform Quantization and Commanding, Differential and adaptive PCM, Noise considerations in PCM, Delta modulation and demodulation, slope overload, Adaptive and Sigma Delta Modulation.

Unit III: Band Pass Transmission
Generation and detection of coherent & non-coherent Amplitude, Frequency & Phase Shift Keying (ASK, FSK, PSK), Quadrature modulation techniques, Minimum Shift Keying (MSK), M-ary FSK, Optimal receiver and Probability of error, comparison of various digital modulation techniques, Coherent reception of ASK, PSK and FSK, Non-Coherent reception of ASK, FSK, PSK and QPSK.

Unit IV: Information Theory and Source Coding
Definition of Information, Entropy, Mutual Information, Properties of Mutual Information, Relationship between entropy and mutual information, Types of Channels, Binary Symmetric Channel, Channel Capacity, Channel Coding Theorem, Shannon-Hartley Theorem, Kraft inequality, Huffman codes, Shannon-Fano coding.

Unit V: Channel Coding
Linear Block Codes, Systematic linear codes and optimum decoding for the binary symmetric channel, Generator and Parity Check matrices, Syndrome decoding on symmetric channels; Hamming codes, Weight enumerators and the Mac-Williams identities; Perfect codes. Cyclic Codes, BCH codes, Convolution codes, Turbo codes.

LIST OF EXPERIMENTS

1. Study of line coding techniques.
2. Study of sampling theorem.
3. Study of pulse code modulation and demodulation.
4. Study of delta modulation and demodulation, the effect of slope overload.
5. Study pulse data coding techniques for NRZ and RZ formats.
6. Study of Binary modulation schemes: ASK, PSK, and FSK.
7. Study of M-ary modulation schemes: QPSK, MSK and QAM modulator and demodulator.
8. Study of channel coding schemes.
9. Study of multiplexing technique such as CDMA.

Total Periods: 40+24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate the understanding of basic concepts in digital communications, including signal processing and random process	K2
CO2	Apply knowledge of various digital modulation techniques to design and implement communication systems	K3
CO3	Analyse the performance of digital communication systems, including metrics such as bit error rate (BER), signal-to-noise ratio (SNR), and bandwidth efficiency	K4
CO4	Explain the concept of information, entropy, and describe various data compression coding techniques	K2
CO5	Design and implement error control coding techniques to detect and correct errors in transmitted data	K5

TEXT BOOKS

1. John G. Proakis and Masoud Salehi, “Fundamentals of Communication Systems,” Pearson Education India, 2007.

2. Simon Haykin, “Communications Systems,” John Wiley, 2008.

3. Peebles, P. Z., “Probability Random Variables and Random Signal Principles”, 4th Edition, Mc-Graw Hill Education, 2018.

REFERENCE BOOKS

1. B.P. Lathi and Z. Ding, “Modern Digital and Analog Communication Systems”, Oxford Press, 5th edition, 2018

2. Taub, H. and Schilling, D. L., “Principles of communication systems,” 4th Edition, Mc-Graw Hill India, 2012.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	2	-	-	-	-	-	3	-	-	-	2
CO2	3	2	2	3	2	-	-	-	-	-	3	3	-	-	2
CO3	3	2	2	3	2	3	-	-	-	-	3	3	-	-	2
CO4	3	2	-	-	2	-	-	-	-	-	3	-	-	-	2
CO5	3	2	2	3	2	-	-	-	-	-	3	-	-	-	2
Score	15	10	6	9	10	3	-	-	-	-	15	6	-	-	10
COM	3	2	2	2	2	-	-	-	-	-	3	3	-	-	2

Course Code	ICAE301
Course Title	Professional Communication and Soft Skills
Number of Credits (L-T-P-C)	0-0-4-2
Course Type	AE

COURSE OBJECTIVES

- To promote theoretical understanding and professional/personal practice of effective and ethical human communication between and within a broad range of contexts and communities.
- To develop awareness about different forms of professional communication and social behavior in a team and as a team.
- To hone the employability related communication skills of the students by empowering them with appropriate language usage for presentation delivery, interviews, and public speaking.
- To deal professionally through the art of negotiation.
- To instill confidence to speak in public and do group discussions.

COURSE CONTENT

Unit I: Introduction to Soft Skills and Professional Ethics
Aspects of Soft Skills: Effective Communication Skills, Personality Development, Importance of Professional Ethics.

Unit II: Team Building
Understanding nature of team: Mapping personal and professional goals of team members, Working effectively in a team through building relations and interpersonal communication. What is negotiation, Ways of negotiating, Understanding the power of language and non-verbal communication.

Unit III: Meetings and Presentation
How to call a meeting, how to organize a meeting, how to design the agenda and prepare minutes of the meeting. Researching for a presentation, structure of presentation, verbs often required, language focus, importance of body language in presentation, preparing an outline of a presentation, ending the presentation.

Unit IV: Stress Management and Time Management
Kinds of stress, Identifying the right reasons of stress, how to handle the pressure, Techniques to cope with the stressful situation at a workplace, Goal setting, Understand the importance of time and how to prepare the time line.

Unit V: Group Discussion and Public Speaking
Nature of discussion, Ways to form and present the arguments, public speaking skills and being successful in it.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Develop awareness about personality development, social behavior and professional ethics.	K3
CO2	Understand and recognize the importance of interpersonal skills and team dynamics, and strengthen individual expression in collaborative peer activities.	K3
CO3	Apply concepts of negotiation to a workplace situation and effectively plan a negotiation using appropriate verbal and nonverbal cues.	K3
CO4	Plan and execute Meetings, and draft minutes, reports and relevant documents.	K3
CO5	Develop coherence, cohesion and competence essential for presentation deliveries.	K3

TEXT BOOKS

1. Rizvi, M. A., *"Effective Technical Communication"* , 2nd edition, McGraw Hill Education, 2017.
2. Mohan, K. and Banerji, M., *"Developing Communication Skills"* , 2nd edition, Laxmi Publications, 2009.
3. Markel, M and Selber, S.A., *"Technical Communication"*, 12th Edition, Macmillan Learning, 2018.

REFERENCE BOOKS

1. Dale, C., *"How to Win Friends and Influence People"*, New York: Simon and Schuster, 1998.
2. Coleman, D., *"Emotional Intelligence"*. Bantam Book, 2006.
3. Thorpe, E., and Thorpe, S., *"Objective English"*, Pearson Education, New Delhi, 2007.
4. Tyagi, V.K. and Tyagi, B, "Soft Skills for Engineers & Professionals, , First Edition, BlueRose Publishers Pvt. Ltd. , 21 July 2022.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-	-
Score	-	-	-	-	-	-	-	3	6	6	-	-	-	-	3
COM	-	-	-	-	-	-	-	3	3	3	-	-	-	-	3

Course Code	ICVA301
Course Title	Professional Ethics
Number of Credits (L-T-P-C)	3-0-0-3
Course Type	VA

COURSE OBJECTIVES

- To enable the students to create on awareness on Engineering Ethics and Human Values, to instil Moral and Social Values and loyalty and to appreciate the rights of others.
- To develop and prompt the accountability and responsibility being engineering professional.
- To enable students to identify and analyze ethical dilemmas in their professional lives and make informed, principled decisions.
- To guide students in adhering to professional codes of conduct and industry standards, ensuring that their behavior positively reflects on their profession and organization.
- To cultivate an appreciation for diverse perspectives, cultures, and practices, ensuring respectful and equitable treatment of all individuals in a professional setting.

COURSE CONTENT

Unit I: Human Values
Morals, values and Ethics, Integrity, Work ethic, Service learning, Civic virtue, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Character, Spirituality: Introduction to Yoga and meditation for professional excellence and stress management.

Unit II: Engineering Ethics
Senses of 'Engineering Ethics', Variety of moral issues, Types of inquiry, Moral dilemmas, Moral Autonomy, Kohlberg's theory, Gilligan's theory, Consensus and Controversy, Models of professional roles, Theories about right action, Self-interest, Customs and Religion, Uses of Ethical Theories

Unit III: Engineering As Social Experimentation
Engineering as Experimentation, Engineers as responsible Experimenters, Codes of Ethics, A Balanced Outlook on Law.

Unit IV: Safety, Responsibilities And Rights
Safety and Risk, Assessment of Safety and Risk, Risk Benefit Analysis and Reducing Risk, Respect for Authority, Collective Bargaining, Confidentiality, Conflicts of Interest, Occupational Crime, Professional Rights, Employee Rights, Intellectual Property Rights (IPR) – Importance of Plagiarism, Discrimination

Unit V: Global Issues
Multinational Corporations, Environmental Ethics, Computer Ethics, Weapons Development, Engineers as Managers, Consulting Engineers, Engineers as Expert Witnesses and Advisors, Moral Leadership, Code of Conduct, Corporate Social Responsibility

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Distinguish between ethical and non-ethical situations.	K4
CO2	Practice moral judgment effectively handling the conditions of dilemma.	K3
CO3	Develop cognitive skills in solving social problems and apply the code of ethics to social experimentation.	K3
CO4	Apply risk and safety measures in various engineering fields.	K3
CO5	Explain corporate social responsibility and understand the concern for ethical contribution for the global society.	K2

TEXT BOOKS

1. Govindarajan M., Natarajan S., Senthilkumar V.S., *"Engineering Ethics"*, Prentice Hall of India, 2013.
2. Martin Mike W., Schinzinger Roland, *"Ethics in engineering"*, Tata Mc Graw Hill, 4th Editin,2005.

REFERENCE BOOKS

1. Ferrell, O. C., Fraedrich, J., & Ferrell, L. (2005). Business ethics: Ethical decision making and cases (13th ed.). Cengage Learning.
2. Singer, P., & Singer, F. A. (2008). Ethics for the real world: Creating a personal code to guide decisions in work and life. Harvard Business Review Press.
3. Brenkert, G. G. (2004). Corporate integrity and accountability. SAGE Publications.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	3	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO4	-	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	2	-	-	-	-	-	-	-	-	-
Score	-	-	-	-	-	2	2	9	-	-	-	-	-	-	3
COM	-	-	-	-	-	2	2	3	-	-	-	-	-	-	3

Course Code	ICHC301
Course Title	Honours Online Course- I*
Number of Credits (L-T-P-C)	X-X-X-3
Course Type	HC

This is an optional course for the students who want to opt for B.Tech. (Honours). The Students must adhere the guidelines mentioned in the applicable UG Ordinance. The students can choose online courses from NPTEL or SWAYAM or MOOCs. They should undergo the online course completely, submit assignments, projects, etc., and appear for the final exam conducted by the online instructor. The awarded marks/grade must be submitted for the award of suitable letter grade in this course.

Course Code	ICOC301
Course Title	Optional Online Course- I*
Number of Credits (L-T-P-C)	X-X-X-(0-3)
Course Type	OC

The Students can opt for Optional Online Course along with their normal courses. The students can choose online courses from NPTEL/SWAYAM/MOOCs. In Optional course the credit will not be counted for the calculation of the final CGPA but the credit will appear in the Grade card and transcript.

SEMESTER VI

Course Code	ECMC302
Course Title	VLSI DESIGN
Number of Credits(L-T-P-C)	3-0-2-4
Course Type	MC

COURSE OBJECTIVES

- To get familiar with emerging VLSI technology and MOS transistor.
- To learn about the various fundamentals used in the design of CMOS logic circuits.
- To understand the concepts related to design of digital CMOS logic circuits.
- To learn about the design techniques for Sequential MOS Logic Circuits.
- To study various low power techniques for CMOS circuits.

COURSE CONTENT

Unit I: Introduction to VLSI and MOS
Need of VLSI, Design Flow, Concept of Regularity, Modularity and Locality, VLSI Design Styles, Computer-Aided Design Technology, MOS Transistor theory: Basic MOS Transistors, MOSFET operation in Enhancement Mode and depletion mode, The MOSFET Transistor regions of operation Accumulation, Depletion and Strong Inversion, The Threshold voltage, I-V Characteristic of MOSFET.

Unit II: MOS Inverters and Its Switching Characteristics
Voltage Transfer Characteristic (VTC), Noise Immunity and Noise Margin, MOS Inverters: Resistive-Load Inverters, CMOS Inverter, Switching Characteristics and Interconnect Effects, Delay-Time Definitions, Rise time, fall time, Calculation of Delay-Times, Switching Power Dissipation of CMOS Inverters.

Unit III: CMOS Combinational Logic Circuit Design
Static CMOS Logic Gates (NAND and NOR), Psuedo NMOS, Dynamic Logic Circuits, Basic Principles of Pass Transistor Circuits, Transmission Gates, High Performance Dynamic CMOS Circuits, Domino Circuits.

Unit IV: CMOS Sequential Logic Circuit Desing
Introduction, Static Latches and Registers, Dynamic Latches and Registers, Flip flop and latch timings, SR, JK, D, and Master-slave flip flop, Pulse based Registers, Sense Amplifier Based registers, MOS Memories, DRAM and SRAM.

Unit V: LOW Power VLSI Design
Need for low power VLSI chips, Sources of power dissipation (Switching, Short and Leakage Power Dissipation), Low-Power Design techniques (VTCMOS, MTCMOS, Adiabatic Logic Circuits).

LIST OF EXPERIMENTS

1. To design and perform the transient analysis and DC analysis (VTC) of Resistive load inverter.
2. To design and perform transient Analysis and DC Analysis (VTC) of CMOS inverter.
3. To design and simulate the NAND and NOR Gate using Static CMOS logic.
4. To design and simulate the XOR and XNOR using pseudo techniques.
5. To design and simulate the Half adder and Full adder using Dynamic CMOS logic.
6. To design and simulate the given boolean expression using Domino CMOS logic.
7. To design and simulate the MUX using the Transmission Gate.
8. To design and simulate the SR and JK Flip Flop.
9. To design and Simulate Master-slave Flip Flop using CMOS logic.
10. To design and simulate D Flip flop using CMOS Logic.
11. To design and simulate the various logic circuits using MTCMOS and VTCMOS techniques.
12. Layout design of CMOS inverter.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	To get familiar with emerging VLSI Design and basic theory of MOS transistors.	K2
CO2	To understand the various characteristics used in the design of CMOS logic circuits.	K2
CO3	To implementation of logic gates using Digital CMOS circuits.	K3
CO4	Develop basic understanding to design basic Analog CMOS circuits.	K3
CO5	Analysis of the low power techniques for CMOS Circuits.	K4

TEXT BOOKS

1. Kang, S. M., Leblebici, Y., and Kim, C., “CMOS Digital Integrated Circuits: Analysis and Design”, 4th Edition, McGraw Hill Education, 2016.
2. Rabaey, J. M., Chandrakasan, A., and Nikolic, B., “Digital Integrated Circuits: A Design Perspective”, 2nd edition, Pearson Education, 2016.

REFERENCE BOOKS

1. Weste, Neil HE, David Harris, “CMOS VLSI design: a circuits and systems perspective”, 3rd edition, Pearson Education India, 2006.

2. Ajit. Pal, “Low-power VLSI circuits and systems”, First Edition, Springer, 2014.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1	-	-	-	-	-	-	-	-	-	2	2
CO2	3	3	1	3	-	-	-	-	-	-	-	-	-	2	2
CO3	3	3	3	3	-	-	-	-	-	-	-	-	-	2	2
CO4	3	3	3	3	-	-	-	-	-	-	-	-	-	2	2
CO5	3	3	3	3	-	2	-	-	-	-	-	-	-	2	2
Score	15	15	11	13	-	2	-	-	-	-	-	-	-	10	10
COM	3	3	3	3	-	2	-	-	-	-	-	-	-	2	2

Course Code	ICAE302
Course Title	Entrepreneurship and Start-ups
Number of Credits (L-T-P-C)	3-0-0-3
Course Type	AE

COURSE OBJECTIVES

- To impart basic entrepreneurial skills and understanding to run a business efficiently and effectively.
- To develop and strengthen motivation and leadership skills in students.
- To make them understand the classification of MSMEs in India and exposing to new business trends for startups.
- To provide the understanding for various financial resources and sustaining the business enterprise.
- To update students with various govt. schemes for the support of sick units and strategies for business growth.

COURSE CONTENT

Unit I: Entrepreneurship
Concept of Entrepreneurship, Entrepreneurial skills, Characteristics of an Entrepreneur, Types of Entrepreneurs, Difference between Entrepreneur and Intrapreneur, Barriers to Entrepreneurship, Risks to an Entrepreneur. Process in Entrepreneurship, Macroeconomic environment, Influence of macroeconomic conditions on Entrepreneurship.

Unit II: Motivation and Leadership
Psychology theories of entrepreneurship, McClelland's Theory of motivation, Rotter's Locus of Control Theory, Action Regulation Theory, Stress Management, Leading mission-driven organizations, Developing Mission, Vision, and Goals statements of the organizations, Developing competitive advantage, Generic competitive strategy, Building Motivated Team. Entrepreneurship Development Programs, Need, Objectives.

Unit III: MSME and Types of New Age Business
MSME – Definition, Classification, Characteristics, Role and Importance of MSMEs in Indian Economy, Ownership Structures, Types of new age business, FinTech, EdTech, Healthcare, AgriTech, Robotics, and Digital Transformation, Emerging markets for Entrepreneurs-Cybersecurity and cloud computing, Tax management, Real Estate, supply chain and logistics. Formulation of Project report, Need, Objectives, and Components of a Project Report, Techno-Economic, Market, and Financial Feasibility Assessment of a project.

Unit IV: Financing and Business Stability
Methods of Financing to Startups, Funding Strategies, Different types of Funding, Sources of

Finance, Term Loans, Capital Structure, Financial Institution, Management of working Capital, Scaling up and stabilization of a business, strategies for scaling business. Data Analytics to advantage, Advantages of Data Analytics.

Unit V: Support to Entrepreneurs
Sickness in small Business, Concept, Magnitude, Causes and Consequences, Corrective Measures, Business Incubators, Government Policy for Small Scale Enterprises, Growth Strategies in small industry, Expansion, Diversification, Joint Venture, Merger and Sub Contracting.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the entrepreneurial mind-set and what it takes to create value.	K2
CO2	Develop motivation and leadership to start one's own business.	K3
CO3	Analyse the business enterprises in order to identify business opportunities in trending new age businesses.	K4
CO4	Learn to secure financial backing and sustain business growth.	K1
CO5	Evaluates the effectiveness of different entrepreneurial strategies.	K5

TEXT BOOKS

1. Khanka. S.S., *"Entrepreneurial Development"* S.Chand & Co. Ltd.,Ram Nagar, New Delhi, 2013.
2. Donald F Kuratko, *"Entrepreneurship – Theory, Process and Practice"*, 9th Edition, Cengage Learning 2014.
3. *"Innovation and Entrepreneurship"* by Peter F. Drucker,2006
4. *"The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses"* by Eric Ries,2011

REFERENCE BOOKS

1. *"Small Business Management: Launching and Growing Entrepreneurial Ventures"* by Justin G. Longenecker, J. William Petty, and Leslie E. Palich,2017
2. *"Entrepreneurship Development and Small Business Enterprises"* by Poornima M. Charantimath, 978-9353066260, 2018
3. *"Entrepreneurship and Startup"* by Institute of Cost Accountants of India, Kolkata,2018
4. *"Disciplined Entrepreneurship: 24 Steps to a Successful Startup"* by Bill Aulet,2013

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	3	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-	3	-	-	2
CO3	-	-	-	-	-	-	2	-	-	-	-	3	-	-	3
CO4	-	-	-	-	-	-	-	-	-	-	3	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	3	3	-	-	-
Score	-	-	-	-	-	-	2	3	3	-	6	15	-	-	5
COM	-	-	-	-	-	-	2	3	3	-	3	3	-	-	3

Course Code	ICPR301
Course Title	Minor Project
Number of Credits (L-T-P-C)	0-0-8-4
Course Type	PR

COURSE OBJECTIVES

- To explore project domain and state objectives clearly and concisely.
- To learn the state-of-art methods for the project domain.
- To survey research problems and derive methodologies to solve the problem.
- To analyze state-of-the-art techniques to solve real-world problems.
- To demonstrate and implement the proposed solution.

COURSE CONTENT

The project work is designed for a one semester duration as a single project involving detailed literature survey, experimentation plan and its implementation.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate a sound technical knowledge of the selected project topic.	K2
CO2	Survey research studies, find research gaps, and formulate a complex engineering problem.	K6
CO3	Experiment with state-of-the-art methods and identify the available solutions.	K4
CO4	Analyze and compare the available solutions.	K4
CO5	Plan, propose, and implement the proposed solution.	K6

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	2	1	2	2	-	3	3	-	3	-	-	3
CO2	3	3	1	3	1	2	2	3	3	3	3	3	3	3	3
CO3	3	3	-	3	1	2	2	-	3	3	3	3	3	3	3
CO4	3	3	2	3	3	-	-	-	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Score	15	14	6	14	9	9	9	6	15	15	12	15	12	12	15
COM	3	3	2	3	2	3	3	3	3	3	3	3	3	3	3

Course Code	ICHC302
Course Title	Honours Online Course- II*
Number of Credits (L-T-P-C)	X-X-X-3
Course Type	HC

This is an optional course for the students who want to opt for B.Tech. (Honours). The Students must adhere the guidelines mentioned in the applicable UG Ordinance. The students can choose online courses from NPTEL or SWAYAM or MOOCs. They should undergo the online course completely, submit assignments, projects, etc., and appear for the final exam conducted by the online instructor. The awarded marks/grade must be submitted for the award of suitable letter grade in this course.

Course Code	ICOC302
Course Title	Optional Online Course- II*
Number of Credits (L-T-P-C)	X-X-X-(0-3)
Course Type	OC

The Students can opt for Optional Online Course along with their normal courses. The students can choose online courses from NPTEL/SWAYAM/MOOCs. In Optional course the credit will not be counted for the calculation of the final CGPA but the credit will appear in the Grade card and transcript.

SEMESTER VII

Course Code	ICPR401
Course Title	Major Project-I
Number of Credits (L-T-P-C)	0-0-10-5
Course Type	PR

COURSE OBJECTIVES

- To explore project domain and state objectives clearly and concisely.
- To learn the state-of-art methods for the project domain.
- To survey research problems and derive methodologies to solve the problem.
- To analyze state-of-the-art techniques to solve real world problems.
- To demonstrate and implement the proposed solution.

COURSE CONTENT

In this course, the student is expected to work on a project to solve a real-world problem un- der the supervision of the faculty member. The project work include detailed literature survey,problem formulation, experimentation plan and its implementation in a form of working proto- type.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate a sound technical knowledge of the selected project topic.	K2
CO2	Survey research studies, find research gaps, and formulate a complex engineering problem.	K6
CO3	Experiment with state-of-the-art methods and identify the available solutions.	K4
CO4	Analyze and compare the available solutions.	K4
CO5	Plan, propose, and implement the proposed solution.	K6

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	2	1	2	2	-	3	3	-	3	-	-	3
CO2	3	3	1	3	1	2	2	3	3	3	3	3	3	3	3
CO3	3	3	-	3	1	2	2	-	3	3	3	3	3	3	3
CO4	3	3	2	3	3	-	-	-	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Score	15	14	6	14	9	9	9	6	15	15	12	15	12	12	15
COM	3	3	2	3	2	3	3	3	3	3	3	3	3	3	3

Course Code	ICHC401
Course Title	Honours Online Course- III*
Number of Credits (L-T-P-C)	X-X-X-3
Course Type	HC

This is an optional course for the students who want to opt for B.Tech. (Honours). The Students must adhere the guidelines mentioned in the applicable UG Ordinance. The students can choose online courses from NPTEL or SWAYAM or MOOCs. They should undergo the online course completely, submit assignments, projects, etc., and appear for the final exam conducted by the online instructor. The awarded marks/grade must be submitted for the award of suitable letter grade in this course.

Course Code	ICOC401
Course Title	Optional Online Course- III*
Number of Credits (L-T-P-C)	X-X-X-(0-3)
Course Type	OC

The Students can opt for Optional Online Course along with their normal courses. The students can choose online courses from NPTEL/SWAYAM/MOOCs. In Optional course the credit will not be counted for the calculation of the final CGPA but the credit will appear in the Grade card and transcript.

SEMESTER VIII

Course Code	ICPR402
Course Title	Major Project-II
Number of Credits (L-T-P-C)	0-0-20-10
Course Type	PR

COURSE OBJECTIVES

- To explore project domain and state objectives clearly and concisely.
- To learn the state-of-art methods for the project domain.
- To survey research problems and derive methodologies to solve the problem.
- To analyze state-of-the-art techniques to solve real-world problems.
- To demonstrate and implement the proposed solution.

COURSE CONTENT

In this course the student may extend the work done in Major Project-I. The extension (if any) should be significant and justifiable with respect to the credits of the course. If extension of the project is not feasible, then the student may work on the new project under the supervision of a faculty member.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Illustrate a sound technical knowledge of the selected project domain.	K2
CO2	Survey research studies, find research gaps, and formulate a complex engineering problem.	K6
CO3	Experiment with state-of-the-art methods and identify the available solutions.	K4
CO4	Demonstrate and compare the available solutions.	K4
CO5	Construct and implement the proposed solution.	K6

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	2	1	2	2	-	3	3	-	-	-	-	3
CO2	3	3	1	3	1	2	2	3	3	3	3	3	3	3	3
CO3	3	3	-	3	1	2	2	-	3	3	3	3	3	3	3
CO4	3	3	2	3	3	-	-	-	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Score	15	14	6	14	9	9	9	6	15	15	12	12	12	12	15
COM	3	3	2	3	2	3	3	3	3	3	3	3	3	3	3

Course Code	ICHC402
Course Title	Honours Online Course- IV*
Number of Credits (L-T-P-C)	X-X-X-3
Course Type	HC

This is an optional course for the students who want to opt for B.Tech. (Honours). The Students must adhere the guidelines mentioned in the applicable UG Ordinance. The students can choose online courses from NPTEL or SWAYAM or MOOCs. They should undergo the online course completely, submit assignments, projects, etc., and appear for the final exam conducted by the online instructor. The awarded marks/grade must be submitted for the award of suitable letter grade in this course.

Course Code	ICOC402
Course Title	Optional Online Course- IV*
Number of Credits (L-T-P-C)	X-X-X-(0-3)
Course Type	OC

The Students can opt for Optional Online Course along with their normal courses. The students can choose online courses from NPTEL/SWAYAM/MOOCs. In Optional course the credit will not be counted for the calculation of the final CGPA but the credit will appear in the Grade card and transcript.

PROGRAM ELECTIVES

PROGRAM ELECTIVE-I

Course Code	ECSE301
Course Title	Data Communication and Networks
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To provide the fundamental concepts of computer networks, reference models and its functionalities.
- To explore the different types of signalling schemes and channel access techniques in physical layer.
- To identify the key design issues of data link and network layer protocols.
- To evaluate the network performance using various services provided by transport and application layers.

COURSE CONTENTS

Unit I: Introduction to Computer Networks
Network, Component and Categories, Topologies, Transmission Media, Reference Models: ISO/OSI Model and TCP/IP Model.

Unit II: Physical Layer
Digital and Analog Signals, Periodic Analog Signals, Transmission Impairments, Digital data transmission techniques, Analog data transmission techniques, Multiplexing and Spread Spectrum.

Unit III: Data Link Layer
Error Detection and Correction, Parity, LRC-CRC, Hamming Code, Flow Control and Error Control, Stop and wait, ARQ, Sliding window, HDLC, Multiple Access Protocols, IEEE 802.3 Ethernet.

Unit IV: Network Layer
Packet Switching and Datagram approach, IP addressing methods, Subnetting, Routing, Distance Vector Routing, Link State Routing, Broadcast and Multicast Routing.

Unit V: Transport and Application Layer
Transport Services, UDP, TCP, Congestion Control, Quality of Services (QOS), Application Layer: Domain Name Space (DNS), Electronic Mail.

LIST OF EXPERIMENTS

1. Study of different typed of Networks Cable and Practically Implement the cross-wired cable and straight through cable using clamping tool.
2. Install and Configure Wired and Wireless NIC and transfer files between systems in LAN and Wireless LAN.
3. Install and configure Network Devices: HUB, Switch and Routers.
4. Connect the Computers in Local Area Network.
5. Configure Host IP, Subnet Mask and Default Gateway in a System in LAN (TCP/IP Configuration).
6. Establish Peer to Peer network connection using two systems using Switch and Router in a LAN.
7. Configure Internet connection and use IPCONFIG, PING / Tracer and Net stat utilities to debug the network issues.
8. Transfer files between systems in LAN using FTP Configuration, install Print server in a LAN and share the printer in a network.
9. Router Configuration Using Packet Tracer.
10. Connection oriented Client server applications with TCP Assignment.
11. Connectionless Client server applications with UDP Assignment.
12. Study of Socket Programming and Client – Server Model.
13. Configure a Network Topology using packet tracer software.
14. Configure a Network using Distance Vector Routing Protocol.
15. Configure a Network using Link State Vector Routing Protocol.
16. To get the MAC or Physical address of the system using Address Resolution Protocol.
17. Simulate the Implementing Routing Protocols using border gateway protocol (BGP) .
18. Simulate the OPEN SHORTEST PATH FIRST routing protocol based on the cost assigned to the path

Total Periods: 40 + 28 = 68

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Infer the layered protocol approach in computer networking.	K2
CO2	Explain and apply the fundamentals of signal processing in the physical layer of the computer networking.	K3
CO3	Make use of various error detection and correction services provided by the data link layer.	K3
CO4	Analyze the various addressing and routing services provided by the network layer and design network layer protocols.	K4
CO5	Infer and apply various services provided by the transport and the application layer.	K3

TEXT BOOKS

1. Foruzan, B. A., “Data Communication and Networking”, 5th Edition, McGraw Hill Education, 2017.
2. Tanenbaum, A. S. and Wetherall, D. J., “Computer Networks”, 5th Edition, Pearson Education, 2013. .

REFERENCE BOOKS

1. Stallings, W., “Data and Computer Communication”, 10th Edition, Pearson Education, 2017.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	-	-	-	-	-	-	-	-	3	-	2
CO2	2	2	2	2	-	-	-	-	-	-	-	-	3	-	2
CO3	2	2	2	2	-	-	-	-	-	-	-	-	3	-	2
CO4	2	2	2	2	-	-	-	-	-	-	-	-	3	-	2
CO5	2	2	2	2	-	-	-	-	-	-	-	-	3	-	2
Score	10	10	10	9	-	-	-	-	-	-	-	-	15	-	10
COM	2	2	2	2	-	-	-	-	-	-	-	-	3	-	2

Course Code	ECSE302
Course Title	Embedded Systems
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To learn about the characteristics and challenges of embedded systems.
- To learn about the typical AVR ATmega32 Microcontroller.
- To learn the assembly language programs for AVR Microcontroller.
- To understand the architecture and functionalities of AVR on-chip peripherals.
- To Interface I/O peripheral devices with AVR microcontroller to develop the embedded system.

COURSE CONTENTS

Unit I: Introduction to Embedded Systems
History of embedded system, Embedded Systems vs General computing systems, Classification of embedded system, Major application areas, Characteristics of embedded system, Constraints, Characteristics and Quality Attributes of Embedded Systems. Introduction to RISC/CISC-based machines. Introduction to 8051 microcontroller.

Unit II: Architecture and instruction set of 8-bit AVR Microcontroller
AVR Microcontroller architecture: Registers, AVR status register, Memory Space, ATmega32 pin-configuration & function of each pin, Addressing mode and instruction set of AVR microcontroller, Data transfer, Arithmetic, Logic and Compare, Rotate and Shift, Branch and Call instructions, Bit manipulation instructions.

Unit III: AVR Assembly and C Programming
AVR data types and assembler directives, AVR assembly language programs, AVR I/O Port Programming, Time delay loop, BCD, ASCII conversion Program, Look-up table, Bit addressability, MACROs, Pros and cons of C and assembly language programming, Data types, Intex Hex file format, Simple C programs for general purpose I/O and bit addressability.

Unit IV: AVR on-chip peripherals and its programming
General purpose I/O Ports, Timers, Interrupts, serial port, Serial port Interfacing protocols, SPI, I2C, UART. Assembly and C Language programming for peripherals.

Unit V: Device interfacing and its programming
LCD and GLCD interfacing, Keyboard Interfacing, TFT interfacing, ADC, DAC and sensor interfacing, Relay, Opto-isolator and Stepper Motor Interfacing, Industrial servo interfacing, Input capture and Wave Generator, PWM programming and DC motor control, SPI protocol and Display interfacing, I2C Protocol and RTC interfacing. Assembly and C Programming.

LIST OF EXPERIMENTS

1. Understand Arduino open source hardware and programming environment and write program to blink LED using Arduino instructions, C language & Assembly language.
2. Interface Digital/Analog input output interfacing module with Arduino board and write programs related to I/O module.
3. Generate PWM waveform and change intensity of LED connected with Arduino board.
4. Write and execute Arduino program for serial communication. Transmit temperature value through serial communication and store it in spreadsheet or text file.
5. Write and execute Arduino program to display message and numbers on LCD, GLCD and TFT.
6. Write and execute Arduino program to read analog value. Sense temperature using LM35 sensor and display temperature value on LCD.
7. Write assembly language programs for ATMega32 Microcontroller and simulate using ATMEL Studio.
8. Understand hardware of ATMega32 Kit. Write program to flash LEDs, read status of switches, Display count values on seven segment display. Upload programs in the kit one by one and execute.
9. Write program to read switch status and display it on LCD. Write program in Assembly as well as C language.
10. Write program to rotate stepper motor in half-step and full-step mode in Assembly as well as C language. Simulate program using PROTEUS software.
11. Write program to rotate DC motor in clockwise and anti-clockwise direction in Assembly as well as C language. Simulate program using PROTEUS software.
12. Observe waveforms of I2C and SPI communication and understand I2C and SPI protocol.
13. Write Arduino program to receive IR Signal from IR remote and operate Electrical device based on switch pressed.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Infer the overall landscape and characteristics of embedded systems.	K2
CO2	Analyze the architecture, pin configuration, addressing modes, and instruction set of the AVR microcontroller, including ATmega32.	K3
CO3	Develop AVR assembly and C programs using data types, assembler directives, and Intel HEX file format, and implement I/O port programming, bit addressability, time delays, and look-up tables to solve practical embedded system tasks.	K4
CO4	Design and implement efficient assembly and C programs to configure and utilize AVR on-chip Peripherals.	K6
CO5	Develop assembly and C programs for device interfacing, including displays, sensors, motors, and communication protocols, to design and implement practical embedded system applications.	K5

TEXT BOOKS

1. Muhammad Ali Mazidi, Sarmad Naimi and Sepehr Naimi,"The AVR Microcontroller and Embedded Systems", Using Assembly and C, Pearson Education, 1st Edition,2012.

2. Dhananjay Gadre, "Programming and Customizing the AVR Microcontroller", TMH, 1st Edition, 2001.

REFERENCE BOOKS

1. Muhammad Ali Mazidi, Janice Gillispie Mazidi and Rolin D. McKinlay,"The 8051 Micro controllers & Embedded Systems Using Assembly and C" 2nd Indian reprint, Pearson education, 2005.

2. William Hohl, "ARM Assembly Language: Fundamentals and Techniques" CRC Press, 2009.

3. Tammy Noergaard, "Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers (Embedded Technology)" Elsevier, Newnes.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	-	-	-	-	-	-	-	-	-	2	2
CO2	3	2	3	3	-	-	-	-	-	-	-	-	-	2	2
CO3	3	2	3	3	-	-	-	-	-	-	-	-	-	2	2
CO4	3	2	3	3	-	-	-	-	-	-	-	-	-	2	2
CO5	3	2	3	3	-	-	-	-	-	-	-	-	-	2	2
Score	15	10	13	13	-	-	-	-	-	-	-	-	-	10	10
COM	3	2	3	3	-	-	-	-	-	-	-	-	-	2	2

PROGRAM ELECTIVE-II

Course Code	ECSE303
Course Title	Digital Design and Synthesis
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To Understand the language constructs and programming fundamentals of Verilog HDL.
- To Construct circuits in Gate level modeling style using Verilog HDL
- To Construct circuits in Data flow modeling style using Verilog HDL
- To Construct circuits in behavioral modeling style using Verilog HDL.
- To Analyze and Verify the functionality of digital circuits/systems using test benches.

COURSE CONTENTS

Unit I: Introduction to Verilog HDL
Verilog as HDL, Levels of Design Description, Concurrency, Program structure, Top-down and Bottomup design methodology, differences between modules and module instances, parts of a simulation, design block, stimulus block, Verilog Data types and Operators, system tasks, compiler directives.

Unit II: Gate level Modeling in Verilog HDL
Modelling using basic Verilog gate Primitives, Description of and/or and buf/not type gates, rise, fall and turn-off delays, min, max, and typical delays, Design of Decoders, Multiplexers, Flip-flops, Registers & Counters in Gate-level Modeling.

Unit III: Data flow modeling in Verilog HDL
Continuous assignments, Delay specification, expressions, operators, Design of Decoders, Multiplexers, Flip-flops, Registers & Counters in dataflow model.

Unit IV: Behavioural Modeling in Verilog HDL
User- Defined Primitives (UDP, System tasks, functions, and compiler directives, Creating test benches for Verilog HDL design verification. Hierarchical design and reusable modules, Parameterized Modules and Macros, Designing Memories and Arithmetic circuits in Verilog HDL, Designing Finite State Machines with Mealy and Moore Models, Verification process and types of simulation, Debugging and error tracing in Verilog HDL design.

Unit V: Components Test and Verification
Test Bench - Combinational Circuits Testing, Sequential Circuits Testing, Test Bench Techniques, Design Verification, Assertion Verification.

LIST OF EXPERIMENTS

1. Write Verilog code to realize all the logic gates
2. Write a Verilog program for the following combinational designs
   a. 2 to 4 decoder
   b. 8 to 3 (encoder without priority & with priority)
   c. 8 to 1 multiplexer.
   d. 4-bit binary to gray converter
3. Write Verilog code to describe the functions of a Full Adder using three modeling styles.
4. Write Verilog code for Code Converter: BCD to Excess-3 code converter, gray code to binary converter.
5. Develop the Verilog code for the following flip-flops, SR, D, JK and T.
6. Design a 4-bit binary, BCD counters (Synchronous reset and Asynchronous reset) and "any sequence" counters, using Verilog code.
7. Write Verilog code for counter with given input clock and check whether it works as clock divider performing division of clock by 2, 4, 8 and 16. Verify the functionality of the code.
8. Write a Verilog code to design a clock divider circuit that generates 1/2, 1/3rd and 1/4thclock from a given input clock. Port the design to FPGA and validate the functionality through oscilloscope.
9. Interface a DC motor to FPGA and write Verilog code to change its speed and direction.
10. Flip-Flop: assemble, test and investigate operation of S-R, D & J-K flip-flops.
11. Write Verilog code using FSM to simulate elevator operation.
12. Write Verilog code to convert an analog input of a sensor to digital form and to display the same on a suitable display like set of simple LEDs, 7-segment display digits or LCD display. Design of Combinational circuits using the Verilog HDL module.
13. Write HDL code to display messages on an alpha numeric LCD display.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Learn Verilog HDL and apply it for simple and complex designs.	K2
CO2	Demonstrate knowledge on HDL design flow, digital circuits design, switch de-bouncing, metastability	K3
CO3	Choose the suitable abstraction level for a particular digital design.	K3
CO4	Construct Combinational and sequential circuits in different modelling styles using Verilog	K4
CO5	Analyse and Verify the functionality of digital circuits/systems using test benches	K2

TEXT BOOKS

1. Samir Palnitkar, ”Verilog HDL: A Guide to Digital Design and Synthesis”, Pearson Education, Second Edition.
2. Michael D. Ciletti, ”Advanced Digital Design with the Verilog HDL” Pearson (Prentice Hall), Second edition.

REFERENCE BOOKS

1. Stephen Brown and Zvonko Vranesic - Fundamentals of Digital Logic with Verilog, 2nd Edition, TMH,
2. Z Navabi - Verilog Digital System Design, 2nd Edition, McGraw Hill, 2005.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	3	0	0	0	0	0	0	0	0	0	0	2
CO2	3	2	2	3	0	0	0	0	0	0	0	0	0	0	2
CO3	3	2	2	3	0	0	0	0	0	0	0	0	0	0	2
CO4	3	2	2	3	0	0	0	0	0	0	0	0	0	0	2
CO5	3	2	2	3	0	0	0	0	0	0	0	0	0	0	2
Score	15	10	9	15	0	0	0	0	0	0	0	0	0	0	10
COM	3	2	2	3	0	0	0	0	0	0	0	0	0	0	2

Course Code	ECSE304
Course Title	Fiber Optic Communication
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To know the principles of Fiber optic communication.
- To learn the concepts and types of optical sources.
- To study the various types of optical detectors and their applications.
- To learn the principle of optical amplification for different types of optical amplifiers.
- To understand the concepts of optical switching and networks.

COURSE CONTENTS

Unit I: Introduction to Optical Fiber
Optical evolution, advantages of optical communication, its representations, Optical waveguides, basic optical laws, acceptance angle, numerical aperture Skew rays, Rays and modes, step-index, graded-index fibers, phase and group velocities, Signal degradation in optical fibers, attenuation units, absorption, scattering dispersions.

Unit II: Optical Sources
Optical source properties, finding the energy from the voltage, finding the frequency from the wavelength of light, operating wavelength of optical sources, semiconductor light-emitting diodes and laser diodes, semiconductor material and device operating principles, light-emitting diodes, surface-emitting LEDS, edge-emitting LEDS, super luminescent diodes, laser diodes, comparison of LED and ILD. Fiber optic transmitters, basic optical transmitters, direct versus external modulation, fiber optic transmitter applications, digital applications, analog applications.

Unit III: Optical Detectors
Light detectors, Role of an optical detector, Detector characteristics: Responsivity, Noise Equivalent Power, Detectivity, Quantum efficiency, Detector response time, Linearity, Spectral response, Noise considerations: Johnson noise, Shot noise, 1/f noise, Photon noise, The PN junction photo diode - PIN photodetector - Avalanche photo diode construction characteristics and properties, APD Specifications, Applications of APD - comparison of performance noise sources - simple model of photo receiver - Its equivalent for circulation of noise SNR, Optical Receivers.

Unit IV: Optical Amplifiers
Basic Concepts, Gain Spectrum and Bandwidth, Gain Saturation, Amplifier Noise, Amplifier Applications, Semiconductor Optical Amplifiers, Raman Amplifiers, Erbium-Doped Fiber

Amplifiers, System Applications - Optical Pre-amplification, Noise Accumulation in Long-Haul Systems, ASE-Induced Timing Jitter, Accumulated Dispersive and Nonlinear Effects, WDM-Related Impairments.

Unit V: Optical Switching & Networks
Transport Networks, Applications, Requirements, Architectures, Technologies, and Solutions, Introduction to Optical Access Networks, Optical TDM, subscriber multiplexing (SCM), WDM and Hybrid multiplexing methods.

LIST OF EXPERIMENTS

1. Handling of Fibers.
2. Index profiles of Optical Fibers
3. Characteristics of LED.
4. Characteristics of Laser Diode.
5. Characteristics of Photo Detector.
6. Characteristics of APD
7. Measurement of Numerical Aperture.
8. Measurement of Attenuation.
9. Measurement of Bending Loss
10. Fiber Dispersion Measurement.
11. Study of BER and Q-Factor.
12. Characteristics of WDM Link.
13. Analog and voice communication through optical link.
14. Simulation & BER calculation using OPTSIM.
15. Responsivity of detector.
16. EDFA Design for D-WDM using OPTSIM.
17. Characteristics of AWGN & BSC channel.
18. BPSK Modulator and Demodulation.
19. Convolutional encoder & decoder.
20. Time Division Multiplexing.
21. Analog communication through optical link.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the basic principle and various types of transmission impairments of Fiber-optic communication.	K2
CO2	Make use of different types of optical sources in various applications.	K3
CO3	Apply the concepts of optical detection techniques for extracting the original signal.	K3
CO4	Develop amplified signals with various types of optical amplifiers.	K3
CO5	Infer the knowledge about WDM and optical networks for optical link design.	K4

TEXT BOOKS

1. Senior, J, M., "Optical Fiber Communications: Principles and Practice", 3rd edition, Pearson Education, 2010.
2. Keiser, G., "Optical Fiber Communications", 5th edition, McGraw Hill Education, 2017.

REFERENCE BOOKS

1. Agrawal, G. P., "Fiber Optic Communication Systems", 5th edition, Wiley, 2021.
2. Liu, M. M. K., "Principles and Applications of Optical Communications", 1st edition, McGraw Hill, 2010.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	3	-	-	-	-	-	-	-	-	-	-	2
CO2	3	2	2	3	-	-	-	-	-	-	-	-	-	-	2
CO3	3	2	2	3	-	-	-	-	-	-	-	-	-	-	2
CO4	3	2	-	3	-	-	-	-	-	-	-	-	-	-	2
CO5	3	2	2	3	-	-	-	-	-	-	-	-	-	-	2
Score	15	10	6	15	-	-	-	-	-	-	-	-	-	-	10
COM	3	2	2	3	-	-	-	-	-	-	-	-	-	-	2

PROGRAM ELECTIVE-III

Course Code	ECSE305
Course Title	Artificial Neural Networks
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To Stay updated with the latest research and advancements in the field of neural networks
- To Understand the basic concepts and architecture of artificial neural networks.
- To Learn different training algorithms and their applications.
- To Explore various types of neural networks including feedforward, convolutional, and recurrent networks.
- To Apply neural networks to solve practical problems.

COURSE CONTENTS

Unit I: Introduction to Artificial Neural Networks
History and evolution of neural networks, Biological inspiration, Basic concepts and terminology, Neuron model: McCulloch-Pitts neuron, Types of activation functions, Introduction to perceptron and multilayer perceptron (MLP)

Unit II: Fundamental Rules and Models of Artificial Neural Networks
Hebbian Learning Rule, Perceptron Learning Rule,Delta Learning Rule (Widrow-Hoff Rule), Competitive Learning Rule, General Architecture and Building Blocks of Pattern Classification Networks, Principle of Linear Separability, ,Hebb Net: Algorithm and Applications, Perceptron: Architecture, Algorithm, Applications, and Convergence Theorem, Adaline: Architecture, Algorithm, and Applications, Madaline: Architecture, MRI and MRII Algorithms.

Unit III: Pattern Association
Hebb Rule and Delta Rule for Pattern Association, Hetero-associative Memory Networks, Auto-associative Memory Networks, Hopfield Networks, Bidirectional Associative Memory Networks

Unit IV: Types of Neural Networks
Feedforward neural networks, Convolutional neural networks: Architecture and layers, convolution, pooling, fully connected, Applications in image processing, Recurrent neural networks (RNNs), Long short-term memory (LSTM), Gated recurrent units (GRUs), Applications in sequential data.

Unit V: Advanced Topics in Neural Networks
Transfer learning, Generative adversarial networks (GANs), Autoencoders and variational autoencoders (VAEs), Attention mechanisms and transformers, Reinforcement learning with neural networks.

LIST OF EXPERIMENTS

1. Implement a single McCulloch-Pitts neuron and demonstrate its ability to perform basic logical operations (AND, OR, NOT).
2. Implement the Perceptron learning algorithm and train it on a linearly separable dataset.
3. Implement the Adaline model using gradient descent and compare its performance with the Perceptron.
4. Implement a simple MLP for classification tasks using backpropagation.
5. Implement Various Activation Functions.
6. Implement Multilayer Feedforward Network.
7. Develop a multilayer perceptron for solving non-linearly separable problems.
8. Implement a Hopfield network and demonstrate its ability to retrieve stored patterns.
9. Implement a simple GAN and generate new data samples from the learned distribution..
10. Implement a recurrent neural network (RNN) or LSTM for time series forecasting.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand Fundamental Concepts of Neural Networks	K2
CO2	Designing and Implementing Neural Networks	K2
CO3	Applying Neural Networks to Real-World Problems	K6
CO4	Analyzing and Evaluating Neural Network Performance	K4
CO5	Exploring Advanced Topics and Emerging Trends	K6

TEXT BOOKS

1. Goodfellow, I., Bengio, Y., and Courville, A., “Deep Learning”, MIT Press, 2020.
2. Aggarwal, C. C., “Neural Networks and Deep Learning: A Textbook”, Springer, 2018.

REFERENCE BOOKS

1. Haykin, S., “Neural Networks and Learning Machines”, 3rd Edition, Pearson Education, 2018.
2. Ng, A., “Machine Learning Yearning”, DeepLearning.AI, 2018.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	3	-	3	-	2	2
CO2	3	3	-	-	2	-	-	-	-	3	-	3	-	-	2
CO3	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3
CO4	3	2	2	3	2	-	-	-	-	3	-	3	-	-	2
CO5	3	3	3	3	3	-	-	-	-	3	-	3	3	2	3
Score	15	13	8	9	10	-	-	-	6	15	-	15	6	4	10
COM	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3

Course Code	ECSE306
Course Title	Digital Image Processing
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To understand and explore the basics of Digital Image Processing.
- To understand concepts of intensity transformations and filtering in spatial domain.
- To implement frequency domain filters for image processing applications.
- To familiarize with the concepts of colour image processing.
- To visualize and implement basic computer vision algorithms.

COURSE CONTENTS

Unit I: Introduction Digital Image Processing
Elements of visual perception, Digital Image fundamentals: Image sensing and acquisition, Image sampling and quantization, Image Processing Toolbox in MATLAB: Digital Image Representation, Reading, Displaying and Writing images, Data Classes, Image Types, Converting between data classes and image types, array indexing, Some important standard arrays.

Unit II: Intensity Transformations and Spatial Filtering
Intensity Transformations Functions, Histogram Processing, Fundamentals of Spatial Filtering, Smoothening Spatial Filters, Sharpening Spatial Filters, Design of various Spatial Filters from Lowpass Filters, Spatial Enhancement Methods.

Unit III: Filtering in Frequency Domain
Sampling and the Fourier Transform of Sampled Functions, The Discrete Fourier Transform, Properties of the 2-D DFT and IDFT, Filtering in the Frequency Domain, Image Smoothing Using Lowpass Frequency Domain Filters, Image Sharpening Using Highpass Filters, Selective Filtering.

Unit IV: Image Transforms and Colour Image Processing
Image Transforms: DCT, DWT, Radon Transforms; Colour Image Processing: Color Models, Pseudocolor Image Processing, Full-Color Image Processing, Color Transformations, Color Image Smoothing and Sharpening, Noise in Color Images.

Unit V: Image Segmentation and Feature Extraction
Point, Line, and Edge Detection, Thresholding, Segmentation by Region Growing and by Region Splitting and Merging, Region Segmentation Using Clustering and Superpixels, Region Segmentation Using Graph Cuts, Principal Components as Feature Descriptors, Whole-Image Features, Scale-Invariant Feature Transform (SIFT).

LIST OF EXPERIMENTS

1. Convert RGB based images into Gray Scale Images using MATLAB.
2. Generate RGB components of the image using MATLAB.
3. Implement the Histogram Equalization using MATLAB.
4. Implement the design techniques of Non-linear Filtering using MATLAB.
5. Detect the Edges using Operators in MATLAB.
6. Implement Discrete Fourier Transform using MATLAB.
7. Implement Discrete Cosine Transform using MATLAB.
8. Implementing techniques of Filtering in frequency domain
9. Implement low pass filtering technique in frequency domain using MATLAB.
10. Implement high pass filtering technique in frequency domain using MATLAB.
11. Developing, creating and display of colour images.
12. Implement conversion between colour spaces.
13. Implement Image Segmentation using watershed transform using MATLAB.
14. Implement PCA algorithm on images.
15. Implement SIFT algorithm on images to extract the features.

Total Periods: 40 + 30 = 70

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate understanding of the fundamentals of digital image processing.	K2
CO2	Develop understanding and underlying techniques of Intensity transformations and filtering.	K3
CO3	Develop understanding of frequency domain processing techniques.	K3
CO4	Perform the colour image processing by applying colour transformations and use colour information in image segmentation	K3
CO5	Execute image segmentation and feature extraction	K3

TEXT BOOKS

1. Gonzalez, Rafael C. Digital image processing. Pearson education India, 2009.
2. Digital Image Processing Using MATLAB by Rafael C. Gonzalez, Richard E. Woods, and Steven L. Eddins

REFERENCE BOOKS

1. Fundamentals of Digital Image Processing by Anil K. Jain
2. Computer Vision: Algorithms and Applications by Richard Szeliski
3. Image Processing, Analysis, and Machine Vision by Milan Sonka, Vaclav Hlavac, and Roger Boyle.
4. Pattern Recognition and Image Analysis by Earl Gose, Richard Johnsonbaugh, and Steve Jost

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	3	-	-	-	-	-	-	-	-	1	3	2	1
CO2	3	2	2	-	2	-	-	-	-	-	-	1	2	1	1
CO3	3	2	2	1	2	-	-	-	-	-	-	2	1	1	2
CO4	3	2	2	2	-	-	-	-	-	-	-	2	3	2	2
CO5	3	3	3	2	2	-	-	-	-	-	-	1	2	2	3
COM	3	2	2	2	2	0	0	0	0	0	0	2	2	2	2

PROGRAM ELECTIVE-IV

Course Code	ECSE401
Course Title	Wireless Communication
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To enable students to acquire in-depth knowledge in the field of wireless communication.
- To familiarize with state-of-the-art standards used in wireless cellular systems.
- To develop students to critically analyze the problems in the field of wireless communication technology and find optimal solution.
- To study the various multiple access techniques.
- To learn the different protocols for GSM and CDMA techniques

COURSE CONTENTS

Unit I: The Cellular Concept- System Design Fundamentals
Evolution of mobile communications, Concept of frequency reuse, Channel Assignment Strategies, Channel and co- channel interference reduction factor, S/I ratio consideration and calculation for Minimum Co-channel and adjacent interference, Handoff strategies, Trunking and Grade of Service, Improving Coverage & Capacity in Cellular System-cell splitting, Cell sectoring, Repeaters, Micro cell zone concept.

Unit II: Mobile Radio Propagation Model, Small Scale Fading and Diversity
Free Space Propagation model, Reflection, Ground reflection model, Diffraction, Scattering, Link budget design, Max. Distance Coverage formula, Empirical formula for path loss, Indoor and outdoor propagation models, Small scale multipath propagation, Impulse model for multipath channel, Small scale multipath measurements, Parameters of mobile multipath channels, Types of small scale fading, Rayleigh and Ricean distribution, Statistical models for multipath fading channels and diversity techniques in brief.

Unit III: Multiple Access Techniques
Introduction, Frequency Division Multiple Access, Time Division Multiple Access, Spread Spectrum Multiple Access, Spread Division Multiple Access, Packer Radio, Carrier Sense Multiple Access Protocols, Capacity of Cellular Systems.

Unit IV: Wireless System
Introduction to CDMA, Fundamentals of CDMA Codes, Correlation Properties of Random CDMA Spreading Sequences, Spread-Spectrum Principles, Direct-Sequence Spread Spectrum (DSSS), DSSS System Model, RAKE Receivers, Frequency-Hopping Spread Spectrum (FHSS).

Unit V: Multicarrier Modulations
Data Transmission Using Multiple Carriers, Multicarrier Modulation with Overlapping Sub channels, Discrete Implementation of Multicarrier Modulation, The Cyclic Prefix, OFDM, Matrix Representation of OFDM, Peak-to-Average Power Ratio (PAPR), Frequency and Timing Offset, SC-FDMA, Subcarrier Mapping in SC-FDMA.

LIST OF EXPERIMENTS

1. Introduction to NI LabVIEW and USRP hardware.
2. M-ary QAM Modulation and Detection with BER performance.
3. Pulse Shaping and Matched Filtering.
4. Channel Estimation & Equalization.
5. Code Division Multiple access
6. Spread Spectrum – DSSS modulation and Detection
7. Synchronization- Frame Detection & Frequency Offset Correction.
8. Simulation of OFDM Modulation & Demodulation.
9. Frequency Domain Equalization.
10. Implementation of OFDM on USRP.

Total Periods: 40 + 28 = 68

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the cellular system design and technical challenges.	K2
CO2	Analyze the mobile radio propagation, fading, diversity concepts and the channel modeling.	K2
CO3	Analyze the design parameters, link design, smart antenna, beam forming and MIMO systems.	K3
CO4	Analyze Multiuser Systems, CDMA, WCDMA network planning and Spread spectrum techniques.	K3
CO5	Analyze Multicarrier Systems, OFDM, Synchronization, and SC-FDMA techniques.	K3

TEXT BOOKS

1. Rappaport, T. S., “Wireless Communication: Principles and Practice”, 2nd Edition, Pearson, 2010..
2. Aditya K J, “Principles of Modern Wireless Communication Systems Theory and Practice”, McGraw Hill Education, 2016.
3. Goldsmith, “Wireless Communications”, Cambridge university press, 2005.

REFERENCE BOOKS

1. Lee, W. C. Y., “Mobile Communication Engineering: Theory and Applications”, 2nd Edition, McGraw Hill Education, 2017.
2. Feher, K., “Wireless Digital Communications: Modulation and Spread Spectrum Applications”, 1st Edition, Prentice Hall India, 1995.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	-	-	-	-	-	-	-	-	-	-	-	-	2
CO2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	2
CO3	3	3	3	-	-	-	-	-	-	-	-	-	3	-	2
CO4	3	3	3	-	-	-	-	-	-	-	-	-	3	-	2
CO5	3	3	3	-	-	-	-	-	-	-	-	-	3	-	2
Score	15	13	11	-	-	-	-	-	-	-	-	-	9	-	10
COM	3	3	3	-	-	-	-	-	-	-	-	-	3	-	2

Course Code	ECSE402
Course Title	Antenna and Wave Propagation
L T P C	3-0-2-4
Course Type	SE

COURSE OBJECTIVES

- To learn basic concepts radiation from antenna element.
- To impart knowledge on principle of pattern multiplication in antenna arrays.
- To study different types of aperture and slot antennas.
- To study various special antennas and their polarization characteristics.
- To learn different wave propagation mechanisms and their applications.

COURSE CONTENTS

Unit I: Fundamentals of Radiation
Introduction to antennas, Definition of antenna parameters, Gain, Directivity, Effective aperture, Radiation Resistance, Band width, Beam width, Input Impedance. Matching, Baluns, Polarization mismatch, Antenna noise temperature, Radiation from oscillating dipole, Half wave dipole, Folded dipole, Yagi array.

Unit II: Antenna Arrays
N element linear array, Pattern multiplication, Broadside and End fire array, Concept of Phased arrays, Adaptive array, Basic principle of antenna Synthesis-Binomial array.

Unit III: Aperture and Slot Antennas
Radiation from rectangular apertures, Uniform and Tapered aperture, Horn antenna, Reflector antenna, Aperture blockage, Feeding structures, Slot antennas, Microstrip antennas, Radiation mechanism, Application, Numerical tool for antenna analysis.

Unit IV: Special Antennas
Principle of frequency independent antennas, Spiral antenna, Helical antenna, Log periodic. Modern antennas- Reconfigurable antenna, Active-antenna, Dielectric antennas, Electronic band gap structure and applications, Antenna Measurements-Test Ranges, Measurement of Gain, Radiation pattern, Polarization, VSWR.

Unit V: Wave Propagation
Propagation in free space. Propagation around the earth, surface wave-propagation, structure of the ionosphere, propagation of plane waves in ionized medium, Determination of critical frequency, MUF, Fading, tropospheric propagation, Super refraction.

LIST OF EXPERIMENTS

1. Introduction to Microwave CST studio suite.
2. Designing, simulation, and analysis of a Wire antennas.
3. Designing, simulation, and analysis of a Half-Wavelength dipole antenna.
4. Designing, simulation, and analysis of a folded dipole antenna.
5. Designing, simulation, and analysis of a Yagi antenna.
6. Designing, simulation, and analysis of a Horn antenna.
7. Designing, simulation, and analysis of a Spiral antenna.
8. Designing, simulation, and analysis of a Helical antenna.
9. Designing, simulation, and analysis of a microstrip patch antenna.
10. Designing, simulation, and analysis of a microstrip patch antenna array.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain fundamentals of radiation and mechanism of radiation from oscillating dipole.	K2
CO2	Develop different types of antenna arrays and study their radiation pattern.	K3
CO3	Apply numerical tools and techniques for analysis of radiation from different aperture and slot antennas.	K3
CO4	Categorize different types of antennas and examine their radiation characteristics.	K4
CO5	Assess and compare different mechanisms for wave propagation in free space.	K5

TEXT BOOKS

1. Balanis, C. A., “Antenna Theory: Analysis and Design”, 4th Edition, Wiley, 2016.
2. Collin, R. E., “Antennas and Radiowave Propagation”, 4th Edition, McGraw-Hill, 1985.

REFERENCE BOOKS

1. Krauss, J. D., “Antennas for All Applications”, 3rd Edition, McGraw-Hill, 2005.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	-	-	-	-	-	-	-	3	-	3	-	-	-
CO2	2	3	3	3	3	-	-	-	3	3	-	3	-	2	-
CO3	2	3	-	-	2	-	-	-	-	3	-	3	-	-	-
CO4	2	2	2	3	2	-	-	-	-	3	-	3	-	-	-
CO5	2	3	3	3	3	-	-	-	3	3	-	3	3	2	3
Score	10	13	8	9	10	-	-	-	6	15	-	15	3	4	3
COM	2	3	3	3	3	-	-	-	3	3	-	3	3	2	3

OPEN ELECTIVES

OPEN ELECTIVE-I

Course Code	SCMS301
Course Title	Artificial Intelligence
Number of Credits(L-T-P-C)	3-0-0-3
Course Type	MS

COURSE OBJECTIVES

- To learn the basics of Artificial Intelligence.
- To be able to mathematically represent knowledge.
- To learn the methods of solving problems in Artificial Intelligence.
- To learn various game playing and planning techniques.
- To learn semantic analysis and its application in NLP.

COURSE CONTENT

Unit I: Introduction to AI
Control strategies, Search strategies, Production system characteristics, Specialized production system, Problem-solving methods, Problem graphs, Matching, Indexing, and Heuristic functions, Hill Climbing, Depth-first and Breadth-first, satisfaction, Related algorithms, Measure of performance and analysis of search algorithms.

Unit II: Knowledge Representation
Knowledge representation, Knowledge representation using Predicate logic, Introduction to predicate calculus, Resolution, Use of predicate calculus, Knowledge representation using other Logic-Structured representation of knowledge.

Unit III: Reasoning
Production-based system, Frame-based system. Inference: Backward chaining, forward chaining, Rule value approach, Fuzzy reasoning, Certainty factors, Bayesian Theory, Bayesian Network, Dempster-Shafer theory.

Unit IV: Game Playing and Planning
Overview, Min-Max search procedure, Alpha-beta cut-offs, Iterative Deepening, Components of planning system, goal stack planning, non-linear planning, hierarchal planning and other planning techniques, reactive systems.

Unit V: Understanding and NLP
Introduction to Understanding, Understanding as constraint satisfaction, Introduction to NLP, Syntactic and Semantic analysis, Statistical NLP, and Spell Checking.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Learn the basic concepts of artificial intelligence.	K2
CO2	Learn the knowledge using predicate calculus.	K2
CO3	Identify problems that are amenably solved by AI methods.	K3
CO4	Apply various game-playing techniques in artificial intelligence.	K3
CO5	Understand semantic information in NLP applications.	K3

TEXT BOOKS

1. Russell, Norvig ”Artificial Intelligence: A Modern Approach”, Fourth Edition, Pearson Education, 2022.
2. Rich, Elaine, Kevin Knight, and Shivashankar B. Nair. *“Artificial intelligence”* . (2010).
3. Winston, Patrick Henry. *“Artificial intelligence”*. Addison-Wesley Longman Publishing Co., Inc., 1992.

REFERENCE BOOKS

1. Kheemani, Deepak, *“A First Course in Artificial Intelligence”*, McGraw Hill Education, First Edition, 2017.
2. Floridi L. *“The ethics of artificial intelligence: Principles, challenges, and opportunities”,2023*.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
Score	15	8	-	-	-	-	-	-	-	-	-	-	8	-	-
COM	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-

Course Code	SEMS301
Course Title	VLSI Technology
Number of Credits(L-T-P-C)	3-0-0-3
Course Type	MS

COURSE OBJECTIVES

- To learn about clean room and safety requirements for VLSI Technology.
- To learn different fabrication and characterization techniques.
- To learn and understand Oxidation and Lithography.
- To learn and understand Chemical Vapour Deposition techniques
- To learn and understand basic concepts of Nano-science and Nanotechnology.

COURSE CONTENT

Unit I: Environment for VLSI Technology
Clean room and safety requirements. Wafer cleaning processes and wet chemical etching techniques.

Unit II: Impurity incorporation
Solid State diffusion modeling and technology; Ion Implantation modeling, technology and damage annealing; characterization of Impurity profiles.

Unit III: Oxidation and Lithography
Kinetics of Silicon dioxide growth both for thick, thin and ultrathin films. Oxidation Technologies in VLSI and ULSI; Characterization of oxide films; High k and low k dielectrics for ULSI. Photolithography, E-beam lithography and newer lithography techniques for VLSI/ULSI; Mask generation.

Unit IV: Chemical Vapour Deposition techniques
CVD techniques for deposition of polysilicon, silicon dioxide, silicon nitride and metal films; Epitaxial growth of silicon; modeling and technology.

Unit V: Introduction To Nanotechnology
Basic principles of Nanoscience and Nanotechnology, Background of nanotechnology, types of nano-technology and nano-machines, top down and bottom up techniques, atomic manipulation-nanodots, semi-conductor quantum dots, self-assembly monolayers, Simple details of characterization tools- SEM, TEM, STM, AFM.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand material processing techniques and Pattern Transfer process	K2
CO2	Evaluate different fabrication and characterization techniques	K3
CO3	Demonstrates the concept behind thin film deposition, and characterization techniques	K3
CO4	Compare metal contact formation, interconnect, bonding and packaging	K3
CO5	Infer the overall landscape and characteristics of nanoscience and nanotechnology	K2

TEXT BOOKS

1. S. M. Sze (Ed), VLSI Technology, 2nd Edition, McGraw-Hill, 2017.
2. Datta, S., "Lessons from Nanoelectronics: A New Perspective on Transport", 2nd edition, World Scientific, 2018.

REFERENCE BOOKS

1. C.Y. Chang and S.M. Sze (Ed), ULSI Technology, McGraw-Hill Companies Inc.
2. S. K. Gandhi, VLSI fabrication Principles, John Wiley Inc.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	0	0	0	0	0	0	0	0	0	0	2
CO2	3	2	3	3	0	0	0	0	0	0	0	0	0	0	2
CO3	3	2	3	3	0	0	0	0	0	0	0	0	0	0	2
CO4	3	2	3	3	0	0	0	0	0	0	0	0	0	0	2
CO5	3	2	3	3	0	0	0	0	0	0	0	0	0	0	2
COM	3	2	3	3	0	0	0	0	0	0	0	0	0	0	2

OPEN ELECTIVE-II

Course Code	SCMS302
Course Title	Soft Computing
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MS

COURSE OBJECTIVES

- To understand the concepts of soft computing techniques.
- To understand the concepts of evolutionary computation.
- To impart adequate knowledge about Neural Networks.
- To understand the general principles of Fuzzy Logic Controllers.
- To apply soft computing algorithms for solving complex problems.

COURSE CONTENT

Unit I: Introduction To Soft Computing
Soft computing vs. Hard computing, Applications of soft Computing, Various types of Soft Computing techniques, Neuron Nerve structure and Synapse, McCulloch Pitts neuron model, Perceptron model, Neural network Architecture, Applications

Unit II: Evolutionary Computation
Historical Development of EC, Genetic programming- Evolutionary Strategies, features of Evolutionary computation, Advantages and Applications of Evolutionary Computation, Basic concept of Genetic algorithm, Genetic Operators, Comparison of Genetic Algorithm with Other Optimization Techniques, Advantages, Applications and Limitations of Genetic Algorithm.

Unit III: Multi-Layer Perceptron
Types of Neural Network, Advantages and Drawbacks, Applications of Single layer and multilayer feed forward networks, back propagation learning methods, effect of learning rule coefficient.

Unit IV: Introduction To Fuzzy Logic
Fuzzy Logic, Applications, Fuzzy set and classical set, membership functions, fuzzy terminologies: Linguistic terms, Support, core, normality, cross over, fuzzy singleton, alpha cut, strong alpha cut, fuzzy set operations (Union, Intersection, Complement, Algebraic product, Algebraic sum, Algebraic Difference, Multiplication of fuzzy sets by a crisp number, Power of Fuzzy Set, Bounded Sum, Bounded Difference, Cartesian Product, Equal Fuzzy Sets

Unit V: Emerging Trends
Particle Swarm Optimization, Applications, Fuzzy control Systems, Applications, Fuzzy controller, Fuzzy Inference System, Mamdani Approach, Defuzzification; Center of sum method,

Advantages, Drawbacks, Applications, Clustering methods, Fuzzy Classification, Clustering and pattern recognition

LIST OF EXPERIMENTS

1. Explore Pandas Library for Data Summarization
2. Demonstrate the basic concept of Perceptron using Python.
3. Demonstrate the use case of different transfer functions (Binary Step, Sigmoid, Relu, Tanh, Softmax) in Artificial Neural Network using Python.
4. Demonstrate the use of Perceptron for deploying Logic Gate: AND, OR, NOR, NAND, XOR
5. Demonstrate the solution of XOR GATE (Non-Linearly Separable Problem) using ANN.
6. Demonstrate Multi-Layer Perceptron Backpropagation using Python.
7. Implement the concept of Fuzzy Logic.
8. Demonstrate basic fuzzy operations using Python (Union, Intersection, Complement)
9. Demonstrate generation of fuzzy values using three different membership functions.
10. Demonstrate Genetic Algorithm using Python.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Experiment with various soft computing techniques and analyse the performance of neural network model using various evaluation metrics.	K4
CO2	Explain evolutionary computation and apply it for solving optimization problems.	K3
CO3	Experiment with genetic algorithm and apply genetic operators in various applications.	K3
CO4	Explain fuzzy systems and contrast fuzzification and defuzzification techniques.	K2
CO5	Classify and compare various soft computing methods.	K2

TEXT BOOKS

1. Deepa, S.N. and Sivanandam, S.N., *"Principles of Soft Computing, 2 nd Edition"*, Wiley India, 2011.
2. Tom, Mitchell, *"Machine Learning"*, McGraw-Hill, 2017.
3. Zimmermann H. J. *"Fuzzy set theory and its Applications"* Springer international edition, 2011.

REFERENCE BOOKS

1. Timothy, J. Ross, *"Fuzzy Logic with Engineering Applications, 3 rd Edition"*, Wiley India, 2010.
2. Shai, Shalev-Shwartz, Shai, Ben-David, *"Understanding Machine Learning from Theory to Algorithms"*, Cambridge University Press, 2014.
3. Malik H, Iqbal A, Yadav AK. *"Soft computing in condition monitoring and diagnostics of electrical and mechanical systems"*. Berlin/Heidelberg, Germany: Springer; 2020.
4. Lesot M, Marsala C. *"Fuzzy Approaches for Soft Computing and Approximate Reasoning: Theories and Applications"*. Switzerland: Springer. 2020.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	2	2	-	-	3	-	-	-	-	-	-	3	-	-	-
CO2	2	2	-	-	3	-	-	-	-	-	-	3	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	3	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Score	10	4	-	-	6	-	-	-	-	-	-	9	-	-	-
COM	2	2	-	-	3	-	-	-	-	-	-	3	-	-	-

Course Code	SEMS302
Course Title	Speech Signal Processing
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MS

COURSE OBJECTIVES

- To learn the basic concepts of articulatory and acoustic phonetics.
- To study different techniques for speech analysis.
- To impart knowledge on speech modelling techniques and their implementation issues.
- To study various speech recognition techniques.
- To learn different speech synthesis techniques and their applications.

COURSE CONTENT

Unit I: Basic Concepts of Speech
Fundamentals of Speech; Articulatory phonetics: production and classification of speech sounds; Acoustic phonetics: acoustics of speech production; Review of Digital Signal Processing; Short Time Fourier Transform (STFT); LPC Methods.

Unit II: Speech Analysis
Introduction to Speech Analysis, Understanding about feature extraction and pattern comparison techniques, Speech distortion measures, mathematical and perceptual, log–spectral distance, Cepstral distances, weighted cepstral distances and filtering, Likelihood distortions, spectral distortion using a warped frequency scale, LPC, PLP and MFCC coefficients, Time alignment and normalization: dynamic time warping, Multiple time: alignment paths.

Unit III: Speech Modelling
Introduction to Speech Modelling, Understanding Hidden Markov Models (HMMs), Markov Process, HMMS-2 evaluation, Optimal state sequence: viterbi search, Baum-Welch parameter re-estimation, Implementation issues.

Unit IV: Speech Recognition
Introduction to Speech Recognition, Architecture of a large vocabulary continuous speech recognition system, Acoustics and language models, N-gram language model, Context dependent sub-word units, Applications and present status.

Unit V: Speech Synthesis
Introduction to Speech Synthesis, Understanding the basics of Text-to-speech synthesis (TTS), Concatenative and waveform synthesis methods, Sub-word units for TTS, Understanding about intelligibility and naturalness, Role of prosody, Applications and present status.

LIST OF EXPERIMENTS

1. Introduction to MATLAB on Speech Signal Processing.
2. Speech acquisition and recording using MATLAB.
3. Analysis of Different Sounds (Phonemes) using MATLAB.
4. Implement of Short Term Time Domain Processing of Speech using MATLAB.
5. Implementation of Linear Predictive Coding (LPC) methods in MATLAB.
6. Fundamental frequency estimation in Speech signal using MATLAB.
7. Format synthesis MFCC extraction from Speech signal using MATLAB.
8. Cepstral Analysis of Speech Signal in MATLAB.
9. LPCC extraction from Speech signal using MATLAB.
10. Implementation of HMM models for Speech Modelling in MATLAB.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain fundamentals of speech processing and classify different speech sounds.	K2
CO2	Apply feature extraction and pattern comparison techniques of speech processing.	K3
CO3	Apply hidden markov model for optimal state sequence search and welch parameter re-estimation.	K3
CO4	Analyze architecture of a large vocabulary continuous speech recognition system.	K4
CO5	Compare and select methods for waveform and text-to-speech synthesis.	K5

TEXT BOOKS

1. Rabiner, L. and Schafer, R., “Theory and Applications of Digital Speech Processing”, 1st edition, Pearson Education, 2010.
2. Rabiner, L. R., Juang, B. H., and Yegnararayana, B., “Fundamentals of Speech Recognition”, 1st edition, Pearson Education, 2008

REFERENCE BOOKS

1. Jurafsky, D. and Martin, J. H., “Speech and Language Processing”, 2nd edition, Pearson Education, 2013.

2. Quatieri, Thomas F. ”Discrete-time speech signal processing: principles and practice”, Pearson Education India, 2002.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	2	2	-	-	-	-	-	-	-	3	-	3	-	-	-
CO2	2	3	3	3	3	2	-	-	-	3	2	3	-	2	-
CO3	2	2	3	3	1	-	-	-	-	3	-	3	-	-	-
CO4	2	2	2	3	2	-	-	-	-	3	-	3	-	-	-
CO5	2	3	3	3	3	-	-	-	3	3	-	3	3	2	3
Score	10	12	11	12	9	2	-	-	3	15	2	15	3	4	3
COM	2	3	3	3	2	-	-	-	3	3	-	3	3	2	3

OPEN ELECTIVE-III

Course Code	SCMS401
Course Title	Deep Learning
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MS

COURSE OBJECTIVES

- To understand the core principles of deep learning and neural networks.
- To develop proficiency in building and training convolutional neural networks (CNNs).
- To gain knowledge of sequence modeling techniques, including recurrent neural networks (RNNs) and LSTM networks.
- To explore advanced deep learning architectures like transformers and auto-encoders.
- To familiarize with the latest advancements in deep learning, including generative adversarial networks (GANs) and large language models.

COURSE CONTENT

Unit I: Basics of Deep Networks
Deep Networks: Gradient-based Learning, Hidden Units, Architecture Design, Back-Propagation, Regularization, Activation functions, Training Deep Models, Hyper-parameters, Optimization.

Unit II: Convolutional Networks
Motivation, Applications, Convolution Operations, CNN layers: Convolutional layer, activation layer, pooling layer, flattening, CNN architecture: 1D-CNN, 2D-CNN, 3D-CNN. Pre-trained models: VGG16, ResNet50.

Unit III: Sequence Modeling
Motivation, Applications, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence architecture, Deep Recurrent Networks, Recursive Neural Networks, Long Short-Term Memory (LSTM), Bi-LSTM.

Unit IV: Transformers and Autoencoders
Transformers: Motivation, Applications, Attention-mechanism, encoder-decoder architecture, BERT Model. Autoencoders: Undercomplete Autoencoders, Regularized Autoencoders, Representational power, Layer Size and Depth, Stochastic Encoders and Decoders, Denoising Encoders., Applications of Autoencoders.

Unit V: Advancements in Deep Learning
Introduction to Generative AI, Generative Adversarial Networks, Large Language Models, Diffusion models, Neuromorphic computing.

LIST OF EXPERIMENTS

1. Implementation of Perceptron Network
2. Implementation of Multilayer Perceptron (MLP) and Hyperparameter Tuning
3. Implementation of Multilayer Neural Network using Keras and Data Augmentation on MNIST Dataset
4. Implementation of Convolutional Neural Network (CNN) on MNIST Dataset.
5. Implementation of Pretrained Transfer Learning Models for Plant Disease Detection
6. Implementation of Autoencoder on MNIST Dataset for Handwritten Digit Recognition
7. Implementation of Recurrent Neural Networks (RNN) for Sentiment Analysis
8. Implementation of LSTM for Text Generation
9. Implementation of GAN for Generating Handwritten Digits on MNIST Dataset
10. Implementation of Capsule Networks for Image Classification
11. Implementation of U-Net for Image Segmentation
12. Implementation of Object Detection Using YOLOv7
13. Implementation of ResNet Architecture for Image Classification
14. Implementation of Spatial and Channel Attention in CNN
15. Implementation of Bidirectional LSTM for Named Entity Recognition (NER)

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Develop and deploy deep neural networks for real-world problem solving, mastering model training and optimization.	K3
CO2	Employ convolutional neural networks to ensure high accuracy and efficiency for different classification/regression task.	K2
CO3	Utilize sequence modeling to capture temporal dependencies and enhance performance in sequential prediction tasks.	K4
CO4	Implement advanced architectures like transformers and autoencoders for tasks such as language understanding, language translation.	K4
CO5	Explore the emerging technologies in deep learning like GANs, LLMs.	K2

TEXT BOOKS

1. Bengio, Yoshua, Ian Goodfellow, and Aaron Courville. *"Deep learning"*. Cambridge, MA, USA: MIT press, 2017.

2. Zhang, Aston, et al. *"Dive into deep learning"*. Cambridge University Press, 2023.

REFERENCE BOOKS

1. Wani, M. Arif, et al. *"Advances in deep learning"*. Springer, 2020.

2. Huang, Kaizhu, et al., *"Deep learning: fundamentals, theory and applications"*. Vol. 2. springer, 2019.

3. Chollet, Francois. *"Deep learning with Python"*. Simon and Schuster, 2021.

4. Foster, David. *"Generative deep learning"*. " O'Reilly Media, Inc.", 2022.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	1	-	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	3	3	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	3	3	-	-	3	3	3
Score	11	-	-	7	3	-	-	-	3	3	-	-	3	3	3
COM	3	-	-	3	3	-	-	-	3	3	-	-	3	3	3

Course Code	SEMS401
Course Title	Introduction to IoT
Number of Credits (L-T-P-C)	3-0-2-4
Course Type	MS

COURSE OBJECTIVES

- To understand the fundamentals of IoT and Networking concepts.
- To familiarize with IoT architecture, sensors, actuators, device design, and the considerations involved in building IoT-enabled systems.
- To provide knowledge about various IoT communication technologies, including constrained devices, low-power networks, and key protocols for data transmission.
- To explore how cloud, fog, and edge computing architectures support IoT applications.
- To examine practical IoT applications in fields such as agriculture, healthcare, and transportation.

COURSE CONTENT

Unit I: IOT Introduction and Fundamentals
Basics of networking (types of networks, layered models, addressing, TCP/IP transport layer), Introduction to the architecture of wireless sensor networks, Machine-to-Machine (M2M) communication and cyber physical systems. Introduction to IoT and its comparison with M2M, WSN and CPS. IoT networking components, Addressing strategies in IoT.

Unit II: IOT Architecture and Networking
Introduction to IoT Sensors and their characteristics, Sensing types and their considerations, Introduction to IoT Actuators, their types and characteristics, IoT processing topologies, their types and its importance, Data formatting, Processing topologies, IoT device design and selection considerations, Processing offloading, IoT connectivity technologies.

Unit III: IOT Communication Technologies
Introduction to nodes, Constrained nodes and network, and the type of devices, Low power and lossy networks, Infrastructure protocols, Discovery protocols, Data protocols, Identification protocols, Device management protocols, Sematic protocols, IoT interoperability standards and frameworks.

Unit IV: Cloud and FOG Computing in IOT
Introduction to cloud computing, Virtualization, Cloud Models, SLA in cloud computing, Cloud implementation in Sensor – Cloud, Introduction to fog computing and its architecture, Fog computing in IoT, Application of fog computing in IoT, Edge computing in IoT.

Unit V: IOT Applications and Data Analytics
IoT applications in agriculture, vehicular networks and healthcare, IoT analytics, Uses of machine learning in IoT, Advantages and challenges of ML in IoT, ML algorithms for IoT applications, Performance metrices for evaluating ML algorithms.

LIST OF EXPERIMENTS

1. Study the fundamental of IOT softwares and components.
2. Familiarization with Arduino/Raspberry Pi and perform necessary software.
3. To interface LED/Buzzer with Arduino/Raspberry Pi and write a program to turn ON LED for 1 sec after every 2 seconds.
4. To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection.
5. To interface DHT11 sensor with Arduino/Raspberry Pi and write a program to print temperature and humidity readings.
6. To interface motor using relay with Arduino/Raspberry Pi and write a program to turn ON motor when push button is pressed.
7. Write a program on Arduino/Raspberry Pi to retrieve temperature and humidity data from thingspeak cloud.
8. To install MySQL database on Raspberry Pi and perform basic SQL queries
9. Write a program to create UDP server on Arduino/Raspberry Pi and respond with humidity data to UDP client when requested.
10. Write a program to create TCP server on Arduino/Raspberry Pi and respond with humidity data to TCP client when requested.

Total Periods: 40 + 24 = 64

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Associate and classify the architecture of various communication systems.	K2
CO2	Elaborate the IoT infrastructure and data processing methodologies.	K2
CO3	Interpret the various networking protocols used in IoT.	K2
CO4	Acquire the concepts of fog and cloud computing in IoT.	K3
CO5	Illustrate the various real-life applications of IoT.	K3

TEXT BOOKS

1. Misra, S., Mukherjee, A. and Roy, A. Introduction to IoT. Cambridge University Press, 2021.

2. Serpanos, D. and Wolf, M. Internet-of-things (IoT) systems: architectures, algorithms, methodologies. Springer, 2017.

REFERENCE BOOKS

1. Xiao, P. Designing Embedded Systems and the Internet of Things (IoT) with the ARM mbed. John Wiley & Sons, 2018.

2. Hersent, O., Boswarthick D., and Elloumi, O., The Internet of Things: Key Applications and Protocols. John Wiley & Sons, 2011.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	2	3	3	3	2	2	2	2	1	3	1	3	3	3	3
CO2	2	3	2	3	3	2	-	-	-	3	2	3	-	2	2
CO3	2	2	3	3	1	-	-	-	-	3	-	3	3	2	2
CO4	2	3	3	3	3	-	-	-	-	3	-	3	-	2	2
CO5	2	3	3	1	-	-	-	-	-	3	-	3	-	2	2
Score	10	14	14	13	9	4	2	2	1	15	3	15	6	11	11
COM	2	3	3	3	3	2	2	2	1	3	3	3	3	3	3

OPEN ELECTIVE-IV

Course Code	SCMS402
Course Title	Cyber Physical Systems
Number of Credits(L-T-P-C)	3-0-0-3
Course Type	MS

COURSE OBJECTIVES

- To understand the basic principles behind cyber physical systems.
- To understand the control theory of cyber physical systems.
- To understand the hardware platform components for Cyber Physical Systems
- To understand the implementation issues related to cyber-physical systems.
- To understand applications of cyber physical systems to expose new opportunities.

COURSE CONTENT

Unit I: Introduction To Cyber-Physical Systems
Cyber-Physical Systems (CPS) in the real world, Basic principles of design and validation of CPS, Industry 4.0 AutoSAR, IIOT implications, Building Automation, Medical CPS, Basic Techniques, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques, Software Tools.

Unit II: Control Systems
Mathematical modelling of electrical systems, mechanical systems, electro-mechanical systems. Laplace transforms, transfer functions, electrical analogues of other dynamical systems. State-space modelling of dynamical systems. Block diagrams, block diagram reductions. Signal flow graph, Mason's gain formula. Linearity, time-invariance versus nonlinearity and time-variance. Linearization. Distributed parameter systems, Feedback Control Systems, Basic Techniques, Controller Timing, Controller Design for resource efficiency

Unit III: Platform Components For Cyber Physical Systems
CPS HW platforms - Processors, Sensors, Actuators CPS Network – Wireless Hart, CAN, Automotive Ethernet Scheduling Real Time CPS tasks: Table-driven and Event driven schedulers Hybrid schedulers. Dynamical Systems and Stability Controller Design Techniques Performance under Packet drop and Noise

Unit IV: CPS Implementation Issues
From features to automotive software components Mapping software components to ECUs. CPS Performance Analysis: Effect of scheduling, bus latency, sense and actuation faults on control performance, network congestion Building real-time networks for CPS Cyber Security Requirements, Attack Model, Countermeasures, Advanced Techniques, System Theoretic Approaches.

Unit V: Cyber Physical Systems Application Domain
Medical Cyber-Physical Systems, Energy Cyber-Physical Systems, Cyber-Physical Systems Built on Wireless Sensor Networks.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Construct the basic models for a given cyber physical system.	K3
CO2	Design a controller system to improve resource efficiency.	K6
CO3	Inspect the hardware requirements of various cyber physical systems.	K4
CO4	Inspect the implementation issues of the CPS.	K2
CO5	Assess the application of cyber physical systems in different domains.	K5

TEXT BOOKS

1. Rajkumar R., Dionisio de Niz and Klein M., *"Cyber-Physical Systems"*, First Edition, Addison-Wesley Professional, 2017.
2. Platzer, A., 2018. Logical foundations of cyber-physical systems (Vol. 662). Cham: Springer.

REFERENCE BOOKS

1. Rajeev Alur, *"Principles of Cyber-Physical Systems"*, MIT Press, 2015.
2. Roy S. and Sajal K. Das, *"Principles of Cyber-Physical Systems: An Interdisciplinary Approach"*, Cambridge University Press, 2020.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	3	3	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	2	-	3	2	-	-	-	-	-	-	-	3	-	-
Score	10	10	3	6	2	-	-	-	-	-	-	-	6	-	-
COM	2	2	3	3	2	-	-	-	-	-	-	-	3	-	-

Course Code	SEMS402
Course Title	Biomedical Signal Processing
Number of Credits(L-T-P-C)	3-0-0-3
Course Type	MS

COURSE OBJECTIVES

- To learn types, sources, acquisition, and analysis of biomedical signals.
- To learn electrocardiography and electromyography, including systems, signals, and processing techniques.
- To learn to identify artifacts and noise in various biosignal processes.
- To understand digital filtering, FIR and IIR filters, and their design.
- To Learn signal averaging, ECG QRS detection, and related analysis techniques.

COURSE CONTENT

Unit I: Introduction to Biomedical Signals
The nature of biomedical signals, examples of biomedical signals, action potential, electroneurogram (ENG), electromyogram signal (EMG), electrocardiogram (ECG), electroencephalogram (EEG), event-related potentials (ERPs), The electrogastrogram (EGG), phonocardiogram (PCG), carotid pulse (CP), speech signal, objectives of biomedical signal analysis, difficulties in biomedical signal processing, what is medical instrument, computer- aided diagnostics.

Unit II: Electrocardiography
Basic electrocardiography, ECG leads systems, ECG signal characteristics, concurrent, coupled and correlated processes.
ELECTROMYOGRAPHY: Muscular system, Electrical signals of the motor unit and gross muscle, Human motor coordination system, Electrodes, Correlation of force and work; EMG integrators, Signals conditioning & processing.

Unit III: Artefacts In Biosignals
Random noise, structured noise, and physiological interference, stationary Vs non-stationary processes. Noise in event-related potentials, high-frequency noise in ECG, motion artifacts in ECG, power-line interference in ECG signals, and maternal interference in fetal ECG.

Unit IV: Basics Of Digital Filtering
Digital filters, z transform, elements of a digital filter, types of digital filters, the transfer function of difference equation, the z-plane pole-zero plot, the rubber membrane concept.
FINITE IMPULSE RESPONSE FILTERS: Characteristics of FIR filters, smoothing filters, notch filters, derivatives, window design, frequency sampling, and minimax design.
INFINITE IMPULSE RESPONSE FILTERS: Generic equations of IIR filters, simple one-pole example, integrators, and design methods for two-pole filters.

Unit V: Signal Averaging
Basics of signal averaging, signal averaging as a digital filter, a typical average, software for signal averaging, limitations of signal averaging. ECG QRS DETECTION: Power spectrum of the ECG, band pass filtering techniques, differentiation techniques, template matching techniques, A QRS detection algorithm. ECG interpretation, ST- segment analyzer, portable arrhythmia monitor.

Total Periods: 40

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Knowledge of fundamental background for biomedical signal processing leading towards automatic disease diagnostics.	K2
CO2	Interpret ECG and EMG systems, signals, and processing techniques effectively.	K2
CO3	Detect and manage random, structured, and physiological artifacts in various biosignal processes.	K6
CO4	Understanding of basic digital filtering techniques and ability to design various types of filters.	K4
CO5	Ability to detect and analyze ECG signals by QRS algorithm.	K4

TEXT BOOKS

1. Rangayyan, R. M., *"Biomedical Signal Analysis"*, 2nd edition, Wiley, 2016.
2. Tompkins, W. J., *"Biomedical Digital Signal Processing: C Language Examples and Laboratory. Experiments for the IBM PC"*, Prentice-Hall India, 1998.
3. D C Reddy, *"Biomedical Digital Signal Processing"*, McGraw Hills, 2005.

REFERENCE BOOKS

1. Bruce, E. N., *"Biomedical Signal Processing and Signal Modeling"*, Wiley, 2007.
2. Semmlow, J. L. and Griffel, B., *"Biosignal and Medical Image Processing"*, 3rd edition, CRC press, 2014.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	3	-	3	-	-	-
CO2	3	3	-	-	2	-	-	-	-	3	-	3	-	-	2
CO3	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3
CO4	3	2	2	3	2	-	-	-	-	3	-	3	-	-	2
CO5	3	3	3	3	3	-	-	-	3	3	-	3	3	2	3
Score	15	13	8	9	10	-	-	-	6	15	-	15	6	4	10

Course Code	SCMS403
Course Title	Introduction to Federated Learning
Number of Credits(L-T-P-C)	3-1-0-4
Course Type	MS

COURSE OBJECTIVES

- To learn distributed machine learning.
- To explore the basics of federated learning.
- To learn with the various types of Federated learnings.
- To apply the federated learning concepts in various applications like transfer learning, computer vision, and natural language processing.
- To explore with different federated learning frameworks for their implementation

COURSE CONTENT

Unit I: Distributed Machine Learning
Introduction to Distributed Learning, Scalability- motivated Distributed Machine Learning, Privacy Motivated Distributed Machine Learning, Privacy-Preserving Distributed Machine Learning, Privacy-Preserving Gradient Descent, Distributed Learning Vs Federated Learning.

Unit II: Federated Learning
Federated Learning as a solution: Motivation and Challenges, Current development in Federated Learning, Classification of Federated Learning, Frameworks of Federated Learning, Privacy-preserving Machine Learning, Privacy-preserving Machine Learning vs Secure Machine Learning, Threats and Security Model.

Unit III: Horizontal and Vertical Federated Learning
Horizontal Federated Learning: Definition, Architecture, Federated Averaging algorithm, Improvement in Federated Averaging algorithm, Research Challenges. Vertical Federated Learning: Definition, Architecture, Algorithms, and Challenges.

Unit IV: Federated Learning for Computer Vision and Natural Language Processing
Federated Transfer Learning (FTL), FTL Framework and Challenges. Federated CV: Introduction, Application, and Challenges. Federated NLP: Introduction, Application, and Challenges. Federated Recommendation: Recommendation Model, Federated Recommendation Systems, Application, and Challenges.

Unit V: Federated Learning: Case Studies and Tools
Case Studies: Finance, Healthcare, Education, Urban Computing, and Smart City. Federated Learning Tools: TensorFlow Federated, Layers of TensorFlow Federated, Installation and Setup, Model Building, Computation Building, and Model Aggregation.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the basis of distributed machine learning.	K1
CO2	Understand the basics of federated learning.	K1
CO3	Apply suitable federated learning techniques for an application.	K3
CO4	Design and develop an efficient federated learning system for an application.	K5
CO5	Implement an effective learning system for the various application domain.	K4

TEXT BOOKS

1. Qiang Yang, Yang Liu, Yong Cheng, Yan Kang, Tianjian Chen, and Han Yu: *"Federated Learning"*, Morgan and Claypool Publishers, 2020.
2. Yaochu Jin, Hangyu Zhu, Jinjin Xu, and Yang Chen: *"Federated Learning: Fundamentals and Advances"*, Springer, 2020.

REFERENCE BOOKS

1. Saravanan Krishnan, A. Jose Anand, R. Srinivasan, R. Kavitha, S. Suresh. *"Handbook on Federated Learning Advances, Applications and Opportunities"*, CRC Press, 2024.
2. Ziller, Alexander, et al. *"Pysyft: A library for easy federated learning."* Federated Learning Systems: Towards Next-Generation AI (2021): 111-139.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	1	-	-	-	-	-	-	-	3	-	3
CO2	3	3	3	3	1	-	-	-	-	-	-	-	3	-	3
CO3	3	2	3	3	1	-	-	-	-	-	-	-	3	-	2
CO4	3	2	3	3	1	-	-	-	-	-	-	-	3	-	2
CO5	3	3	3	3	1	-	-	-	-	-	-	-	3	-	3
Score	15	13	15	15	5	-	-	-	-	-	-	-	15	-	13
COM	3	3	3	3	1	-	-	-	-	-	-	-	3	-	3

Course Code	SEMS403
Course Title	Quantum Computing
Number of Credits (L-T-P-C)	3-1-0-4
Course Type	MS

COURSE OBJECTIVES

- To learn the mathematical foundations of quantum computing.
- To understand the basic principles of model of computations in quantum world.
- To learn the key concepts of quantum mechanics for solving computing problems.
- To apply quantum mechanics concepts for solving significant computing problems.
- To understand quantum noise and correction techniques.

COURSE CONTENT

Unit I: Foundation
Overview of traditional computing, Church-Turing Thesis, Circuit model of computation, reversible computation, Dirac notation, complex inner and Hilbert spaces, Hermitian and Unitary operators, Dual vectors, tensor products, polar and singular value decompositions, the spectral theorem, Schmidt decomposition theorem.

Unit II: Quantum Model of Computation
Postulates of quantum mechanics, state space, time evolution of a closed system, measurement, dynamics, composite systems, Bits and Qubits as vector spaces, unary quantum operators – single qubit gates, Bloch sphere, mixed states and density matrices, multiple qubits and multi-qubit gates, Universal quantum gates, quantum black-box model, quantum circuit complexities – gates, qubits, depth, query.

Unit III: Quantum Computations
No-cloning theorem, dense coding, quantum teleportation, quantum parallelism and entanglement, phase kickback, Deutsch algorithm, Deutsch-Jozsa algorithm, Bernstein-Vazirani algorithm, Simon's algorithm, Quantum Key Distribution – BB84 protocol, Simulation of quantum systems

Unit IV: Quantum Algorithms Quantum integral transform, quantum Fourier transform (QFT), application of QFT – Period-finding, Shor's factorization algorithm, quantum phase estimation, discrete logarithm solution, hidden subgroup solution, Grover's search, optimality of quantum search, amplitude amplification, quantum counting, minimum and maximum finding algorithms.

Unit V: Quantum Error Correction
Decoherence, open quantum system, errors – bit-flip, phase-flip, depolarizing, amplitude-damping,

Lindblad equation, Quantum Error Correction Codes (QECC) – 3-qubit bit-flip QECC, 3-qubit phase-flip QECC, Shor's 9-qubit QECC, Hamming QECC, 7-qubits and 5-qubits QECC.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the fundamentals of quantum computing and relevant mathematical ideas.	K2
CO2	Understand formal model of computation for quantum computing and components for constructing quantum circuits.	K2
CO3	Explain primary concepts of quantum computing used for solving mathematical and computing problems.	K3
CO4	Solve significant computing problems using quantum mechanical phenomenon.	K3
CO5	Understand decoherence and error correction techniques for fault tolerant quantum systems.	K2

TEXT BOOKS

1. M. A. Nielsen and I. L. Chuang, "Quantum Computation and Quantum Information," Cambridge University Press, 2010.

2. N. D. Mermin, "Quantum Computer Science: An Introduction (1st ed.)," Cambridge University Press, 2007.

REFERENCE BOOKS

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CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	3	3	-	-	-	-	-	-	-	2	3	2	-
CO3	3	2	-	3	-	-	-	-	-	-	-	-	-	2	-
CO4	-	2	-	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	2	3	3	3	-	-	-	-	-	-	2	3	2	-
Score	11	12	9	9	3	-	-	-	-	-	-	4	-	9	8
COM	3	3	3	3	-	-	-	-	-	-	-	2	-	3	2


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